Honours Analysis 2 (Math 255) Tutorial Notes

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1 First Tutorial

Let us begin with a brief recall of the key concepts seen in class. A metric space is a pair (X, d) where X is a non-empty set and $d : X \times X \to [0, \infty)$ is a *metric* (or distance-function) satisfying the following:

(i) $d(x, y) = 0$ if and only if $x = y$ (nondegeneracy);

(ii) $d(x, y) = d(y, x)$ for all $x, y \in X$ (symmetry);

(iii) $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in X$.

We note that (iii) is called the *triangle inequality*. As seen in class, any normed vector space V inherits a metric structure from its norm. Indeed, V is automatically a metric space when equipped with the metric $d(x, y) = \|x - y\|$. The most frequently encountered metric space is $\mathbb{R}^n$ with its usual norm and induced metric. However, you've now seen, there are more "exotic" instances of such spaces. For instance, on any non-empty set X, one may define the so-called *discrete* metric.

Example 1.1 (The Discrete Metric)**.** Given any non-empty set X, we can define the *discrete* metric on X by putting

$$d(x, y) := \begin{cases} 1 & \text{if } x \neq y, \\ 0 & \text{if } x = y. \end{cases}$$

Even on $\mathbb{R}$, which naturally comes equipped with the familiar Euclidean metric (the absolute value metric you worked with in Analysis 1: $d(x, y) := |x - y|$), the discrete metric defined above will produce a space with very different convergent sequences, continuous functions, etc. This will be explored in greater detail as the course progresses. In fact, a major goal in this course is to familiarize students with abstract metric spaces, aiming to ensure students are comfortable working in spaces where a "point" might be a sequence, or a function. Before we begin exploring anything of this type, however, let us begin with an exercise on $\mathbb{R}$, in which we are asked to decide whether or not a given function $\mathbb{R} \times \mathbb{R} \to [0, \infty)$ is a valid metric.

Example 1.2. Decide whether each of the functions $d : \mathbb{R} \times \mathbb{R} \to [0, \infty)$ defines a valid metric on $\mathbb{R}$. In all cases, prove your answer.

(1) $d(x, y) := (x - y)^2$,

(2) $d(x, y) := \sqrt{|x - y|}$,

(3) $d(x, y) := \left|x^2 - y^2\right|$,

(4) $d(x, y) := |x - 2y|$,

(5) $d(x, y) := \dfrac{|x - y|}{1 + |x - y|}$.

Remark 1.1. If we are trying to show that a function is *not* a metric, it suffices to show that even *one* of the axioms fails. The solution we give below is overkill, as it checks every single property in all cases, regardless of the claim we are making. This would not be necessary if you were writing the solution for an assignment or an exam, but we nonetheless include the full details for the sake of exposition.

Solution.

(1) First, note that this function satisfies the nondegeneracy condition $d(x, y) = 0$ if and only if $x = y$. Indeed,

$$d(x, y) = 0 \iff (x - y)^2 = 0 \iff x - y = 0 \iff x = y.$$

Moreover, the function d is symmetric:

$$d(x, y) = (x - y)^2 = (-(y - x))^2 = (-1)^2(y - x)^2 = (y - x)^2 = d(y, x).$$

However, d is *not* a metric because the triangle inequality fails. Indeed, notice that

$$4 = (-1 - 1)^2 = d(-1, 1) \not\leq d(-1, 0) + d(0, 1) = (-1 - 0)^2 + (0 - 1)^2 = 1 + 1 = 2$$

(2) We claim that d is a metric. First, for any $x, y \in \mathbb{R}$ we have

$$d(x, y) = 0 \iff \sqrt{|x - y|} = 0 \iff |x - y| = 0 \iff x = y.$$

In addition, $d(x, y) = \sqrt{|x - y|} = \sqrt{|y - x|} = d(y, x)$ and so symmetry holds as well. To see that the triangle inequality holds, let $x, y, z \in \mathbb{R}$ be arbitrary and notice that

$$\begin{aligned} d(x, y) = \sqrt{|x - y|} &= \sqrt{|x - z + z - y|} \\ &\leq \sqrt{|x - z| + |z - y|} \\ &\leq \sqrt{|x - z|} + \sqrt{|z - y|}. \end{aligned}$$

Here, we have used both that $\sqrt{x}$ is increasing on $[0, \infty)$, and that this function is subadditive, i.e. $\sqrt{x + y} \leq \sqrt{x} + \sqrt{y}$ for all $x, y \geq 0$.

(3) Obviously, $d(x, x) = \left|x^2 - x^2\right| = 0$, but the converse does not hold. Indeed,

$$d(1, -1) = \left|1^2 - (-1)^2\right| = |1 - 1| = 0$$

and, yet, $-1 \neq 1$. Thus, the nondegeneracy condition fails and d is *not* a metric. Nonetheless, d satisfies the symmetry requirement and the triangle inequality. To see the first, simply observe that $d(x, y) = \left|x^2 - y^2\right| = \left|y^2 - x^2\right| = d(y, x)$. As for the triangle inequality, given $x, y, z \in \mathbb{R}$ we have

$$\begin{aligned} d(x, y) = \left|x^2 - y^2\right| &= \left|x^2 - z^2 + z^2 - y^2\right| \\ &\leq \left|x^2 - z^2\right| + \left|z^2 - y^2\right| \\ &= d(x, z) + d(z, y). \end{aligned}$$

(4) Notice that $d(1,1) = |1 - 2\cdot 1| = |1-2| = |-1| = 1 \neq 0$, and so d automatically fails the nondegeneracy condition, and is *not* a metric. This function is actually a very poor candidate for a metric, as it fails each of the three conditions. Notice that $d(x,y) = d(y,x)$ is equivalent to

$$\begin{aligned} |x-2y| = |y-2x| &\iff (x-2y)^2 = (y-2x)^2 \iff x^2 - 4xy + 4y^2 = y^2 - 4xy + 4x^2 \\ &\iff x^2 = y^2 \\ &\iff |x| = |y|\,. \end{aligned}$$

Thus, $d(x,y) = d(y,x)$ if and only if $|x| = |y|$. Since a metric requires that $d(x,y) = d(y,x)$ for *all* $x, y \in \mathbb{R}$, we also see that d fails the symmetry requirement. Finally, the triangle inequality fails as well because

$$d(10,0) = |10 - 2\cdot 0| = 10 \not\leq 5 = |10 - 2\cdot 5| + |5 - 2\cdot 0| = d(10,5) + d(5,0).$$

(5) We claim that this function d is a valid metric on $\mathbb{R}$. First, given $x, y \in \mathbb{R}$, we write

$$d(x,y) = 0 \iff \frac{|x-y|}{1+|x-y|} = 0 \iff |x-y| = 0 \iff x = y.$$

Consequently, the first condition (nondegeneracy) is valid. As for symmetry, we have

$$d(x,y) = \frac{|x-y|}{1+|x-y|} = \frac{|y-x|}{1+|y-x|} = d(y,x)$$

for all $x, y \in \mathbb{R}$. Hence, d is symmetric. Finally, for the triangle inequality we argue via a similar strategy as for part (2) of this problem. To achieve this, we first make some observations on the function

$$f : [0,\infty) \to [0,1), \qquad f(t) = \frac{t}{1+t} = 1 - \frac{1}{1+t}.$$

It is straightforward to check that f is an increasing function. Moreover, given $s, t \geq 0$ there holds

$$f(s) + f(t) = \frac{s}{1+s} + \frac{t}{1+t} \geq \frac{s}{1+s+t} + \frac{t}{1+s+t} = \frac{s+t}{1+s+t} = f(s+t). \tag{1.1}$$

Consequently, f is also *subadditive*. Equipped with these observations, for any $x, y, z \in \mathbb{R}$, we may write

$$\begin{aligned} d(x,y) = \frac{|x-y|}{1+|x-y|} = f(|x-y|) &= f(|x-z+z-y|) \\ &\leq f(|x-z| + |z-y|) && \text{since } f \text{ increases} \\ &= f(|x-z|) + f(|z-y|) && \text{by (1.1)} \\ &= \frac{|x-z|}{1+|x-z|} + \frac{|z-y|}{1+|z-y|} \\ &= d(x,z) + d(z,y). \end{aligned}$$

Thus, the triangle inequality has also been verified and, as such, d does define a valid metric on $\mathbb{R}$.

□

Remark 1.2. In the tutorial, there were questions about when this trick with a function is successful in producing a metric on $\mathbb{R}$. Namely, when does the rule $d(x,y) := f(|x-y|)$ produce a valid metric on $\mathbb{R}$? Which conditions on f are sufficient to ensure this is a metric? Clearly, since $|x-y| \geq 0$ no matter which $x, y \in \mathbb{R}$ we choose, we restrict the domain of f to $[0,\infty)$ without losing any generality. In addition, we require that f only take values in $[0,\infty)$, since otherwise $d(x,y) := f(|x-y|)$ could achieve negative values, which is not allowed. Beyond this, if we require that $f : [0,\infty) \to [0,\infty)$ satisfy

(1) $f(x) = 0$ if and only if $x = 0$;

(2) f is increasing on $[0,\infty)$, i.e. $f(x) \leq f(y)$ whenever $0 \leq x \leq y$;

(3) f is subadditive, i.e. $f(x+y) \leq f(x) + f(y)$ for all $x, y \geq 0$;

then $d(x,y) := f(|x-y|)$ is a metric. Indeed, if $d(x,y) = 0$, then $f(|x-y|) = 0$ implies by condition (1) that $|x-y| = 0$ so that $x = y$. Of course, $d(x,x) = 0$ by inspection and so the nondegeneracy condition of the metric is satisfied. As for symmetry:

$$d(x,y) = f(|x-y|) = f(|y-x|) = d(y,x).$$

Thus, d inherits it's symmetry from the argument $|x-y|$. Finally, for the triangle inequality, we use properties (2)-(3) as in the previous example:

$$\begin{aligned} d(x,y) = f(|x-y|) = f(|x-z+z-y|) &\leq f\left(|x-z| + |z-y|\right) \\ &\leq f(|x-z|) + f(|z-y|) \\ &= d(x,z) + d(z,y). \end{aligned}$$

It follows that d is a valid metric on $\mathbb{R}$.

Remark 1.3 (A Condition for Subadditivity). Within the tutorials, there was significant interest in this subadditivity property used in Example 1.2 and Remark 1.2, and for good reason! Notably, someone asked which conditions on a function can guarantee subadditivity. As I described, the notion of subadditivity is related to *concavity*. Formally, if we assume that $f : [0,\infty) \to [0,\infty)$ is *concave*, i.e. that

$$f(tx + (1-t)y) \geq tf(x) + (1-t)f(y), \quad \forall x, y \geq 0,\ t \in [0,1], \tag{1.2}$$

and that $f(0) \geq 0$, the function f will be subadditive. To see this, let $a, b \geq 0$ be arbitrary. To establish the subadditivity of f, we must demonstrate that

$$f(a+b) \leq f(a) + f(b).$$

Notice that if $a = b = 0$ then the above is simply $f(0) \leq 2f(0)$, which is valid being that $f(0) \geq 0$. Thus, we may assume that one of a or b is non-zero, which implies that $a+b > 0$. Then, invoking (1.2) for $x = a+b$ and $y = 0$, and $t = a/(a+b) \in [0,1]$, we find that

$$\begin{aligned} f(a) = f(t \cdot (a+b) + (1-t)0) &\geq tf(a+b) + (1-t)f(0) \\ &\geq tf(a+b) \\ &= \frac{a}{a+b} f(a+b). \end{aligned}$$

In a similar vein, an application of (1.2) with $x = a+b$ and $y = 0$ but using $t = b/(a+b) \in [0,1]$, we obtain

$$\begin{aligned} f(b) = f\left(t(a+b) + (1-t)\cdot 0)\right) &\geq tf(a+b) + (1-t)f(0) \\ &\geq tf(a+b) \\ &= \frac{b}{a+b} f(a+b). \end{aligned}$$

Thus, we have both

$$f(a) \geq \frac{a}{a+b} f(a+b) \quad \text{and} \quad f(b) \geq \frac{b}{a+b} f(a+b).$$

Summing these inequalities yields

$$f(a) + f(b) \geq \frac{a+b}{a+b} f(a+b) = f(a+b).$$

Being that $a, b \geq 0$ were arbitrary, it follows that f is subadditive.

For the purposes of providing some more interesting examples which challenge our understanding of distances, let us consider two different metrics on the space of sequences.

Example 1.3. Consider the space $X = \mathbb{R}^{\mathbb{N}}$ of real sequences [1], i.e. let

$$X := \{(x_n) : (x_n) \text{ is a sequence of real numbers}\}.$$

[1] Writing $\mathbb{R}^{\mathbb{N}}$ is simply a matter of notation, and not an "algebraic operation" on sets as one may first believe. In the same way that $\mathbb{R}^2$ simply means the collection of all pairs (x, y), with $x, y \in \mathbb{R}$, we write $\mathbb{R}^{\mathbb{N}}$ to mean the set of all infinite tuples with coordinates counted by the natural numbers, i.e.

$$\mathbb{R}^{\mathbb{N}} := \left\{(x_1, x_2, x_3, \ldots, x_n, \ldots) : x_j \in \mathbb{R}, \forall j \in \mathbb{N}\right\}.$$

Equivalently, $\mathbb{R}^{\mathbb{N}} = \prod_{n=1}^{\infty} \mathbb{R}$ in much the same way that $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ and

$$\mathbb{R}^n = \underbrace{\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}}_{n \text{ times.}}$$

That is, an element $x \in X$ is a sequence $x = (x_n)_{n\in\mathbb{N}}$ of real numbers. We show that

$$d(x,y) = \sum_{n=1}^{\infty} 2^{-n} \frac{|x_n - y_n|}{1 + |x_n - y_n|}$$

is a metric on X. First, observe that d is indeed a well defined mapping $X \times X \to [0,\infty)$ because

$$0 \leq \sum_{n=1}^{\infty} 2^{-n} \frac{|x_n - y_n|}{1 + |x_n - y_n|} \leq \sum_{n=1}^{\infty} 2^{-n} = 1.$$

(See Remark 1.4 for more details if this is not clear).

(i) It is clear that $d(x,x) = 0$ for any $x = (x_n) \in X$. Conversely, if $d(x,y) = 0$ for any two sequences $x = (x_n), y = (y_n)$ of real numbers then

$$0 = d(x,y) = \sum_{n=1}^{\infty} 2^{-n} \frac{|x_n - y_n|}{1 + |x_n - y - n|}$$

implies that every term appearing in the series above, is necessarily 0. That is,

$$2^{-n} \frac{|x_n - y_n|}{1 + |x_n - y_n|} = 0$$

for each $n \in \mathbb{N}$. Thus $|x_n - y_n| = 0$ for all $n \in \mathbb{N}$ so $x_n = y_n$ at each index n; i.e. $x = y$.

(ii) The equality $d(x,y) = d(y,x)$ for all $x, y \in X$ is trivially true by symmetry of the absolute value function. Certainly,

$$d(x,y) = \sum_{n=1}^{\infty} 2^{-n} \frac{|x_n - y_n|}{1 + |x_n - y_n|} = \sum_{n=1}^{\infty} 2^{-n} \frac{|y_n - x_n|}{1 + |y_n - x_n|} = d(y,x).$$

(iii) Finally, to establish the triangle inequality, we borrow some observations made in Example 1.2-(5). Recalling the observations made about the function f declared therein, we see that for arbitrary $x, y, z \in X$ there holds

$$\begin{aligned}
d(x,y) &= \sum_{n=1}^{\infty} 2^{-n} f(|x_n - y_n|) \\
&\leq \sum_{n=1}^{\infty} 2^{-n} f(|x_n - z_n| + |z_n - y_n|) && \text{since } f \text{ increases} \\
&\leq \sum_{n=1}^{\infty} 2^{-n} \left(f(|x_n - z_n|) + f(|z_n - y_n|)\right) && \text{by (1.1)} \\
&= \sum_{n=1}^{\infty} 2^{-n} f(|x_n - z_n|) + \sum_{n=1}^{\infty} 2^{-n} f(|z_n - y_n|) \\
&= d(x,z) + d(z,y).
\end{aligned}$$

Summarizing, we have shown that d is a valid metric on $X = \mathbb{R}^{\mathbb{N}}$, the space of real sequences.

Remark 1.4. Recall from analysis 1 that an infinite series $\sum_{n=1}^{\infty} a_n$, where (a_n) is a *sequence* of real numbers, is said to converge if the sequence of partial sums $\left(\sum_{n=1}^{N} a_n\right)_{N\in\mathbb{N}}$ converges as $N \to \infty$. In this case, for every pair $(x, y) \in X \times X$, and each $n \in \mathbb{N}$, there holds

$$0 \leq 2^{-n} \frac{|x_n - y_n|}{1 + |x_n - y_n|} \leq 2^{-n}$$

whence

$$0 \leq \sum_{n=1}^{N} 2^{-n} \frac{|x_n - y_n|}{1 + |x_n - y_n|} \leq \sum_{n=1}^{N} 2^{-n}$$

for all $N \in \mathbb{N}$. Since the geometric series $\sum_{n=1}^{N} 2^{-n}$ is known to converge (with limit 1), it is bounded. By the above, it follows that $0 \leq \sum_{n=1}^{N} 2^{-n} \frac{|x_n - y_n|}{1+|x_n - y_n|}$ is bounded independently of N. Since the sequence of partial sums

$$\sum_{n=1}^{N} 2^{-n} \frac{|x_n - y_n|}{1 + |x_n - y_n|}$$

is monotone increasing with respect to N, it thereby follows from the monotone convergence theorem that

$$d(x, y) = \sum_{n=1}^{\infty} 2^{-n} \frac{|x_n - y_n|}{1 + |x_n - y_n|} = \lim_{N\to\infty} \sum_{n=1}^{N} 2^{-n} \frac{|x_n - y_n|}{1 + |x_n - y_n|}$$

is well defined.

Example 1.4. Consider once more the space $X = \mathbb{R}^{\mathbb{N}}$ of real valued sequences. Given two *distinct* sequences $x = (x_n)$ and $y = (y_n)$ in X, we define

$$k(x, y) = k((x_n), (y_n)) := \min \left\{ j \in \mathbb{N} : x_j \neq y_j \right\}.$$

Notice that, since (x_n) and (y_n) are distinct (i.e. not equal), the well ordering principle ensures that $k(x, y)$ is a well defined natural number. Following this, we define our metric $d : X \times X \to [0, \infty)$ via the rule

$$d(x, y) := \begin{cases} 0 & \text{if } x = y, \\ \frac{1}{k(x,y)} & \text{if } x \neq y. \end{cases}$$

We claim that d is a valid metric on X. To this end, let us verify our three essential axioms.

(i) By definition, we have that $d(x, x) = 0$ for all $x \in X$. Conversely, we show that $d(x, y) \neq 0$ whenever $x = (x_n)$ and $y = (y_n)$ are distinct elements of X, i.e. sequences of real numbers. Indeed, if (x_n) and (y_n) are distinct sequences of real numbers, then there exists an index

$n \in \mathbb{N}$ for which $x_n \neq y_n$. Letting $n \in \mathbb{N}$ be the minimal such index (this exists by the well ordering principle), we have $k(x,y) = n$ whence

$$d(x,y) = \frac{1}{k(x,y)} = \frac{1}{n} \neq 0.$$

Thus, $d(x,y) = 0$ if and only if $x = y$.

(ii) Next, we show that $d(x,y) = d(y,x)$ for all $x = (x_n)$ and $y = (y_n)$ in X. If $x = y$, this is trivial. Otherwise, $x, y \in X$ are distinct and we have, by definition, that

$$d(x,y) = \frac{1}{k(x,y)} \quad \text{and} \quad d(y,x) = \frac{1}{k(y,x)}.$$

Now, observe that k is symmetric in x and y because

$$k(x,y) = \min\left\{j : x_j \neq y_j\right\} = \min\left\{j \in \mathbb{N} : y_j \neq x_j\right\} = k(y,x).$$

Consequently,

$$d(x,y) = \frac{1}{k(x,y)} = \frac{1}{k(y,x)} = d(y,x).$$

(iii) Finally, we verify that d satisfies the triangle inequality, i.e. that

$$d(x,y) \leq d(x,z) + d(z,y), \quad \forall x, y, z \in X.$$

To this end, let $x = (x_n)$, $y = (y_n)$, and $z = (z_n)$ be arbitrary elements of X. Observe that this inequality is trivially satisfied if $x = z$ since the inequality reduces to the equality $d(z,y) = d(z,y)$. Therefore, we may assume without harm to the proof that $x \neq z$. In addition, if $x = y$, the inequality above is trivially satisfied. Thus, we are reduced to the case $x \neq y$ as well. We now distinguish two possible cases:

(1) If $k(x,z) \leq k(x,y)$, then

$$d(x,y) = \frac{1}{k(x,y)} \leq \frac{1}{k(x,z)} = d(x,z).$$

Especially, $d(x,y) \leq d(x,z) \leq d(x,z) + d(y,z)$.

(2) If instead $k(x,y) < k(x,z)$, then $y_{k(x,y)} \neq x_{k(x,y)} = z_{k(x,y)}$. Furthermore, it holds that $x_n = y_n = z_n$ for all $n \in \mathbb{N}$ with $n < k(x,y)$. By definition, it follows that $k(y,z) = k(x,y)$ so that

$$d(x,y) = \frac{1}{k(x,y)} = \frac{1}{k(y,z)} = d(y,z)$$

whence $d(x,y) = d(y,z) \leq d(x,z) + d(y,z)$.

In either case, we see that the triangle inequality is satisfied. This completes the proof that d is a valid metric on X.

Remark 1.5. Something interesting occurs here with the triangle inequality in this example. Notice that in our proof, we demonstrated more than

$$d(x,y) \leq d(x,z) + d(z,y), \quad \forall x,y,z \in X.$$

Certainly, our proof of the triangle inequality shows that $d(x,y) \leq \max(d(x,z), d(y,z))$ for all $x,y,z \in X$. Metrics that satisfy this strengthened inequality are called *ultrametrics* and a metric space (X,d) where d is an ultrametric is called an *ultrametric space*. Although our particular case is somewhat extravagant, ultrametric spaces arise naturally in number theory. Indeed, the field $\mathbb{Q}_p$ of p-adic numbers is an ultrametric space for every prime p.

Let X be a set and $d_1, d_2 : X \times X \to [0,\infty)$ be two metrics on X. We say that d_1 and d_2 are *strongly* equivalent provided there exists a constant $C > 0$ such that

$$C^{-1}d_1(x,y) \leq d_2(x,y) \leq Cd_1(x,y), \quad \forall x,y \in X. \tag{1.3}$$

It can be shown that being strongly equivalent defines an equivalence relation for the metrics on X, but that is not important for our current purposes. It is, however, something to be kept in mind, particularly as Problem 5 in your first assignment asks you to prove that any two strongly equivalent metrics on X generate the same topology on X, i.e. induces the same open sets.

Example 1.5. Let $X = \mathbb{R}^{\mathbb{N}}$ be the space of real sequences we discussed in the last two examples Let d_p be the metric from Example 1.3 and let d_m denote the metric from Example 1.4. We shall show that these metrics are *not* strongly equivalent on X. Arguing by contradiction, suppose that d_p and d_m are strongly equivalent. Then, there exists a constant $C > 0$ such that

$$C^{-1}d_p(x,y) \leq d_m(x,y) \leq Cd_p(x,y), \quad \forall x,y \in X. \tag{1.4}$$

Fix $k \in \mathbb{N}$; letting $y = \mathbf{0} = (0,0,\dots,0,\dots)$ be the 0-sequence and letting $e_k = (0,0,\dots,1,0,\dots)$ be the sequence which is always 0 except for the kth-term, which is 1, it follows from (1.4) that

$$C^{-1}d_p(e_k,\mathbf{0}) \leq d_m(e_k,\mathbf{0}) \leq Cd_p(e_k,\mathbf{0}), \quad \forall k \in \mathbb{N}.$$

Substituting in the definitions of d_p and d_m, the above reduces to the statement

$$C^{-1}2^{-k}\frac{|1-0|}{1+|1-0|} \leq \frac{1}{k} \leq C2^{-k}\frac{|1-0|}{1+|1-0|}, \quad \forall k \in \mathbb{N}.$$

That is,

$$C^{-1}2^{-(k+1)} \leq \frac{1}{k} \leq C2^{-(k+1)}, \quad \forall k \in \mathbb{N}.$$

Especially, this implies that

$$\frac{2^{k+1}}{k} \leq C, \quad \forall k \in \mathbb{N}$$

which is absurd since

$$\frac{2^{k+1}}{k} \xrightarrow{k\to\infty} \infty.$$

1.1 Normed Vector Spaces

Let us first begin with a brief recollection. A vector space (over $\mathbb{R}$) is a set V equipped with addition and scalar multiplication operations. Formally, this means that a vector space over $\mathbb{R}$ is a triple $(V, +, \cdot)$ where V is a set satisfying the following axioms:

(V1) $u + v \in V$ for all $u, v \in V$;

(V2) $u + (v + w) = (u + v) + w$ for all $u, v, w \in V$;

(V3) $u + v = v + u$ for all $u, v \in V$;

(V4) there exists an element $0 \in V$ such that $v + 0 = v$ for all $v \in V$;

(V5) for every $v \in V$ there exists an element $(-v) \in V$ such that $v + (-v) = 0$;

(V6) for every $v \in V$ and every $\alpha \in \mathbb{R}$ we have that $\alpha v \in V$;

(V7) for every $v \in V$ and $\alpha, \beta \in \mathbb{R}$ there holds $\alpha(\beta v) = (\alpha\beta)v$;

(V8) $1v = v$ for all $v \in V$;

(V9) $\alpha(u + v) = \alpha u + \alpha v$ for all $\alpha \in \mathbb{R}$ and $u, v \in V$;

(V10) $(\alpha + \beta)v = \alpha v + \beta v$ for all $\alpha, \beta \in \mathbb{R}$ and $v \in V$.

Here, we write αv to mean the expression $\alpha \cdot v$. You have already seen various examples of (normed) vector spaces in class, and are already comfortable with the familiar vector space $\mathbb{R}^n$. In practice, vector spaces often come equipped with an additional structure called a *norm*. Although we have seen this in class, I reiterate the definition below for clarity and completeness.

Definition 1.1. Let V be a vector space over $\mathbb{R}$. A *norm* on V is a non-negative function

$$\|\cdot\| : V \to [0, \infty)$$

satisfying the following properties:

(i) $\|v\| = 0$ if and only if $v = 0$;

(ii) $\|\alpha v\| = |\alpha| \, \|v\|$ for all $\alpha \in \mathbb{R}$ and $v \in V$;

(iii) $\|u + v\| \leq \|u\| + \|v\|$ for all $u, v \in V$.

A normed vector space is a vector space V together with a norm.

As you know, a normed vector space automatically inherits a metric space structure via the norm. Indeed, given a normed space V, we *define* the induced metric by the rule $d(u, v) := \|u - v\|$. An important class of norms on $\mathbb{R}^n$, which we have discussed in the lectures, are the following p-norms:

Example 1.6. Let $1 \leq p < \infty$ and define

$$\|x\|_p := \left(\sum_{j=1}^{n} |x_j|^p \right)^{1/p}$$

for all $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$. Then, $\|\cdot\|_p$ is a norm on $\mathbb{R}^n$ called the p-norm. Notice that if $n = 1$, then $\|x\|_p = |x|$ for all x. Furthermore, when $p = 2$, this is simply the familiar Euclidean norm

$$\|x\|_2 = \left(\sum_{j=1}^{n} |x_j|^2 \right)^{1/2} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Example 1.7. For $p = \infty$, given $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, we define

$$\|x\|_p = \|x\|_\infty := \max_{1 \leq i \leq n} |x_i| .$$

Example 1.8. Consider once more the metric space (X, d) from Example 1.4. It is left as an exercise to show that $X = \mathbb{R}^{\mathbb{N}}$ is a vector space over $\mathbb{R}$ when endowed with coordinate-wise addition and scalar multiplication, i.e.

$$\begin{aligned} (x_1, x_2, \ldots, x_n, \ldots) + (y_1, y_2, \ldots, y_n, \ldots) &:= (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n, \ldots), \\ \lambda(x_1, x_2, \ldots, x_n, \ldots) &:= (\lambda x_1, \lambda x_2, \ldots, \lambda x_n, \ldots), \end{aligned}$$

for all $(x_1, x_2, \ldots, x_n, \ldots), (y_1, y_2, \ldots, y_n, \ldots) \in X$ and $\lambda \in \mathbb{R}$. The purpose of this exercise is to show that the metric d is *not* induced by a norm $\|\cdot\|$. Arguing by contradiction, suppose there is a norm

$$\|\cdot\| : X \longrightarrow [0, \infty)$$

inducing the metric d, i.e. such that $d(x, y) = \|x - y\|$ for all $x, y \in X$. Then, especially, for all $\lambda > 0$ we have

$$d(\lambda \mathbf{e}_1, \mathbf{0}) = \|\lambda \mathbf{e}_1 - \mathbf{0}\| = \|\lambda v \mathbf{e}_1\| = \lambda \|\mathbf{e}_1\| , \tag{1.5}$$

where $e_1 = (1, 0, 0, \ldots, 0, \ldots)$ and $\mathbf{0} = (0, 0, 0, \ldots, 0, \ldots)$ is the 0-vector. Observe that

$$d(\lambda \mathbf{e}_1, \mathbf{0}) = 1, \quad \forall \lambda > 0$$

since $\lambda \mathbf{e}_1 = (\lambda, 0, 0, 0, \ldots)$ only differs from $\mathbf{0}$ in the first term/coordinate. Especially, $d(\mathbf{e}_1, \mathbf{0}) = 1$. Consequently, we infer from (1.5) that

$$1 = \lambda \|\mathbf{e}_1\| = \lambda d(\mathbf{e}_1, \mathbf{0}) = \lambda.$$

However, this is an absurdity, since $\lambda > 0$ was arbitrary.

Naturally, one could ask if the metric d_p we discussed in (1.3) is induced by a norm. However, the answer is once again negative.

Example 1.9. The metric d on the space of sequences $\mathbb{R}^{\mathbb{N}}$ given in Example 1.3 is not induced by any norm. To prove this, we argue by contradiction and assume there is a norm $\|\cdot\|$ on X for which $d(x, y) = \|x - y\|$, for all sequences $x, y \in X$. Let $\mathbf{e}_1 := (1, 0, 0, \dots,) = (1, \mathbf{0})$ be the sequence whose first term is 1 and all others are 0. If $d(x, y) = \|x - y\|$, then for each $\lambda > 0$ we obtain

$$\begin{aligned} d(\lambda \mathbf{e}_1, \mathbf{0}) = \|\lambda \mathbf{e}_1 - \mathbf{0}\| &= \|\lambda \mathbf{e}_1\| \\ &= \lambda \|\mathbf{e}_1\| \\ &= \lambda d(\mathbf{e}_1, \mathbf{0}). \end{aligned}$$

That is,

$$d(\lambda \mathbf{e}_1, \mathbf{0}) = \lambda d(\mathbf{e}_1, \mathbf{0}). \tag{1.6}$$

Calculating both sides of this equation using the definition of d, we see that

$$2^{-1} \frac{|\lambda - 0|}{1 + |\lambda - 0|} = \lambda 2^{-1} \frac{1}{1 + 1}$$

which reduces to

$$\frac{\lambda}{1 + \lambda} = \frac{\lambda}{2}.$$

Simplifying further, we find that $\lambda + 1 = 2$ whence $\lambda = 1$. However, the scaling identity in (1.6) must be valid for all $\lambda > 0$ and hence we have a contradiction.

Notice that, in each of these examples, it is the scaling property of the norm that the metric fails to satisfy.

2 Second Tutorial

In this tutorial, we shall mainly explore the notion of an inner product space, and point out how it both relates (and sometimes differs) from the more general normed spaces. In much the same way that a normed space is naturally a metric space, an inner product space automatically inherits the structure of a normed space. However, it is not necessarily the case that a normed vector space is also an inner product space. Luckily, there is a criterion that guarantees the existence of an inner product generating a given norm.

2.1 Inner Product Spaces

An inner product space V (over $\mathbb{R}$) is a vector space over $\mathbb{R}$ equipped with a symmetric, positive definite, bilinear form

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}.$$

In other words, a (real) inner product space is a real vector space V together with a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying the following:

(i) $\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle u, w \rangle$ for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{R}$;

(ii) $\langle u, v \rangle = \langle v, u \rangle$ for all $u, v \in V$;

(iii) $\langle v, v \rangle \geq 0$ for all $v \in V$, with equality if and only if $v = 0$.

Remark 2.1. Some authors instead assume linearity in the second argument of the inner product (i.e. they assume $\langle u, \alpha v + \beta w \rangle = \alpha \langle u, w \rangle + \beta \langle u, w \rangle$ instead of (i)). Rest assured that this is a matter of convention, and is perfectly equivalent. Indeed, if we assume (i), then for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{R}$ we have by symmetry that

$$\langle u, \alpha v + \beta w \rangle = \langle \alpha v + \beta w, u \rangle = \alpha \langle v, u \rangle + \beta \langle w, u \rangle = \alpha \langle u, v \rangle + \beta \langle u, w \rangle .$$

Conversely, one can similarly check that linearity in the second variable implies linearity in the first argument.

As seen in the lectures, an inner product naturally induces a norm on V via the rule

$$\|v\| := \sqrt{\langle v, v \rangle}.$$

Thus, every inner product space is naturally a normed vector space, and consequently also a metric space in its own right. Furthermore, the norm induced by this inner product must satisfy the so-called "parallelogram law".

Proposition 2.1. *Let V be a real vector space with inner product $\langle \cdot, \cdot \rangle$ and let $\|\cdot\|$ be the induced norm. Then,*

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2 \tag{2.1}$$

for all $u, v \in V$.

Proof. See your assignment! □

It turns out that the parallelogram law is not just a by-product of an inner product, but also a condition one can impose on a norm that ensures there exists an inner-product inducing this norm. Thus, a norm is generated by an inner product if and only if this norm satisfies the parallelogram law. This is known as the Jordan-Neumann theorem, which we prove below.

Theorem 2.2 (Jordan-Neumann)**.** *Let $(V, \|\cdot\|)$ be a normed space over $\mathbb{R}$. There exists an inner product*

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$$

such that $\|x\| = \sqrt{\langle x, x \rangle}$ if and only if the norm satisfies the parallelogram law (2.1).

Proof. By Proposition 2.1, we know that any norm induced by an inner product forcibly satisfies the parallelogram law. Therefore, we need only show that any norm on V satisfying (2.1) is induced by an inner product. To this end, let $(V, \|\cdot\|)$ be a normed vector space whose norm satisfies (2.1) and *define*

$$\langle u, v\rangle := \frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2\right)$$

for all $u, v \in V$. Notice that for all $v \in V$ there holds

$$\langle v, v\rangle = \frac{1}{4}\left(\|2v\|^2 + \|v-v\|^2\right) = \frac{1}{4}\left(2^2\|v\|^2\right) = \|v\|^2.$$

In particular, since $\|v\|^2 \geq 0$ with equality if and only if $v = 0$, it follows that $\langle v, v\rangle \geq 0$ for all $v \in V$ with equality if and only if $v = 0$. Moreover, $\langle u, v\rangle = \langle v, u\rangle$ by inspection. In addition to this,

$$\begin{aligned}\langle -u, v\rangle = \frac{1}{4}\left(\|-u+v\|^2 - \|-u-v\|^2\right) &= \frac{1}{4}\left(\|u-v\|^2 - \|u+v\|^2\right)\\ &= -\frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2\right)\\ &= -\langle u, v\rangle\end{aligned}$$

for all $u, v \in V$. That is,

$$\langle -u, v\rangle = -\langle u, v\rangle, \quad \forall u, v \in V. \tag{2.2}$$

It now remains only to verify that $\langle\cdot,\cdot\rangle$ is indeed a valid inner product on V. We demonstrate this in a series of steps, which we outline below.

Step 1. We claim that $\langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$ for all $u, v, w \in V$.

Proof of Step 1. Using the parallelogram law (2.1) successively, we directly evaluate

$$\begin{aligned}\langle u+v, w\rangle &= \frac{1}{4}\left[\|u+v+w\|^2 - \|u+v-w\|^2\right]\\ &= \frac{1}{4}\left[\|u+v+w\|^2 + \|u-v-w\|^2 - \|u-v-w\|^2 - \|u+v-w\|^2\right]\\ &= \frac{1}{4}\left[\|u+v+w\|^2 + \|u-v-w\|^2 - \left(\|u-w+v\|^2 + \|u-w-v\|^2\right)\right]\\ &= \frac{1}{4}\left[2\|u\|^2 + 2\|v+w\|^2 - \left(2\|u-w\|^2 + 2\|v\|^2\right)\right]\\ &= \frac{1}{4}\left[2\|u\|^2 + 2\|w\|^2 + 2\|v+w\|^2 - 2\|u-w\|^2 - 2\|w\|^2 - 2\|v\|^2\right]\\ &= \frac{1}{4}\left[\|u+w\|^2 + \|u-w\|^2 + 2\|v+w\|^2 - 2\|u-w\|^2 - \|v+w\|^2 - \|v-w\|^2\right]\\ &= \frac{1}{4}\left[\|u+w\|^2 - \|u-w\|^2 + \|v+w\|^2 - \|v-w\|^2\right].\end{aligned}$$

Therefore,

$$\begin{aligned}\langle u+v, w\rangle &= \frac{1}{4}\left[\|u+w\|^2 - \|u-w\|^2 + \|v+w\|^2 - \|v-w\|^2\right] \\ &= \frac{1}{4}\left[\|u+w\|^2 - \|u-w\|^2\right] + \frac{1}{4}\left[\|v+w\|^2 - \|v-w\|^2\right] \\ &= \langle u, w\rangle + \langle v, w\rangle .\end{aligned}$$

Since $u, v, w \in V$ were arbitrary, this completes the proof of the first step.

Step 2. The Cauchy-Schwarz inequality $|\langle u, v\rangle| \leq \|u\| \, \|v\|$ holds true for all $u, v \in V$.

Proof of Step 2. Let $u, v \in V$ and write

$$\langle u, v\rangle = \frac{1}{4}\left[\|u+v\|^2 - \|u-v\|^2\right] \tag{2.3}$$

$$= \frac{1}{4}\left[2\|u+v\|^2 - \left(\|u+v\|^2 + \|u-v\|^2\right)\right] \tag{2.4}$$

$$= \frac{1}{4}\left[2\|u+v\|^2 - 2\|u\|^2 - 2\|v\|^2\right] \tag{2.5}$$

$$\leq \frac{1}{4}\left[2\left(\|u\| + \|v\|\right)^2 - 2\|u\|^2 - 2\|v\|^2\right] \tag{2.6}$$

where we have used the parallelogram law in (2.5), and the triangle inequality in (2.6). Continuing, we obtain

$$\begin{aligned}\langle u, v\rangle &\leq \frac{1}{4}\left[2\|u\|^2 + 4\|u\| \, \|v\| + 2\|v\|^2 - 2\|u\|^2 - 2\|v\|^2\right] \\ &= \frac{1}{4}\left(4\|u\| \, \|v\|\right) \\ &= \|u\| \, \|v\| .\end{aligned}$$

Since this is valid for all $u, v \in V$, plugging in $-u$ instead of u yields that

$$-\langle u, v\rangle = \langle -u, v\rangle \leq \|-u\| \, \|v\| = \|u\| \, \|v\|$$

whence $\langle u, v\rangle \geq -\|u\| \, \|v\|$ as well. Combining all of this, it follows that $|\langle u, v\rangle| \leq \|u\| \, \|v\|$.

Step 3. We claim that $\langle tu, v\rangle = t\langle u, v\rangle$ for all $t \in \mathbb{R}$.

Proof of Step 3. Fix $u, v \in V$. We prove first by induction that $\langle nu, v\rangle = n\langle u, v\rangle$ for all $n \in \mathbb{N}$. Clearly, the base case $n = 1$ is obvious. Assuming that $\langle nu, v\rangle = n\langle u, v\rangle$ for some $n \in \mathbb{N}$, we have by Step 1 that

$$\langle (n+1)u, v\rangle = \langle nu + u, v\rangle = \langle nu, v\rangle + \langle u, v\rangle = n\langle u, v\rangle + \langle u, v\rangle = (n+1)\langle u, v\rangle .$$

Thus, it follows by induction that $\langle nu, v\rangle = n\langle u, v\rangle$ for all $n \in \mathbb{N}$. Since $\langle -u, v\rangle = -\langle u, v\rangle$, this also implies that $\langle nu, v\rangle = n\langle u, v\rangle$ for all $n \in \mathbb{Z}$. Next, given a rational number $\frac{p}{q} \in \mathbb{Q}$, we may assume that $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. Then, observe that

$$q\left\langle \frac{p}{q}u, v\right\rangle = qp\left\langle \frac{1}{q}u, v\right\rangle = pq\left\langle \frac{1}{q}u, v\right\rangle = p\left\langle \frac{q}{q}u, v\right\rangle = p\langle u, v\rangle .$$

Consequently,

$$\left\langle \frac{p}{q}u, v\right\rangle = \frac{p}{q}\langle u, v\rangle .$$

Thus, we have shown that $\langle tu, v\rangle = t\langle u, v\rangle$ for all $t \in \mathbb{Q}$. We now extend this to all $t \in \mathbb{R}$ by a density argument. Obviously, the mapping $t \mapsto t\langle u, v\rangle$ is continuous in t on $\mathbb{R}$. Furthermore, by Step 1 and the Cauchy-Schwarz inequality proven in Step 2, we have that

$$\begin{aligned} |\langle tu, v\rangle - \langle su, v\rangle| = |\langle tu - su, v\rangle| = |\langle (t-s)u, v\rangle| &\leq \|(t-s)u\|\,\|v\| \\ &= |t-s|\,\|u\|\,\|v\| \end{aligned}$$

whence we see that the mapping $t \mapsto \langle tu, v\rangle$ is (Lipschitz) continuous on $\mathbb{R}$. Since sums of continuous functions are again continuous, we infer that

$$f : \mathbb{R} \to \mathbb{R}, \quad t \mapsto \langle tu, v\rangle - t\langle u, v\rangle$$

is continuous on $\mathbb{R}$. However, we know that $\langle tu, v\rangle = t\langle u, v\rangle$ for $t \in \mathbb{Q}$, so $f(t) = 0$ for all $t \in \mathbb{Q}$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, this implies (why?) that $f(t) = 0$ for all $t \in \mathbb{R}$. Thus, $\langle tu, v\rangle = t\langle u, v\rangle$ for all $t \in \mathbb{R}$. This proves Step 3.

Combining Steps 1-2-3, together with our original observations, shows that $\langle \cdot, \cdot \rangle$ is a valid inner product on V whose generated norm is precisely the original norm we started with. This completes the proof of the Jordan-Neumann theorem. □

3 Third Tutorial

Our goal in this tutorial is to practice with the abstract definitions of open and closed sets within metric spaces. In particular, we shall discuss interior, closure, and boundary of sets. In doing so, we aim to provide examples that challenge our usual intuition regarding the properties of open balls. To achieve this, we solve exercises within the context of ultrametric spaces (see Example 1.3 and Remark 1.5) which generalize some of the strange properties you have seen in the lectures for discrete metric spaces (e.g. the open ball is closed and its boundary is empty). We hope that this convinces the reader that their ingrained intuition about balls in $\mathbb{R}^n$ (or, more generally, normed spaces) may not be true when handling arbitrary metric spaces.

3.1 Open and Closed Sets in Metric Spaces

Let us begin by reviewing the definitions of open and closed sets, within a metric space. To this end, let (X, d) be a metric space. As seen in class, the metric d induces a notion of open sets. That is, we declare a subset U of X to be open if, for each $x \in U$, there exists $\varepsilon > 0$ such that

$$B(x, \varepsilon) := \{y \in X : d(x, y) < \varepsilon\} \subseteq U.$$

In contrast, a set $F \subseteq X$ is said to be closed if its complement $F^c = X \setminus F$ is open in X. Open and closed sets satisfy the familiar properties we saw in $\mathbb{R}$, namely:

(i) $X, \varnothing$ are open in X;

(ii) arbitrary unions of open sets are open;

(iii) finite intersections of open sets remain open.

For closed sets, we instead have

(i) $X, \varnothing$ are closed in X;

(ii) arbitrary intersections of closed sets are closed;

(iii) finite unions of closed sets remain closed.

Let us now give a simple example. We remark that this next example remains true in certain more general spaces, even without the presence of a metric.

Example 3.1. Let (X, d) be a metric space. All finite subsets of X are closed in X.

Proof. We first show that every singleton is closed in X. Indeed, let $x \in X$ and consider $\{x\}$. Given any $y \in X \setminus \{x\}$, we have $y \neq x$ whence $\varepsilon := d(x, y) > 0$. Then, the ball $B(y, \varepsilon)$ does not contain x and thus $B(y, \varepsilon) \subseteq X \setminus \{x\}$. Since $y \in X \setminus \{x\}$ was arbitrary, it follows that $X \setminus \{x\}$ is open. To see that this implies that every finite set is closed, notice that every finite subset of X may be written as $\{x_1, \ldots, x_n\}$ for some points $x_1, \ldots, x_n \in X$. Then, because $\{x_1, \ldots, x_n\} = \bigcup_{j=1}^{n} \{x_j\}$, we conclude that $\{x_1, \ldots, x_n\}$ is closed since it is the *finite* union of closed singletons. □

Recall that last tutorial we introduced a metric on the space $\mathbb{R}^{\mathbb{N}}$ of real sequences that turned out the satisfy the additional property $d(x, y) \leq \max\{d(x, z), d(z, y)\}$ for all $x, y, z \in X$ (see Example 1.4 and Remark 1.5 thereafter). Metrics that satisfied this strengthened triangle inequality are called *ultrametrics*. For the sake of completeness and clarity, we formally define ultrametric spaces here.

Definition 3.1 (Ultrametric Space)**.** An ultrametric space is a metric space (X, d) such that the metric d is an ultrametric on X, i.e. such that $d(x, y) \leq \max\{d(x, z), d(z, y)\}$ for all $x, y, z \in X$.

These metrics are relatively common and need not be defined on exotic spaces. Indeed, on any non-empty set X, the discrete metric defines an ultrametric.

In order to familiarize ourselves with the definitions and properties of open and closed sets within abstract spaces, let us work through some counterintuitive exercises within the category of ultrametric spaces. We begin with a surprising result which states that, in an ultrametric space, every point in an open ball is the "center" of this ball. Since every ultrametric space is itself a metric space, one must be cautious when working intuitively within an arbitrary metric space.

Example 3.2. Let (X, d) be an ultrametric space and let $B(x, \varepsilon)$ be an open ball in X. Then, for all $y \in B(x, \varepsilon)$, there holds that $B(x, \varepsilon) = B(y, \varepsilon)$. Informally, we can interpret this statement as saying that every point in an open ball is the center of this ball.

Proof. We first show that $B(y, \varepsilon) \subseteq B(x, \varepsilon)$. Indeed, let $z \in B(y, \varepsilon)$ be arbitrary and observe that

$$d(x, z) \leq \max\{d(x, y), d(y, z)\}$$

where $d(x, y) < \varepsilon$ and $d(y, z) < \varepsilon$ (since $y \in B(x, \varepsilon)$ and $z \in B(y, \varepsilon)$). Therefore, $d(x, z) < \varepsilon$ and so $z \in B(x, \varepsilon)$. Since $z \in B(y, \varepsilon)$ is arbitrary, we find that $B(y, \varepsilon) \subseteq B(x, \varepsilon)$. For the reverse inclusion, observe that $y \in B(x, \varepsilon) \iff x \in B(y, \varepsilon)$, whence the same argument gives $B(x, \varepsilon) \subseteq B(y, \varepsilon)$. □

This next example is relatively famous as an exercise, despite having a short proof. Perhaps it is so well known because it has an interesting "cute" interpretation.

Example 3.3. Let (X, d) be an ultrametric space and let $x, y, z \in X$ be arbitrary. Then, at least two of the quantities $d(x, y), d(y, z), d(x, z)$ are equal.

Proof. Suppose by way of contradiction that all three of $d(x, y), d(y, z), d(x, z)$ are distinct. Then, they may be placed in strictly increasing order. In fact, without loss of generality, we may assume that

$$d(x, y) < d(y, z) < d(x, z).$$

However, the strong triangle inequality of the ultrametric forces

$$d(x, z) \leq \max\{d(x, y), d(y, z)\} < d(x, z)$$

which gives a contradiction. □

Remark 3.1. This last example's conclusion is sometimes summarized by saying "all triangles are isosceles in an ultrametric space".

Example 3.4. Let (X, d) be an ultrametric space and let $B(x, \delta)$ and $B(y, \varepsilon)$ be two open balls in X. If $B(x, \delta) \cap B(y, \varepsilon) \neq \varnothing$, then $B(x, \delta) \subseteq B(y, \varepsilon)$ or $B(y, \varepsilon) \subseteq B(x, \delta)$.

Proof. Without loss of generality, we may assume $\delta \leq \varepsilon$. Choose a point $z \in B(x,\delta) \cap B(y,\varepsilon)$. By the previous example,

$$B(z,\delta) = B(x,\delta) \quad \text{and} \quad B(z,\varepsilon) = B(y,\varepsilon).$$

Therefore, $B(x,\delta) = B(z,\delta) \subseteq B(z,\varepsilon) = B(y,\varepsilon)$. □

Before proceeding with more examples, let us recall the definition of closure and interior. Let A be a subset of a metric space (X,d) and recall that the *interior* of A, denoted $\mathring{A}$ or $\text{int}(A)$, is the subset of A consisting of all $x \in A$ such that there exists an open ball $B(x,\varepsilon)$, centered at x, that is contained in A. In other words,

$$\mathring{A} = \text{int}(A) := \{x \in A : \exists \varepsilon > 0 \text{ such that } B(x,\varepsilon) \subseteq A\}.$$

As seen in the lectures, $\mathring{A} \subseteq A$ is always an open subset of X and may be characterized by the following:

$$\mathring{A} = \bigcup_{\substack{O \subseteq A \\ O \text{ is open in } X}} O.$$

Intuitively, we can think of $\mathring{A}$ as the largest open subset of A in X. We define the closure of A, denoted $\overline{A}$ or $\text{cl}(A)$, to be the subset of X consisting of all $x \in X$ such that every ε-ball centered at x intersects A, i.e.

$$\overline{A} = \text{cl}(A) = \{x \in X : B(x,\varepsilon) \cap A \neq \emptyset,\ \forall \varepsilon > 0\}.$$

Equivalently,

$$\overline{A} = \bigcap_{\substack{F \subseteq A \\ F \text{ is closed in } X}} F,$$

i.e. $\overline{A}$ is the smallest closed subset of X that contains A. We then define the *boundary* of a the set A as $\partial A = \overline{A} \setminus \mathring{A} = \overline{A} \cap \overline{X \setminus A}$. Notice that

$$\partial A = \{x \in X : B(x,\varepsilon) \cap A \neq \emptyset \text{ and } B(x,\varepsilon) \cap (X \setminus A) \neq \emptyset,\ \forall \varepsilon > 0\}.$$

Example 3.5. Let (X,d) be an ultrametric space. Every open ball in X is also closed in X. Thus, $\overline{B(x,\varepsilon)} = B(x,\varepsilon)$ for every open ball $B(x,\varepsilon)$ in X.

Proof. Let $B(x,\varepsilon)$ be an open ball; we show that $B(x,\varepsilon)$ is closed in X by showing that its complement $B(x,\varepsilon)^c = X \setminus B(x,\varepsilon)$ is open in X. To this end, let $y \in B(x,\varepsilon)^c$ so that $d(x,y) \geq \varepsilon$. We claim that $B(y,\varepsilon) \subseteq B(x,\varepsilon)^c$ which would prove that $B(x,\varepsilon)^c$ is open. To this end, let $z \in B(y,\varepsilon)$ and observe that $z \notin B(x,\varepsilon)$, for $d(x,z) < \varepsilon$ would imply that

$$d(x,y) \leq \max\{d(x,z), d(z,y)\} < \varepsilon,$$

contradicting the fact that $d(x,y) \geq \varepsilon$. Therefore, $z \in B(x,\varepsilon)^c$ whence $B(y,\varepsilon) \subseteq B(x,\varepsilon)^c$ for all $y \in B(x,\varepsilon)^c$. In particular, $B(x,\varepsilon)^c$ is open whence $B(x,\varepsilon)$ is closed in X. □

Corollary 3.1. *If (X, d) is an ultrametric space, then the boundary of any open ball is empty.*

Proof. Let $B = B(x, r)$ be an open ball in X. Since B is open, we have $\mathring{B} = B$. Since B is closed, it holds in addition that $\overline{B} = B$. Consequently, $\partial B = \overline{B} \setminus \mathring{B} = B \setminus B = \varnothing$. □

Remark 3.2. Despite every open ball being closed, and every open set being the union of open balls, it does not follow that every open set is closed in an ultrametric space. By way of a counter example, consider the metric space (X, d) of real sequences from Example 1.4. Letting $\mathbf{0} = (0, 0, 0, \dots, 0, \dots)$ be the 0-sequence, it follows that $X \setminus \{\mathbf{0}\}$ is open since $\{\mathbf{0}\}$ is closed in X. However, $X \setminus \{0\}$ is not closed because $\{\mathbf{0}\}$ is not open in (X, d). To see this, notice that if $\{\mathbf{0}\}$ were open, there would exist $\varepsilon > 0$ such that $B(\mathbf{0}, \varepsilon) \subseteq \{\mathbf{0}\}$. By the Archimedean property, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$. Let now

$$x = (0, 0, \dots, 1, 0, \dots)$$

be the sequence obtained by letting every term be 0 except for the nth-term which we define to be 1. Then, clearly, $x \neq \mathbf{0}$ but

$$d(x, \mathbf{0}) = \frac{1}{n} < \varepsilon$$

whence $x \in B(\mathbf{0}, \varepsilon) \subseteq \{\mathbf{0}\}$ which forces $x = \mathbf{0}$.

3.2 Vector Subspaces

Let V be a vector space over $\mathbb{R}$, and let $W \subseteq V$ be a non-empty subset. Recall that we say W is a vector subspace of V provided

$$\alpha u + \beta v \in W,$$

for all $\alpha, \beta \in \mathbb{R}$ and $u, v \in W$. Note that W is then a vector space over $\mathbb{R}$, in its own right.

Consider now three dimensional Euclidean space, i.e. $\mathbb{R}^3$ with the standard norm (i.e. the p-norm for $p = 2$). We can identify $\mathbb{R}^2$ with the subspace of $\mathbb{R}^3$ given by $\{(x, y, 0) : x, y \in \mathbb{R}\}$. Intuitively and informally, $\mathbb{R}^2$ then looks like an infinitely thin sheet of paper, sitting in the "middle" of 3-dimensional space $\mathbb{R}^3$. Now, any open ball in $\mathbb{R}^3$ can be pictured as a set that has a certain "thickness" to it, in the sense that it is itself a 3-dimensional object. In this interpretation, the reader may convince themselves that no three dimensional ball could be contained within this infinitely flat sheet that is $\mathbb{R}^2$. If we try to formalize what this means, and make the claim rigorous, one might ask whether a *proper* vector subspace of a normed space can ever contain a ball from the larger ambient space. The answer is no, and is the next result we establish.

Proposition 3.2. *Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$ and $W \subseteq V$ a vector subspace. If W has non-empty interior, then $W = V$.*

Proof. Suppose that there exists a point x belonging to the interior of W. Then, there exists $\varepsilon > 0$ such that $B(x, 2\varepsilon) \subseteq W$. Being that W is a vector subspace of V, it must automatically contain 0 (why?). Given $y \neq 0$, we observe that

$$x + \frac{\varepsilon y}{\|y\|} \in B(x, 2\varepsilon) \subseteq W$$

because

$$\left\| x + \frac{\varepsilon y}{\|y\|} - x \right\| = \frac{\varepsilon}{\|y\|} \|y\| = \varepsilon < 2\varepsilon.$$

However, since $x \in W$, we must have

$$\frac{\varepsilon y}{\|y\|} = x + \frac{\varepsilon y}{\|y\|} - x \in W$$

as well. Finally, it follows that

$$y = \frac{\|y\|}{\varepsilon} \left(\frac{\varepsilon y}{\|y\|} \right) \in W.$$

In short, we have shown that $y \in W$ for any $y \in V$. This proves $V \subseteq W$ which establishes the desired equality. □

Corollary 3.3. *A normed vector space possesses no proper open vector subspaces.*

3.3 Separable Metric Spaces

In your second assignment, you are asked to prove a result regarding separable metric spaces. Formally, let us recall that a subset D of a metric space (X, d) is called *dense* in X provided $\overline{D} = X$. That is, if $B(x, \varepsilon) \cap D \neq \varnothing$ for all $x \in X$ and every $\varepsilon > 0$. The metric space (X, d) is called separable if there exists a countable dense subset D of X. In the next example, we discuss the separability of sequences spaces equipped with the metric from Example 1.4.

Example 3.6. Let (X, d) be the metric space from Example 1.4. Show that (X, d) is *not* separable.

Proof. The statement is equivalent to the claim that every dense subset of X is uncountable. To this end, suppose $D \subseteq X$ is a dense subset. This is precisely the same as saying that $\overline{D} = X$, i.e.

$$B(x, \varepsilon) \cap D \neq \varnothing, \quad \forall x \in X, \, \forall \varepsilon > 0.$$

Now, given $x \in \mathbb{R}$, we associate to it the sequence

$$s_x := (x, 0, 0, \ldots, 0, \ldots)$$

whose first term is x, followed entirely by 0 terms. Clearly, s_x is a sequence of real numbers, and hence $s_x \in X$. Being that $\overline{D} = X$, we see that $s_x \in \overline{D}$ which ensures that

$$B(s_x, 1) \cap D \neq \varnothing.$$

Choose now some d_x) from this non-empty intersection; then $d_x \in D$ and $d(d_x, s_x) < 1$. By definition of the metric d from Example 1.4, the inequality $d(d_x, s_x) < 1$ can only be possible if the first term of d_x equals the first term of s_x, which is precisely the real number x.

For each $x \in \mathbb{R}$, we have selected a point $d_x \in D$ whose first term is x. This allows us to define a function

$$f : \mathbb{R} \to D, \quad f(x) = d_x$$

Moreover, this mapping is injective because $f(x) = f(y)$ would give $d_x = d_y$, where these two sequences being equal means they share, in particular, their first terms, implying $x = y$. Consequently, f is an injection of the uncountable set $\mathbb{R}$ into the dense subset D, thereby showing that D is itself uncountable. □

Example 3.7. Let $X = \{0, 1\}^{\mathbb{N}}$ be the space of all binary sequences[2] equipped with the metric

$$d(x, y) := \begin{cases} 0 & \text{if } x = y, \\ \frac{1}{k(x,y)} & \text{if } x \neq y, \end{cases}$$

where we let k be as in Example 1.4. Show that (X, d) is separable.

Solution. We have already seen in Example 1.4 that d defines a valid metric on the larger space $\mathbb{R}^{\mathbb{N}}$ of all real valued sequences, so its restriction to X is obviously a valid metric on X. Thus, we must only show that there exists a countable dense subset of (X, d). To this end, let us define for each $N \in \mathbb{N}$ the set $F_N \subset X$ which consists of all binary sequences $x = (x_n) \in X$ such that $x_n = 0$ for all $n \geq N$, i.e.

$$F_N := \{x = (x_n) \in X : x_n = 0, \, \forall n \geq N\}.$$

Observe that F_N is finite for each $N \in \mathbb{N}$ (with cardinality 2^{N-1}) and is hence countable. Letting $F \subset X$ be the subset consisting of all binary sequences that are eventually 0, we must have that $F = \bigcup_{N \in \mathbb{N}} F_N$ whence F is itself countable. To conclude the proof, we shall prove that $\overline{F} = X$.

To justify this, we show that every open ball $B(x, \varepsilon)$ intersects F at some point. Indeed, given $x \in X$ and $\varepsilon > 0$, we may choose (by the Archimedean property) some $N \in \mathbb{N}$ such that $\frac{1}{N} < \varepsilon$. Writing $x = (x_n)$, we define a new binary sequence $y = (y_n) \in X$ according to the rule

$$y_n := \begin{cases} x_n & \text{if } n < N, \\ 0 & \text{if } n \geq N. \end{cases}$$

Then, $y = (y_n) \in F_N$ and $x_n = y_n$ for all indices $n < N$. If $x = y$, then trivially $x \in B(x, \varepsilon) \cap F_N \subseteq B(x, \varepsilon) \cap F$. If $x \neq y$ then, since $x_n = y_n$ for all $n < N$, we must have $k(x, y) \geq N$ whence

$$d(x, y) = \frac{1}{k(x, y)} \leq \frac{1}{N} < \varepsilon$$

so that $y \in B(x, \varepsilon)$. Because $y \in F_N \subseteq F$, it follows that $y \in B(x, \varepsilon) \cap F$. To summarize, we have shown that $B(x, \varepsilon) \cap F \neq \varnothing$ for each $\varepsilon > 0$ and every $x \in X$. This means that $\overline{F} = X$, as was asserted. □

[2] A binary sequence is a sequence (x_n) such that $x_n \in \{0, 1\}$ for all indices $n \in \mathbb{N}$.

4 Fourth Tutorial

In this tutorial, we return to our familiar examples of sequence spaces, with the two metrics we introduced upon it. For these two examples, we illustrate equivalent characterizations of sequential convergence and deduce from this that the space of sequences is complete with respect to both of these metrics.

Example 4.1. Let $X = \mathbb{R}^{\mathbb{N}}$ and equip X with the metric from Example 1.4. Given a point $x \in X$, we shall write $x(k)$ to denote the kth-term (or coordinate) of x.

(a) Prove that a sequence (x_n) in X converges to $x \in X$ if and only if, for every $K \in \mathbb{N}$, there exists $N \in \mathbb{N}$ such that

$$x_n(k) = x(k)$$

for all $k = 1, \ldots, K$ and $n \geq N$.

(b) Deduce from this that (X, d) is complete.

Proof.

(a) Suppose first that $x_n \to x$ in X and let $K \in \mathbb{N}$ be arbitrary. Since $x_n \to x$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x) < \frac{1}{K}, \quad \forall n \geq N.$$

Therefore, if $n \geq N$, we must have $x_n(k) = x(k)$ for all $k = 1, \ldots, K$ for otherwise there would exist a minimal $j \in \{1, \ldots, K\}$ such that $x_n(j) \neq x(j)$ whence

$$d(x_n, x) = \frac{1}{j} \geq \frac{1}{K},$$

contradicting our choice of N. It follows that $x_n(k) = x(k)$ for all $k = 1, \ldots, K$ and $n \geq N$, as was asserted.

Conversely, we assume this property and seek to show that $x_n \to x$ with respect to this metric d. Indeed, given $\varepsilon > 0$, we may choose $K \in \mathbb{N}$ such that $\frac{1}{K} < \varepsilon$. Then, by assumption, there exists $N \in \mathbb{N}$ such that $x_n(k) = x(k)$ for all $k = 1, \ldots, K$ and $n \geq N$. Consequently, if $n \geq N$ and $x_n \neq x$, then

$$\min \{j \in \mathbb{N} : x_n(j) \neq x(j)\} > K$$

whence

$$d(x_n, x) < \frac{1}{K} < \varepsilon.$$

As $\varepsilon > 0$ was arbitrary, it follows that $x_n \to x$ in (X, d).

(b) We now claim that X is complete with this metric. To this end, let (x_n) be a Cauchy sequence in X and let $k \in \mathbb{N}$ be arbitrary. Since (x_n) is Cauchy, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x_m) < \frac{1}{k}, \quad \forall n, m \geq N.$$

By definition of the metric, this forces $x_n(k) = x_m(k)$ for all $n, m \geq N$. Especially, we obtain $x_n(k) = x_N(k)$ for all $n \geq N$. In other words, the kth-term of (x_n) is eventually a constant, which we shall dub $x(k)$. Letting $x = (x(1), x(2), \ldots, x(k), \ldots)$, we obtain a point $x \in X$ such that, for each $K \in \mathbb{N}$, there exists $N \in \mathbb{N}$ with the property that $x_n(k) = x(k)$ for all $k = 1, \ldots, K$ and $n \geq N$. By (a), this means that $x_n \to x$ in X which concludes the proof.

□

Example 4.2. Let $X = \mathbb{R}^{\mathbb{N}}$ be the space of real sequences endowed with the metric d from Example 1.3. Given an element $x \in X$, we shall write $x(k)$ to denote the kth-term of the sequence x. Given a sequence (x_n) in X and a point $x \in X$, show that $x_n \to x$, as $n \to \infty$, if and only if

$$\lim_{n \to \infty} x_n(k) = x(k). \quad \forall k \in \mathbb{N}.$$

Proof. First, suppose that $x_n \to x$ in (X, d). Fix an index $k \in \mathbb{N}$ and let $\varepsilon > 0$ be arbitrary. Since

$$\lim_{\delta \to 0^+} \frac{2^k \delta}{1 - 2^k \delta} = 0,$$

there exists $\delta \in (0, 2^{-k})$ such that

$$0 < \frac{2^k \delta}{1 - 2^k \delta} < \varepsilon.$$

Now, since $x_n \to x$ in X, there exists $N \in \mathbb{N}$ such that

$$2^{-k} \frac{|x_n(k) - x(k)|}{1 + |x_n(k) - x(k)|} \leq d(x_n, x) < \delta$$

for all $n \geq N$. Rewriting this, we have for all $n \geq N$ that

$$(1 - 2^k \delta)\, |x_n(k) - x(k)| < 2^k \delta$$

whence

$$|x_n(k) - x(k)| < \frac{2^k \delta}{1 - 2^k \delta} < \varepsilon,$$

for all $n \geq N$. Since $\varepsilon > 0$ was arbitrary, we have shown that $x_n(k) \to x(k)$ in the sense of real numbers, for every index $k \in \mathbb{N}$.

Conversely, suppose that for all indices $k \in \mathbb{N}$ there holds that

$$\lim_{n \to \infty} x_n(k) = x(k). \tag{4.1}$$

We claim that $x_n \to x$ in (X, d). To this end, let $\varepsilon > 0$ and choose $K \in \mathbb{N}$ such that $2^{-K} < \frac{\varepsilon}{2}$. By virtue of (4.1), there exists $N \in \mathbb{N}$ such that for each $k = 1, \ldots, K$ there holds

$$|x_n(k) - x(k)| < \frac{\varepsilon}{2}, \quad \forall n \geq N\,.$$

Therefore, for all $n \geq N$, we have

$$\begin{aligned}
d(x_n, x) &= \sum_{k=1}^{\infty} 2^{-k} \frac{|x_n(k) - x(k)|}{1 + |x_n(k) - x(k)|} \\
&= \sum_{k=1}^{K} 2^{-k} \frac{|x_n(k) - x(k)|}{1 + |x_n(k) - x(k)|} + \sum_{k=K+1}^{\infty} 2^{-k} \frac{|x_n(k) - x(k)|}{1 + |x_n(k) - x(k)|} \\
&\leq \sum_{k=1}^{K} 2^{-k} |x_n(k) - x(k)| + \sum_{k=K+1}^{\infty} 2^{-k} \\
&= \sum_{k=1}^{K} 2^{-k} |x_n(k) - x(k)| + 2^{-K} \\
&< \frac{\varepsilon}{2} \sum_{k=1}^{K} 2^{-k} + \frac{\varepsilon}{2} \\
&< \frac{\varepsilon}{2} \sum_{k=1}^{\infty} 2^{-k} + \frac{\varepsilon}{2} \\
&= \varepsilon.
\end{aligned}$$

□

Example 4.3. The metric space (X, d) from Example 4.2 is complete.

Proof. Let (x_n) be a Cauchy sequence in X and let $\varepsilon > 0$ be arbitrary. Given any index $k \in \mathbb{N}$, we may choose (as in the last example) some $\delta \in (0, 2^{-k})$ such that

$$\frac{2^k \delta}{1 - 2^k \delta} < \varepsilon.$$

Since (x_n) is Cauchy in X, there exists $N \in \mathbb{N}$ having the property that

$$2^{-k} \frac{|x_n(k) - x_m(k)|}{1 + |x_n(k) - x_m(k)|} \leq d(x_n, x_m) < \delta$$

for all $n \geq N$. Following the same rearrangements from the previous example, it follows that

$$|x_n(k) - x_m(k)| < \frac{2^k \delta}{1 - 2^k \delta} < \varepsilon, \quad \forall n \geq N.$$

Since $\varepsilon > 0$ was arbitrary, we infer that the kth-term sequence $(x_n(k))_{n\in\mathbb{N}}$ is Cauchy in $\mathbb{R}$ with respect to the standard metric. Being that this space is complete, this means there exists some real number, which we call $x(k)$, such that

$$\lim_{n\to\infty} x_n(k) = x(k).$$

Letting $x \in X$ be the sequence of real numbers whose k^{th}-term is $x(k)$, it follows from our last example that $x_n \to x$ in X since $x_n(k) \to x(k)$, as $n \to \infty$, for all $k \in \mathbb{N}$. □

By the sequential characterization of closed sets, we may now easily compare the open sets induced by the two metrics in Examples 1.3 and 1.4.

Example 4.4. Let $X = \mathbb{R}^{\mathbb{N}}$ be the space of real sequences, let d_1 be the metric from Example 1.3, and d_2 be the metric from Example 1.4. If $U \subseteq X$ is open with respect to d_1, then it is open with respect to d_2.

Solution. Let $U \subseteq X$ be open with respect to d_1, so that U^c is closed with respect to d_1. To show that U^c is closed with respect to d_2, we shall apply the sequential characterization of closed sets. To this end, let (x_n) be a sequence of points in U^c that converges to some $x \in X$ with respect to d_2. By Example 4.1, this means that every coordinate sequence $(x_n(k))_{n\in\mathbb{N}}$ is eventually constant (equal to $x(k)$), with respect to n. Obviously, this means that

$$\lim_{n\to\infty} x_n(k) = x(k), \quad \forall k \in \mathbb{N}.$$

By virtue of Example 4.2, this means that $x_n \to x$ with respect to d_1. Now, since U^c is closed with respect to d_1, the sequential characterization of closed sets asserts that $x \in U^c$. But, (x_n) was an arbitrary convergent sequence (with respect to d_2) from U^c, and we demonstrated that this implied $x = \lim x_n \in U^c$. Consequently, the sequential characterization of closed sets implies that U^c is closed with respect to d_2, i.e. U is open with respect to d_2. □

Remark 4.1. Notice that convergence $x_n \to x$ with respect to d_1 is equivalent to the convergence (in $\mathbb{R}$) of the individual coordinate sequences $(x_n(k))_{n\in\mathbb{N}}$, whilst convergence with respect to d_2 requires the stronger property that every coordinate sequence $(x_n(k))_{n\in\mathbb{N}}$ eventually be constant (and equal to $x(k)$). Thus, convergence with respect to d_2 is more restrictive than convergence with respect to d_1 and one can interpret this as meaning that there are *more* open sets in (X, d_2) than (X, d_1). This is precisely what the last example showed.

5 Fifth Tutorial

In this tutorial, we continue to discuss completeness and begin to address compactness. Regarding completeness, we shall prove (as an exercise) that the metric spaces for which closed *disjoint sets* are separated by positive distance (this will be made precise shortly) are necessarily complete. We

also remark that completeness is not sufficient to guarantee this, by way of a counterexample. Following this, we shall use the open cover property of compact sets to solve another exercise involving a compact set and points in its complement. Before we may do this however, we require a simple lemma which also helps provide some review/practice involving closure.

Lemma 5.1. *Let (X, d) be a metric space and let $A \subseteq X$. If $x \in \overline{A} \setminus A$, then for any $\varepsilon > 0$ the intersection $B(x, \varepsilon) \cap A$ is infinite.*

Remark 5.1. Notice that if we may select a point x satisfying the assumptions of this lemma (i.e. $x \in \overline{A} \setminus A$), then A must be infinite, and so it is reasonable to expect the intersection $B(x, \varepsilon) \cap A$ to be infinite. Certainly, if A is finite, then A is closed whence $\overline{A} = A$ and $\overline{A} \setminus A = \varnothing$ (see Example 3.1). Especially, if X is finite, then there will never exist a point $x \in \overline{A} \setminus A$ for any $A \subseteq X$.

Proof. Since $x \in \overline{A}$, it is obvious that $B(x, \varepsilon) \cap A$ is non-empty. Arguing by contradiction, if $B(x, \varepsilon) \cap A$ were finite, then we would have $B(x, \varepsilon) \cap A = \{x_1, \ldots, x_k\}$ for some $x_1, \ldots, x_k \in X$. Because $x \notin A$, we have $x_j \neq x$ for all $j = 1, \ldots, k$. Therefore,

$$\delta := \min_{1 \leq j \leq k} d(x_j, x) > 0$$

and so $B(x, \delta) \cap A$ is non-empty and contains a point z. But then,

$$z \in B(x, \delta) \cap A \subseteq B(x, \varepsilon) \cap A = \{x_1, \ldots, x_n\}$$

since $\delta < \varepsilon$ (note that $d(x_j, x) < \varepsilon$ for all $j = 1, \ldots, k$). This is a contradiction since

$$d(z, x) < \delta \leq d(x_j, x)$$

for all $j = 1, \ldots, k$. □

Recall that a metric space (X, d) is said to be complete if every Cauchy sequence (x_n) in X converges to a point $x \in X$. Below, we solve an interesting exercise which states that *any* metric space for which disjoint closed sets are "separated" by positive distance, is necessarily complete. The converse, however, does not hold despite it seeming intuitive a priori. See Remark 5.2 below for more details.

Proposition 5.2. *Let (X, d) be a metric space such that $d(A, B) > 0$ for all non-empty disjoint closed subsets $A, B \subseteq X$. Then, (X, d) is complete. Here,*

$$d(A, B) := \inf \{d(a, b) : a \in A,\ b \in B\}$$

is the distance *between the sets A and B.*

Remark 5.2. We should point out here that it is not true in general that $d(A,B) > 0$ for all disjoint non-empty closed sets A, B, even if the metric space (X,d) is assumed to be complete. Consider, for instance, $\mathbb{R}$ with the usual metric and let

$$A := \mathbb{N}, \quad B := \left\{ n + \frac{1}{2n} : n \in \mathbb{N} \right\}.$$

Since $\mathbb{N}^c = (-\infty, 1) \cup \bigcup_{n=1}^{\infty} (n, n+1)$, it is obvious that $A = \mathbb{N}$ is closed in $\mathbb{R}$. To see that B is closed, we use the sequential criterion for closedness. If (x_n) is a sequence of points in B converging to some $x \in \mathbb{R}$, then we may write each x_n in the form

$$x_n := y_n + \frac{1}{2y_n}, \quad y_n \in \mathbb{N}.$$

Since (x_n) is convergent, it is bounded and therefore so is the sequence (y_n). Being bounded, it then follows from Bolzano-Weierstrass that there exists a convergent subsequence (y_{n_k}) of (y_n) converging to some limit $y \in \mathbb{R}$. Since (y_{n_k}) is a sequence of natural numbers, which is closed in $\mathbb{R}$, we must have that $y \in \mathbb{N}$. But then,

$$x = \lim x_n = \lim x_{n_k} = \lim \left(y_{n_k} + \frac{1}{2y_{n_k}} \right) = y + \frac{1}{2y} \in B.$$

Moreover, if there exists $z \in A \cap B$ then we would have that

$$m = n + \frac{1}{2n}$$

for some $m, n \in \mathbb{N}$. Of course, since $n + \frac{1}{2n} > n$, this would imply that $m > n$ whence

$$\frac{1}{2n} = m - n \in \mathbb{N},$$

which is absurd. Thus, A and B are non-empty disjoint closed subsets of $\mathbb{R}$. However, $d(A,B) = 0$ because, for all $n \in \mathbb{N}$, we have

$$0 \leq d(A,B) \leq \left| n - \left(n + \frac{1}{2n} \right) \right| = \frac{1}{2n} \xrightarrow{n \to \infty} 0.$$

Similarly, if A and B are both open and disjoint, the distance between them may be 0. Consider $A = (-1, 0)$ and $B = (0, 1)$ for example. Then, $a_n := -1/n$ is a sequence in A and $b_n := 1/n$ a sequence in B for which

$$d(A,B) \leq |a_n - b_n| = \left| -\frac{1}{n} - \frac{1}{n} \right| = \frac{2}{n}, \quad \forall n \in \mathbb{N}.$$

Thus,

$$0 \leq d(A,B) \leq \lim \frac{2}{n} = 0.$$

Proof. Let (x_n) be a Cauchy sequence in X. Notice that if (x_n) has a convergent subsequence, then (x_n) must automatically converge. Indeed, suppose (x_{n_k}) is a convergent subsequence with limit $x \in X$. Then, given $\varepsilon > 0$, there exists $K \in \mathbb{N}$ such that $d(x_{n_k}, x) < \frac{\varepsilon}{2}$ for all $k \geq K$. Next, using that (x_n) is Cauchy, we may find $N \in \mathbb{N}$ such that $d(x_n, x_m) < \frac{\varepsilon}{2}$ whenever $n, m \geq N$. Letting $M := \max(K, N)$, we see that for all $n \geq N$ there holds

$$d(x_n, x) \leq d(x_n, x_{n_M}) + d(x_{n_M}, x) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

because $n_M \geq M \geq N$ and $M \geq K$. Since $\varepsilon > 0$ was arbitrary, this would imply that $\lim x_n = x$.

Having demonstrated the above, it now suffices to show that (x_n) must have a convergent subsequence. First, if the sequence (x_n) has only finitely many distinct terms, then it must possess a constant subsequence, which certainly converges. Therefore, it remains only to handle the case where (x_n) has infinitely many distinct values. In this case, there exist two subsequences (x_{n_k}) and (x_{m_j}), each with non-repeating distinct terms, such that $x_{n_k} \neq x_{m_j}$ for all $k, j \in \mathbb{N}$. Let now

$$S := \left\{x_{n_k} : k \in \mathbb{N}\right\} \quad \text{and} \quad T := \left\{x_{m_j} : j \in \mathbb{N}\right\}$$

and observe that $S \cap T = \varnothing$ by construction. Furthermore, since (x_n) is Cauchy, given any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $d(x_n, x_m) < \varepsilon$ whenever $n, m \geq N$. So, for any $n_k, m_j \geq N$, we have that

$$d(S, T) \leq d(x_{n_k}, x_{m_j}) < \varepsilon$$

whence it follows that $0 \leq d(S, T) < \varepsilon$ for all $\varepsilon > 0$, i.e. $d(S, T) = 0$. By hypothesis, this implies that one of S or T is *not* closed. Without loss of generality, we may assume S is not closed so that there exists a point $x \in \overline{S} \setminus S$. By Lemma 5.1, this means that

$$B\left(x, \frac{1}{\ell}\right) \cap S$$

is infinite for all $\ell \in \mathbb{N}$, which means that there exists a subsequence $(x_{n_{k_\ell}})$ of (x_{n_k}) (and hence of (x_n)) converging to x as $\ell \longrightarrow \infty$. Consequently, (x_n) has a convergent subsequence and must therefore be itself convergent. □

Finally, let us solve an exercise involving the use of open covers. Given a metric space (X, d), recall that a subset K of X is called *compact* provided every sequence (x_n) in K has a subsequence (x_{n_k}) converging to a point in K. In the lectures, it was shown that compact sets necessarily satisfy the open cover property, which states that every open cover of K admits a finite subcover. That is, if $K \subseteq X$ is compact, then given any family (potentially uncountable) of open sets $\{U_\alpha\}_{\alpha \in I}$ whose union contains K, there exists finitely many $U_{\alpha_1}, \ldots, U_{\alpha_n}$ from the collection $\{U_\alpha\}_{\alpha \in I}$ whose union still contains K. We shall explore compactness even more within the next tutorial, while also focusing on properties of continuous functions.

Lemma 5.3. *Let (X, d) be a metric space and let $K \subseteq X$ be compact. For any $x \in X \setminus K$, there exists an open set $U \supseteq K$ and an open set $V \ni x$ such that $U \cap V = \varnothing$.*

Proof. If K is empty, then the claim is trivial as $U := K = \varnothing$ and $V = X$ will satisfy the required conditions. Otherwise, given any $y \in K$, we must have that $x \neq y$. Thus, we can find open sets $U_y \ni y$ and $V_y \ni x$ such that $U_y \cap V_y = \varnothing$.[3] Now, because $K \subseteq \bigcup_{y \in K} U_y$, it follows that $\{U_y\}_{y \in K}$ is an open cover of K. Being that K is compact, there exists a finite subcover $U_{y_1}, \ldots, U_{y_n}$ such that $K \subseteq U_{y_1} \cup \cdots \cup U_{y_n}$. Define

$$U := U_{y_1} \cup \cdots \cup U_{y_n} \quad \text{and} \quad V := V_{y_1} \cap \cdots \cap V_{y_n}$$

and notice that both U and V are open (finite intersections of open sets remain open). Moreover, $U \supseteq K$ and $V \ni x$, since every V_{y_j} contains x by construction. Finally, because $U_{y_j} \cap V_{y_j} = \varnothing$ for all $j = 1, \ldots, n$, it follows that $U \cap V = \varnothing$. □

Previously, in Example 3.1, we demonstrated that in any metric space (X, d), every finite subset $F \subseteq X$ is necessarily closed in X. Below, we sharpen this example, and show that finite sets are in fact *compact*. Note that in a metric space (X, d), compact sets are automatically closed, so there is no contradiction in the slightest.

Proposition 5.4. *Let (X, d) be a metric space and $F \subseteq X$ a finite set. Then, F is compact.*

Proof. We must show that every sequence (x_n) in F possesses a convergent subsequence. To this end, let (x_n) be a sequence of functions taking values in F, i.e. $x_n \in F$ for each index $n \in \mathbb{N}$. Then, owing to the fact that F is finite, the sequence (x_n) can only take a finite number of distinct values. Consequently, (x_n) has a subsequence (x_{n_k}) which is *constant*. Since constant sequences are automatically convergent, it follows that (x_n) has a convergent subsequence. Moreover, because (x_{n_k}) is a constant sequence in F, we must have $\lim x_{n_k} \in F$. Summarizing, we have shown that every sequence (x_n) in F has a convergent subeqeunce, with limit in F. This establishes precisely the definition of (sequential) compactness, and we are done. □

Although we have defined open covers, and shown that compacta satisfy the *open cover property*[4], we have not proven that this *open cover compactness* is equivalent to the sequential definition given. Nonetheless, the equivalence has been stated and working with open covers (particularly that sequential compactness *implies* open cover compactness is very useful. As such, we demonstrate below that finite sets satisfy the open cover definition of compactness. This is not done as an alternative proof to the above, but rather to ensure we have some practice using open covers.

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[3] Certainly, since $x \neq y$, we have that $d(x, y) > 0$. Then, the open sets $U_y := B\left(y, \frac{d(x,y)}{2}\right)$ and $V_y := B\left(x, \frac{d(x,y)}{2}\right)$ contain y and x (respectively) and have empty intersection by the triangle inequality. In other words, every metric space satisfies the *Hausdorff* condition.

[4] Let (X, d) be a metric space and $K \subseteq X$ a compact set. Recall from the lectures that an open cover of K is a (possibly uncountable) family of open sets $\{U_{\alpha \in I}\}$ in X such that $K \subseteq \bigcup_{\alpha \in I} U_\alpha$. It was shown in class that the compact set K must then satisfy the (finite) open cover property, i.e. there exists a finite subcollection $\{U_{\alpha_1}, U_{\alpha_2}, \ldots, U_{\alpha_N}\}$ from our original open cover $\{U_\alpha\}_{\alpha \in I}$ such that $K \subseteq \bigcup_{k=1}^{N} U_{\alpha_k}$. We remark that every open cover of K having a finite subcover is a property called *open cover compactness*.

Example 5.1. Let (X, d) be a metric space and let $F \subseteq X$ be finite. Prove directly from the definitions that F is open cover compact.

Solution. Let $\{U_\alpha\}_{\alpha \in I}$ be an open cover of F, i.e. each U_α is open in X and $\bigcup_{\alpha \in I} U_\alpha \supseteq F$. If F is empty, then it is automatic that F is contained in a finite subcover of $\{U_\alpha\}_{\alpha \in I}$. If $F \neq \varnothing$, then we may write it as $F = \{x_1, \ldots, x_n\} \subseteq X$ for some $x_1, \ldots, x_n \in X$. For each index $j = 1, \ldots, n$ we have that $x_j \in F \subseteq \bigcup_{\alpha \in I} U_\alpha$ so there must exist some U_{α_j} for which $x_j \in U_{\alpha_j}$. Then, $\{U_{\alpha_1}, U_{\alpha_2}, \ldots, U_{\alpha_n}\}$ is a finite subcollection of $\{U_\alpha\}_{\alpha \in I}$ such that

$$F = \{x_1, \ldots, x_n\} \subseteq U_{\alpha_1} \cup \cdots \cup U_{\alpha_n}.$$

Therefore, F is open cover compact. □

Although finite sets are always compact, there is no need in general for a compact set to be finite. Indeed, as we know from our study of $\mathbb{R}$ with the usual metric, the closed and bounded intervals of the form $[a, b]$ are compact, and are uncountable whenever $a < b$. However, there is some insight on compactness to be gained by trying to understand the nature of compactness in more exotic spaces. For example, suppose $X \neq \varnothing$ is a set and d is the discrete metric on X. Which subsets of (X, d) are compact? Before answering this question and providing a proof of our conjecture, let us first try and decide what the answer to this question *should be*.

First, as seen in the lectures and the earlier examples of these notes, a discrete metric space has the property that *every* subset of X is open in X.[5] Now, if $K \subseteq X$ is compact, then *every* open cover of K must have a finite subcover. Notice that this is a statement about *every* collection of open subsets of X whose union contains K. In addition, increasing the number of sets that are open in X, increases the number of possibilities we have in constructing/selecting open covers of the set K. Put otherwise, the more open sets a space has, the more *expensive* compactness becomes. Since a discrete space (X, d) is a space with the most possible open sets (indeed, every subset of X is open), this implies that compactness is most expensive within a discrete space. In light of our previous example, it is not absurd to suspect that the only compact sets in a discrete space are the finite sets. This is indeed a truth, and we prove this below.

Proposition 5.5. *Let (X, d) be a discrete metric space. Then, a set $K \subseteq X$ is compact if and only if K is finite.*

Proof. By the previous example, it is already known that K being finite implies that K is compact. Conversely, let K be compact. Given any $x \in X$, we associate to it the open set $B_x := B(x, 1) = \{x\}$. Then, $\{B_x\}_{x \in K}$ is an open cover of K. Because K is comapct, and $K = \bigcup_{x \in K} B_x$, there must exist finitely many $B_{x_1}, \ldots, B_{x_N}$ from this family for which $K \subseteq B_{x_1} \cup \cdots \cup B_{x_N}$. But then, we must have

$$K \subseteq B_{x_1} \cup \cdots \cup B_{x_N} = \{x_1, \ldots, x_N\}.$$

□

[5] Indeed, this follows from the fact that, for every $x \in X$, we have $B(x, 1) = \{x\}$. Consequently, given a set $U \subseteq X$, we can write $U = \bigcup_{x \in U} \{x\}$ as the union of open sets. Hence, every $U \subseteq X$ is open. This also implies that every subset of X is closed.

6 Sixth Tutorial

In this tutorial, we continue to concern ourselves with applications of compactness and exercises involving continuous functions on metric spaces. In several instances, we offer multiple solutions to examples with the intention of explicitly demonstrating the applicability of various equivalences (e.g. equivalent formulations of continuity). As a first example, let us verify directly that continuous images of compacta satisfy the open cover property (every open cover admits a finite subcover). Naturally, this is automatic if we assume the result from the lectures that continuous images of compact sets remain compact, however our intention here is to practice with the notion of open coverings without appealing to this result. As a final note, we point out that although we have not proven the open cover property is equivalent to compactness, we *have* shown that all compact sets satisfy this property. Therefore, it is worthwhile to be comfortable working with open covers as it can sometimes be very useful in solving certain problems.

Example 6.1. Let (X, d) be a metric space and $K \subseteq X$ satisfy the open cover property. Prove directly that, if $f : X \to Y$ is continuous, then $f(K)$ also satisfies the open cover property.

Proof. Let $\{U_\alpha\}_{\alpha \in i}$ be an open cover of $f(K)$, i.e. a family of open subsets of Y whose union contains $f(K)$. We aim to show that $f(K)$ can be contained within finitely members from this collection. To see this, observe that $f(K) \subseteq \bigcup_{\alpha \in I} U_\alpha$ implies that $K \subseteq \bigcup_{\alpha \in I} f^{-1}(U_\alpha)$, where each preimage $f^{-1}(U_\alpha)$ is open X by the continuity of f.[6] Therefore, $\{f^{-1}(U_\alpha)\}_{\alpha \in I}$ is an open covering of the compact set K. Being that K is compact, there exist finitely many $f^{-1}(U_{\alpha_1}), \ldots, f^{-1}(U_{\alpha_n})$ such that $K \subseteq \bigcup_{i=1}^n U_{\alpha_i}$ whence $f(K) \subseteq \bigcup_{i=1}^n U_{\alpha_i}$. In summary, we have shown that $\{U_\alpha\}_{\alpha in I}$ admits a finite subcovering of $f(K)$. □

Example 6.2. Endow $\mathbb{R}^n$ and $\mathbb{R}$ with their usual norms and let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuous function such that there exists $c > 0$ with the property that

$$|f(x)| \geq c \|x\|, \quad \forall x \in \mathbb{R}^n. \tag{6.1}$$

Show that $f^{-1}(K)$ is compact in $\mathbb{R}^n$ for every compact set $K \subseteq \mathbb{R}$.[7]

First Proof. Since $f^{-1}(K)$ is a subset of $\mathbb{R}^n$, with the usual norm, it suffices (by Heine-Borel) to show that $f^{-1}(K)$ is closed and bounded. To see that it is closed, notice that K is closed in $\mathbb{R}$ (it is compact) and therefore, by the continuity of f, the preimage $f^{-1}(K)$ must be closed in $\mathbb{R}^n$. Finally, since K is compact it is bounded in $\mathbb{R}$, meaning there exists $M > 0$ such that $|y| \leq M$ for all $y \in K$. Thus, given $x \in f^{-1}(K)$, we have by assumption that

$$\|x\| \leq \frac{|f(x)|}{c} \leq \frac{M}{c}$$

[6] We proved in the lectures that a function $f : X \to Y$ between metric spaces is continuous if and only if $f^{-1}(V)$ is open in X for every open set $V \subseteq Y$. Analogously, f is continuous if and only if $f^{-1}(C)$ is closed in X for each closed set $C \subseteq Y$.

[7] A mapping with this property is called a proper map.

whence $f^{-1}(K)$ is also bounded in $\mathbb{R}^n$. Having shown that $f^{-1}(K)$ is closed and bounded, it follows from Heine-Borel that $f^{-1}(K)$ is compact. □

Second Proof. If $f^{-1}(K)$ is empty, there is nothing to prove. Otherwise, let (x_n) be an arbitrary sequence in $f^{-1}(K)$. By (6.1), applied to each x_n, we have that

$$\|x_n\| \leq \frac{|f(x_n)|}{c},$$

where $f(x_n) \in K$ for every $n \in \mathbb{N}$. Since K is compact, it is closed and bounded and it therefore follows from the above that (x_n) is a bounded sequence in $\mathbb{R}^n$. Consequently, there exists a convergent subsequence of (x_n), denoted (x_{n_k}), such that x_{n_k} converges to some $x \in \mathbb{R}^n$ as $k \to \infty$. But, by continuity, we have that $f(x_{n_k}) \to f(x)$ as $k \to \infty$, and $(f(x_{n_k}))$ is a sequence belonging to the compact set $K \subset \mathbb{R}$. Since K is closed (it is actually compact), we must have $f(x) \in K$ whence $x \in f^{-1}(K)$. Therefore, we have found a subsequence of (x_n) converging to a point $x \in f^{-1}(K)$. This verifies that $f^{-1}(K)$ is sequentially compact. □

Continuity of Inverse Functions

Proposition 6.1. *Let (X, d_X) and (Y, d_Y) be metric spaces, and $K \subseteq X$ be compact. Let $f : K \to Y$ be continuous and injective. Then, the inverse function $f^{-1} : f(K) \to X$ is continuous on $f(K)$.*

Proof. Notice that f^{-1} is indeed defined on $f(K)$ since f is injective on K (and hence is a bijection $K \to f(K)$). Fix $y \in f(K)$; we claim that f^{-1} is continuous at y. Arguing by contradiction, suppose that f^{-1} is not continuous at y. Then, by the sequential criterion for continuity, there exists a sequence (y_n) in $f(K)$ such that $y_n \to y$ and $f^{-1}(y_n) \not\to f^{-1}(y)$, as $n \to \infty$. Consequently, there exists a subsequence (y_{n_k}) and $\varepsilon_0 > 0$ such that

$$d_X(f^{-1}(y_{n_k}), f^{-1}(y)) \geq \varepsilon_0, \quad \forall k \in \mathbb{N}. \tag{6.2}$$

Write $x_{n_k} := f^{-1}(y_{n_k})$ and $x := f^{-1}(y)$, and observe that (x_{n_k}) is a sequence in the compact set K. Therefore, there exists a further subsequence $(x_{n_{k_j}})$ such that $x_{n_{k_j}} \to x^* \in K$. By the continuity of f on K, it must hold that

$$f(x^*) = \lim_{j\to\infty} f(x_{n_{k_j}}).$$

On the other hand, $f(x_{n_{k_j}}) = y_{n_{k_j}}$ for each $j \in \mathbb{N}$, which is a subequence of (y_n) whence $f(x_{n_{k_j}}) \to y$ as $j \to \infty$. Since limits are unique, this forces $f(x^*) = y = f(x)$ so that $x^* = x$. In short, this means that

$$\lim_{j\to\infty} f^{-1}(y_{n_{k_j}}) = \lim_{j\to\infty} x_{n_{k_j}} = x = f^{-1}(y)$$

which contradicts (6.2). It follows that f^{-1} is continuous at $y \in f(K)$ and, since $y \in f(K)$ was arbitrary, this means f^{-1} is continuous on $f^{-1}(K)$. □

Corollary 6.2. *Let (X, d_X) be a compact metric space and (Y, d_Y) any metric space. If $f : X \to Y$ is a continuous bijection, then $f^{-1} : Y \to X$ is continuous.*[8]

Remark 6.1. We offer an alternative proof of Corollary 6.2 using the fact that continuous functions map compacta to compacta; this proof extends to more general spaces, but utilizes the fact that the continuous image of a compact set is always compact in the target space. Recall that f^{-1} is continuous if and only if $(f^{-1})^{-1}(C)$ is closed in Y for each closed set $C \subseteq X$. Put otherwise $f^{-1} : Y \to X$ is continuous , if and only if $f(C)$ is closed in Y for each closed set $C \subseteq X$ (i.e. if and only if f is a *closed* map). Now, if $C \subseteq X$ is closed, then it is compact whence $f(C)$ is compact in Y. But then, $f(C)$ must be closed in Y.

Example 6.3. Let (X, d_X), (Y, d_Y) be metric spaces and suppose $f : X \to Y$ is continuous. If f is injective then $f(\partial A) \subseteq \partial f(A)$

Solution. Suppose f is injective and let $y \in f(\partial A)$. Then $y = f(x_0)$ for some $x \in \partial A$. Let $\varepsilon > 0$ be arbitrary; by continuity of f we may select $\delta > 0$ such that if $x \in X$ then

$$d_X(x, x_0) < \delta \implies d_Y(f(x), y) = d_Y(f(x), f(x_0)) < \varepsilon \,. \tag{6.3}$$

Because $x_0 \in \partial A$, we may select $x_1 \in B_\delta(x_0) \cap A$ and $x_2 \in B_\delta(x_0) \cap A^c$.

Now, let $y_1 = f(x_1)$ and $y_2 = f(x_2)$. Because $x_1, x_2 \in B_\delta(x_0)$ we have $d_X(x_1, x_0) < \delta$ and $d_X(x_2, x_0) < \delta$. It then follows from (6.3) that $d_Y(y_1, y) < \varepsilon$ and $d_Y(y_2, y) < \varepsilon$. Thus, $y_1, y_2 \in B_\varepsilon(y)$.

Moreover, because $x_1 \in A$ we see that $y_1 = f(x_1) \in f(A)$ so the set $B_\varepsilon(y) \cap f(A)$ is non-empty. Similarly, because $x_2 \in A^c$ we have $y_2 = f(x_1) \in f(A^c)$. Since f is injective, it follows[9] that $y_2 \in f(A)^c$ so, since $y_2 \in B_\varepsilon(y)$, we deduce that $B_\varepsilon(y) \cap f(A)^c$ is non-empty. So, for an arbitrary $\varepsilon > 0$ we have shows that $B_\varepsilon(y) \cap f(A)$ and $B_\varepsilon(y) \cap f(A)^c$ are non-empty. It follows that $y \in \partial f(A)$. □

7 Seventh Tutorial

This tutorial was a review session, wherein we reviewed important class results, including the relationship between boundedness, closedness, completeness and compactness. We went over solutions to problems from your assignments, and covered examples from previous tutorials that were either important, and some problems from previous sections we did not have the time to cover explicitly during previous tutorials. No new material or problems were covered here.

[8] A continuous bijection between metric spaces whose inverse is also continuous is called a *homeomorphism.* A homeomorphism between metric spaces (or, more generally, topological spaces) is a bijective continuous map such that its inverse $f^{-1} : Y \to X$ is also continuous. These are the isomorphisms in the category of topological spaces.

[9] If $y_2 \in f(A)$ then $y_2 = f(\tilde{x})$ for some $\tilde{x} \in A$. Then, $f(x_2) = y_2 = f(\tilde{x})$ so, by injectivity, $x_2 = \tilde{x}$. However, this is a contradiction because $x_2 \in A^c$ while $\tilde{x} \in A$. We conclude that $y_2 \in f(A)$ is impossible so $y_2 \in f(A)^c$.

8 Eighth Tutorial

In this tutorial, our focus will be problems involving continuous functions. Especially, we aim to solve several exercises involving continuous functions on defined on compact metric spaces (or, more generally, compact subsets of metric spaces). Depending on the nature of the given problem, we may use different (but equivalent) notions of continuity. Thus, we should recall that the $\varepsilon - \delta$, sequential, and open/closed set characterizations of continuity are all equivalent by class results. Furthermore, we shall liberally invoke the fact that a continuous function

$$f : K \subseteq X \to \mathbb{R}$$

achieves its maximum (respectively minimum) on K provided K is a compact subset of a metric space (X, d). Finally, given a metric space (X, d) and a function $f : X \to X$, a point $\xi \in X$ is said to be a *fixed point* of f provided $f(\xi) = \xi$. With regards to fixed points, the important cornerstone result to keep in mind is the Banach Fixed Point theorem, which states that every contraction mapping $X \to X$ possesses a unique fixed point $\xi \in X$ if (X, d) is complete. That is, if (X, d) is a complete metric space and $f : X \to X$ satisfies

$$d(f(x), f(y)) \leq cd(x, y)$$

for some constant $c \in [0, 1)$ and all $x, y \in X$, then there exists a unique point $\xi \in X$ such that $f(\xi) = \xi$.

In our first example, we demonstrate that the assumption that f be a contraction may be replaced by the requirement that some composition of f be contractive.

Example 8.1. Let (X, d) be a complete metric space and let $f : X \to X$ be given. Given $k \in \mathbb{N}$, let $f^{(k)}$ denote the k-fold composition

$$f^{(k)}(x) := \underbrace{(f \circ \cdots \circ f)}_{k-times}(x).$$

Suppose there exists $k \in \mathbb{N}$ such that $f^{(k)}$ is a contraction. Then, f still possesses a unique fixed point in X. Indeed, to see the existence of such a point, observe that $f^{(k)}$ has a fixed point $\xi \in X$ by virtue of the Banach Fixed Point Theorem. But then,

$$f^{(k)}(\xi) = \xi \implies f\left(f^{(k)}(\xi)\right) = f(\xi) \iff f^{(k)}(f(\xi)) = f(\xi),$$

meaning that $f(\xi)$ is a fixed point for $f^{(k)}$ whenever ξ is. By the uniqueness of the fixed point, we infer that $f(\xi) = \xi$. Hence, f has a fixed point. Let now $\eta \in X$ be any fixed point of f, i.e. $f(\eta) = \eta$. Clearly,

$$f^{(k)}(\eta) = f^{(k-1)}(f(\eta)) = f^{(k-1)}(\eta) = \cdots = f(\eta) = \eta$$

whence η is a fixed point of $f^{(k)}$. Since $f^{(k)}$ has a unique fixed point, it follows that f has a unique fixed point.

8.1 Fixed Points in Compact Spaces

Part of the power of the Banach Fixed Point theorem is the fact that it only requires that the domain space (X, d) be complete – not necessarily compact. In turn, it makes a relatively strong assumption about the nature of the function f. Indeed, by requiring that f be a contraction mapping, we are automatically assuming that f is Lipschitz continuous and that it "shrinks" distances by a uniform factor (the constant $c > 0$ is independent of the points $x, y \in X$). In light of these observations, it is natural to ask whether the conclusion of the Banach Fixed Point theorem will continue to hold if we relax the contraction assumption on f, bur strengthen the assumptions on the space X (e.g. requiring that (X, d) be compact instead of merely complete).

Proposition 8.1. *Let (X, d) be a compact metric space and $T : X \to X$ a continuous function such that*

$$d(T(x), T(y)) < d(x, y), \quad \forall x, y \in X,\ x \neq y.$$

Then, T has a unique fixed point in X.

Remark 8.1. Completeness would not be sufficient in this case. Indeed, consider the function

$$T : \mathbb{R} \to \mathbb{R}, \quad T(x) = \sqrt{x^2 + 1}$$

where $\mathbb{R}$ is given the standard metric. Evidently, $\mathbb{R}$ is complete and T is continuous. However, the function T cannot achieve a fixed point in $\mathbb{R}$. To see this, suppose by way of contradiction that there exists $x \in \mathbb{R}$ with $T(x) = x$, i.e. $x = \sqrt{x^2 + 1}$. Then, $x > 0$ and $x^2 = x^2 + 1$ which implies $0 = 1$ – a clear absurdity. However, it does hold that

$$\begin{aligned} |T(x) - T(y)| = \left|\sqrt{x^2+1} - \sqrt{y^2+1}\right| &= \frac{\left|x^2 - y^2\right|}{\sqrt{x^2+1} + \sqrt{y^2+1}} \\ &\leq \left(\frac{|x| + |y|}{\sqrt{x^2+1} + \sqrt{y^2+1}}\right)|x - y| \\ &< \left(\frac{|x| + |y|}{|x| + |y|}\right)|x - y| \\ &= |x - y| \end{aligned}$$

for all $x, y \in \mathbb{R}$ with $x \neq y$. In this counter example, it is important to understand that the Banach Fixed Point Theorem does *not* apply to T since the assumption $d(T(x), T(y)) < d(x, y)$ for all $x \neq y$ does not imply the existence of a uniform constant $c \in (0, 1)$ such that $d(T(x), T(y)) \leq cd(x, y)$. That is, T need not be a contraction mapping.

Proof. Consider the function

$$f : X \to \mathbb{R}, \quad x \mapsto d(x, T(x)).$$

We claim first that f is continuous on X. Indeed, given any distinct $x, y \in X$ it holds that

$$\begin{aligned} d(x, T(x)) \leq d(x, y) + d(y, T(x)) &\leq d(x, y) + d(y, T(y)) + d(T(y), T(x)) \\ &< 2d(x, y) + d(y, T(y)) \end{aligned}$$

whence

$$f(x) - f(y) = d(x, T(x)) - d(y, T(y)) \leq 2d(x, y).$$

Reversing the roles of x and y in the argument above, we also obtain $f(y) - f(x) \leq 2d(x, y)$ so that

$$|f(x) - f(y)| \leq 2d(x, y)$$

for all $x \neq y$. Since this clearly holds when $x = y$, we conclude that f satisfies the Lipschitz condition on all of X. Being that X is compact, it follows that $f(X)$ is a compact subset of $\mathbb{R}$ and therefore has a minimum $\alpha \geq 0$. That is, there exists $x_0 \in X$ such that $f(x_0) = \alpha$, where $\alpha = \min f(X)$. Now, if $\alpha > 0$, then observe that $T(T(x_0)) \in f(X)$ satisfies

$$f(T(x_0)) = d(T(x_0), T(T(x_0))) < d(x_0, T(x_0)) = f(x_0) = \alpha$$

contradicting the fact that α is the minimum of $f(X)$. Thus, we must have $\alpha = 0$ whence $0 = f(x_0) = d(x_0, T(x_0))$ and x_0 is a fixed point of T. To see that x_0 is unique, suppose $z \in X$ is a different fixed point of T in x_0. Then,

$$d(x_0, z) = d(T(x_0), T(z)) < d(x_0, z)$$

which is absurd. □

With the help of the Mean Value Theorem, it follows that suitably differentiable functions $[a, b] \to [a, b]$ whose first derivative is bounded by 1 in absolute value admit a unique fixed point in $[a, b]$. This provides a nice condition for the existence of a unique fixed point in $\mathbb{R}$ that is relatively easy to verify by inspection and helps us to "realize" the last proposition in a more familiar context.

Corollary 8.2. *Let $[a, b] \subset \mathbb{R}$ be an interval and $f : [a, b] \to [a, b]$ be continuous on $[a, b]$ and differentiable on (a, b). Assume in addition that $|f'(x)| < 1$ for all $x \in (a, b)$. Then, f possesses a unique fixed point in $[a, b]$.*

Proof. Observe that $[a, b]$ is a compact metric space with the usual metric it inherits from $\mathbb{R}$. We claim that f satisfies the requirements of Proposition 8.1, whence it will follow that f admits a unique fixed point in $[a, b]$. To this end, let $x, y \in [a, b]$ with $x \neq y$. Without loss of generality, we may assume that $x < y$. Since f is continuous on $[x, y]$ and differentiable on (x, y), the Mean Value Theorem asserts that there exists $c \in (x, y)$ such that

$$\frac{f(y) - f(x)}{y - x} = f'(c)$$

whence $|f(x) - f(y)| = |f'(c)|\,|x - y| < |x - y|$. Therefore, f satisfies the conditions of the previous proposition and must admit a unique fixed point on $[a, b]$. □

The following proposition treats the existence of fixed points for certain functions $f : B \to B$, where $B \subset \mathbb{R}^n$ is the closed unit ball in $\mathbb{R}^n$. As before, we would like to point out that we once again weaken the assumption on the function f in favour of imposing requirements on the domain space. In this particular case, we are implicitly imposing some geometric requirements on the domain (it is a ball!), beyond the merely topological notion of compactness.

Proposition 8.3. *Give $\mathbb{R}^n$ the usual Euclidean norm, let $B \subset \mathbb{R}^n$ be the closed unit ball, and let $f : B \to B$ be Lipschitz continuous with Lipschitz constant 1, i.e.*

$$\|f(x) - f(y)\| \leq \|x - y\|, \quad \forall x, y \in B.$$

Then, f has a fixed point.

Proof. We claim that

$$\inf_{x \in B} \|f(x) - x\| = 0.$$

This will be sufficient since B is compact (the infimum is achieved at some point on B).[10] To this end, let $\varepsilon \in (0, 1)$ be arbitrary and consider the function

$$f_\varepsilon : B \to B, \quad x \mapsto (1 - \varepsilon) f(x).$$

For all $x, y \in B$, we have

$$\|f_\varepsilon(x) - f_\varepsilon(y)\| = (1 - \varepsilon) \|f(x) - f(y)\| \leq (1 - \varepsilon) \|x - y\|$$

whence f_ε is a contraction of B. Since B is compact, it is also complete. Consequently, the Banach Fixed Point Theorem guarantees the existence of a unique point $x_\varepsilon \in B$ such that $f_\varepsilon(x_\varepsilon) = x_\varepsilon$. That is, there exists a unique $x_\varepsilon \in B$ such that

$$(1 - \varepsilon) f(x_\varepsilon) = x_\varepsilon.$$

Therefore, for each $\varepsilon \in (0, 1)$,

$$\inf_{x \in B} \|f(x) - x\| \leq \|f(x_\varepsilon) - x_\varepsilon\| = \|\varepsilon f(x_\varepsilon)\| \leq \varepsilon,$$

where we have used that $\|f(x_\varepsilon)\| \leq 1$ because f takes values in the unit ball B. This means that $\inf_{x \in B} \|f(x) - x\| \leq \|f(x_\varepsilon) - x_\varepsilon\| \leq \varepsilon$ for all $\varepsilon \in (0, 1)$ and so

$$\inf_{x \in B} \|f(x) - x\| = 0,$$

as was asserted. By our previous comments, the proof is complete. □

[10] Indeed, since $f : B \to B$ is continuous, the mapping $x \mapsto \|f(x) - x\|$ is a continuous map $B \to [0, \infty)$ by composition. Being that the domain B is compact, this map achieves its absolute minimum at a point $\xi \in B$. If we know that $\inf_{x \in B} \|f(x) - x\| = 0$, then we must have that

$$\|f(\xi) - \xi\| = \inf_{x \in B} \|f(x) - x\| = 0$$

which forces $f(\xi) = \xi$.

Remark 8.2. As it turns out, this statement remains true if we merely assume that f is continuous $B \to B$. This is known as Brouwer's fixed point theorem, and its proof is significantly harder and far beyond the scope of this course.

Remark 8.3. It is not always true that 1-Lipschitz functions have fixed points over compacta. For instance, let us treat the unit sphere $\mathbb{S}^2$ in $\mathbb{R}^3$. Consider the mapping $\rho : \mathbb{S}^2 \to \mathbb{S}^2$ which reflects the sphere across the xy-plane (e.g. sending the north pole to the south pole and vice versa) composed with a small rotation (e.g rotation by an angle $\theta < \pi/4$) about the z-axis. This function is an *isometry* (i.e. distance preserving mapping – in particular 1-Lipschitz) of the sphere that leaves no points fixed, since the first operation leaves only the equator fixed which is undone by the rotation.

One can improve Proposition 8.3 further so that the statement does not apply only to balls. In fact, it is possible to improve the statement to include more general subsets of $\mathbb{R}^n$, while minimally altering the proof. This is done below, where we show that the conclusions of Proposition 8.3 remain valid when the domain (and range) of f is *star shaped* with respect to the origin.

Definition 8.1 (Star Shaped)**.** A non-empty set $S \subseteq \mathbb{R}^n$ is called *star shaped* (with respect to the origin) if $x \in S$ implies that $tx \in S$ for all $0 \leq t \leq 1$.

Intuitively speaking, a non-empty set $S \subseteq \mathbb{R}^n$ is star shaped (with respect to 0) if for each $x \in S$, the straight line connecting the origin to the point x is entirely contained within S. In particular, this means that $0 = 0 \cdot x \in S$. Thus, S necessarily contains the origin. Examples of star shaped domains include all balls centred at the origin, and all convex sets containing the origin.

Proposition 8.4. *Let $S \subseteq \mathbb{R}^n$ be closed and bounded, and star shaped (with respect to the origin). Suppose $f : S \to S$ is a function such that*

$$\|f(x) - f(y)\| \leq \|x - y\|$$

for all $x, y \in S$. Then, f has a fixed point in S.

Proof. We alter the proof slightly from Proposition 8.3, not out of necessity, but to take the opportunity to demonstrate a different approach.

Given arbitrary $k \in \mathbb{N}$, consider the function

$$f_k : S \to \mathbb{R}^n, \quad f(x) := \left(1 - \frac{1}{k}\right) f(x).$$

Because f is a function $S \to S$, we have $f(x) \in S$ for each $x \in S$. Now, since $0 \leq 1 - \frac{1}{k} \leq 1$ for each $k \in \mathbb{N}$, and S is star shaped, it follows that $f_k(x) = (1 - \frac{1}{k}) f(x) \in S$. Therefore, f_k is a

mapping $S \to S$. Moreover, for any $x, y \in S$, we have

$$\begin{aligned} \|f_k(x) - f_k(y)\| &= \left\| \left(1 - \frac{1}{k}\right) f(x) - \left(1 - \frac{1}{k}\right) f(y) \right\| \\ &= \left(1 - \frac{1}{k}\right) \|f(x) - f(y)\| \\ &\leq \left(1 - \frac{1}{k}\right) \|x - y\| . \end{aligned}$$

Since $0 \leq 1 - \frac{1}{k} < 1$ for all $k \in \mathbb{N}$, we infer that every f_k is a contraction $S \to S$. Owing to the fact that S is closed, and $\mathbb{R}^n$ is complete, we see that S is complete. Thus, the Banach fixed point theorem guarantees the existence of a (unique) point $x_k \in S$ such that $f_k(x_k) = x_k$.

Finally, as $k \in \mathbb{N}$ was arbitrary, this produces a sequence (x_k) in S such that $f_k(x_k) = x_k$ for all $k \in \mathbb{N}$. Because S is closed and bounded, it is compact by Heine-Borel and there exists a subsequence (x_{k_j}) of (x_k) converging to some $x^* \in S$. By continuity, it follows that

$$\begin{aligned} x^* = \lim x_{k_j} = \lim f_k(x_{k_j}) &= \lim \left[\left(1 - \frac{1}{k_j}\right) f(x_{k_j}) \right] \\ &= \lim \left(1 - \frac{1}{k_j}\right) \lim \left(f(x_{k_j}) \right) \\ &= 1 \cdot f(x^*) \\ &= f(x^*). \end{aligned}$$

Thus, x^* is the desired fixed point of f we were seeking. □

8.2 Isometries

We now take a brief segue to introduce (formally) the concept of an isometry, which we have previously touched upon. Essentially, an isometry is a mapping between metric spaces that preserves distances, rather than shrinking them. Surjective isometries may be thought of as the "isomorphisms in the category of metric spaces", which is fancy jargon for saying that two metric spaces (X, d_X) and (Y, d_Y) are "the same" if there exists a surjective isometry $X \to Y$.

Definition 8.2 (Isometry)**.** Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \to Y$ is called an isometry provided

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2)$$

for all $x_1, x_2 \in X$.

The main purpose of this last definition is to set the framework for the next exercise, which we state as a proposition. It states that an isometry from a compact metric space X to itself is necessarily bijective and continuous. In fact, as one can show, it's inverse will *also* be continuous.[11]

[11] Hence, such a mapping is an automorphism of X, i.e. a homeomorphism of X onto itself.

Proposition 8.5. *Let (X, d) be a compact metric space and suppose that $f : X \to X$ is an isometry of X. Then, f is bijective.*

Proof. First, we note that f is automatically injective. Certainly, given $x, y \in X$, if $f(x) = f(y)$ then

$$0 = d(f(x), f(y)) = d(x, y)$$

forces $x = y$ by the non-degeneracy of the metric. Thus, f is injective. Moreover, f is continuous as a mapping $X \to X$. To see this, fix $y \in X$ and let $\varepsilon > 0$ be given. Taking $\delta := \varepsilon > 0$, it follows that

$$d(f(x), f(y)) = d(x, y) < \delta = \varepsilon$$

whenever $d(x, y) < \delta$. Therefore, by the $\varepsilon - \delta$ definition of continuity, f is continuous on all of X. Since X is compact, we infer that $f(X)$ is a compact subspace of X.

Now, to show that f is bijective, it remains only to verify that f is surjective. If f is not surjective, then $f(X) \subsetneq X$ and there exists a point $x_0 \in X \setminus f(X)$. Define a sequence (x_n) in X recursively by the rule $x_n := f(x_{n-1})$, for $n \in \mathbb{N}$. Clearly, $x_n \in f(X)$ for each $n \in \mathbb{N}$ whence

$$d(x_0, x_n) \geq d(x_0, f(X)) =: \delta > 0, \quad \forall n \in \mathbb{N}$$

because $f(X)$ is compact, and hence closed, in X. Then, for all $m > n \geq 1$ we find that

$$d(x_m, x_n) = d(f(x_{m-1}), f(x_{n-1})) = d(x_{m-1}, x_{n-1}) = \cdots = d(x_{m-n}, x_0) \geq \delta > 0.$$

It follows from the above that (x_n) can have no Cauchy subsequences, which directly contradicts the sequential compactness of the space (X, d). Ergo, $f(X) = X$ whence f is surjective. □

As an exercise, the leave the following.

Exercise 8.1. Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ a surjective isometry. Justify that $f^{-1} : Y \to X$ exists and demonstrate that f^{-1} is itself a surjective isometry. Conclude that f^{-1} is continuous.

8.3 Fixed Point Methods in Expanding Maps

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \to Y$ is called an *expansion* map, or an expanding map, provided there exists a constant $C \geq 1$ such that

$$d_Y(f(x), f(y)) \geq C d_X(x, y)$$

for all $x, y \in X$. As it turns out, the case where $(X, d_X) = (Y, d_Y)$ is a compact metric space and $f : X \to X$ is an expansion for $C = 1$, is particularly interesting; in the sense that it is not interesting at all. This is what the next example establishes, which should be thought of as an additional exercise in which we simply practice out fixed point (iterative) proof methods.

Proposition 8.6. *Let (X, d) be a compact metric space and $f : X \to X$ a function for which*

$$d(f(x), f(y)) \geq d(x, y), \quad \forall x, y \in X. \tag{8.1}$$

Then, f is an isometry $X \to X$. In particular, it is Lipschitz continuous, and invertible with continuous inverse.

Remark 8.4. Note that we are not assuming that the function f is continuous, a priori. We are simply assuming that X is compact, and that f satisfies the expanding condition (8.1). From this, we are deducing continuity (in fact, a very special form of continuity). Moreover, as you shall see, although the statement has nothing to do with fixed points, and the proof does not invoke fixed point theory directly, it *does* follow an iterative scheme which is highly reminiscent of fixed point methodology.

Proof. We will prove that $d(f(x), f(y)) \leq d(x, y)$ for all $x, y \in X$. This, combined with the hypothesis in (8.1), will imply that f is an isometry. Then, the remaining assertions will follow from previous results (see Proposition 8.5, Corollary 6.2 and Exercise 8.1).

To this end, let $x, y \in X$ be arbitrary and fixed. Define for each $n \in \mathbb{N}$ the points x_n and y_n given below by

$$x_n := f^{(n)}(x) \quad \text{and} \quad y_n := f^{(n)}(y).$$

Here, $f^{(n)} = f \circ \cdots \circ f$ denotes the n-fold composition of f with itself. Of course, since f is a mapping $X \to X$, this procedure produces sequences (x_n) and (y_n) in the space X. Owing to the compactness of X, we may find subsequences (x_{n_k}) and (y_{n_k}) which converge in (X, d) as $k \to \infty$. Note that we may select the same indices $(n_k)_{k\in\mathbb{N}}$ for both subsequences (why[12] is this?).

Consequently, (x_{n_k}) and (y_{n_k}) are Cauchy in (X, d). Defining $m_k := n_{k+1} - n_k \in \mathbb{N}$, we show that $x_{m_k} \to x$ and $y_{m_k} \to y$, as $k \to \infty$. To see this, note that for each $k \in \mathbb{N}$ there holds

$$\begin{aligned} d(x_{m_k}, x) = d\left(f^{(m_k)}(x), x\right) &= d\left(f^{(n_{k+1}-n_k)}(x), x\right) \\ &\leq d\left(f^{n_{k+1}}(x), f^{(n_k)}(x)\right) \\ &= d(x_{n_{k+1}}, x_{n_k}) \xrightarrow{k\to\infty} 0. \end{aligned}$$

Here, we have iteratively applied (8.1) n_k-times and used in the last step that (x_{n_k}) is Cauchy. Thus, $x_{m_k} \to x$ as $k \to \infty$, and $y_{m_k} \to y$ by a verbatim argument. Finally, as $m_k \geq 1$ for all $k \in \mathbb{N}$, we have from successive applications of (8.1) (yet again) that

$$d(f(x), f(y)) \leq d\left(f^{(m_k)}(x), f^{(m_k)}(y)\right) = d(x_{m_k}, y_{m_k}).$$

[12] By compactness, every sequence in X has a convergent subsequence. Thus, (x_n) has a convergent subsequence (x_{n_j}). Next, (y_{n_j}) is itself a sequence in the compact space (X, d), and therefore it has a subsequence $(y_{n_{j_k}})$ converging in X. As (x_{n_j}) is convergent, so is $(x_{n_{j_k}})$. Defining $n_k := n_{j_k}$ yields two converging subsequences (x_{n_k}) and (y_{n_k}) of (x_n) and (y_n), respectively, with the same indices and converging as $k \to \infty$.

Therefore, by your midterm exam,

$$d(f(x), f(y)) \leq \lim d(x_{m_k}, y_{m_k}) = d(x, y)$$

because $x_{m_k} \to x$ and $y_{m_k} \to y$ as $k \to \infty$. By our earlier remarks, this completes the proof. □

8.4 A Glimpse Into the World of Linear Operators

In light of the problems on Assignment 4 involving linear mappings between normed spaces, we thought it instructive and/or helpful to include a subsection shedding some light on these newer notions encountered. This especially applies to the "operator norm" introduced within the first question of the aforementioned assignment. However, given the rather restrictive time frame of the tutorials, it is very unlikely we will have the time to cover all this material. In addition, the contents of the final subsection is purely for additional practice and goes beyond what could be asked of you. As such, the final subsection is marked as optional.

Definition 8.3. Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces. A linear operator $T : X \to Y$ is said to be bounded if

$$\|T\|_{X,Y} := \sup \{\|T(x)\|_Y : \|x\|_X = 1\} < \infty.$$

By assignment 4, this is equivalent to there existing $C \geq 0$ such that $\|T(x)\|_Y \leq C \|x\|_X$ for all $x \in X$. Furthermore, a linear operator between normed spaces is continuous if and only if it is bounded (see again Assignment 4). Consequently, you will have proven the following:

Theorem 8.7. *Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces and $T : X \to Y$ linear. The following statements are equivalent.*

(1) T is continuous;

(2) $\|T\|_{X,Y}$ is a well defined real number, i.e. T is bounded;

(3) there exists $C \geq 0$ such that $\|T(x)\|_Y \leq C \|x\|_X$ for all $x \in X$.

Remark 8.5. The term "bounded" is a bit of a misnomer here. A function $f : X \to Y$ between normed spaces is usually said to be bounded if there exists $M \geq 0$ such that $\|f(x)\|_Y \leq M$ for all $x \in X$. However, in the particular case of linear operators, this is a useless definition. Indeed, if $f : X \to Y$ is linear and satisfies $\|T(x)\|_Y \leq M$ for all $x \in X$, then $f = 0$ on all of X. Indeed, suppose for a contradiction that there exists $x \in X$ such that $f(x) \neq 0$. Since f is linear, $x \neq 0$. Then, let $u := (M + 1)x/\|T(x)\|_Y$ and notice that $\|T(u)\|_Y \leq M$ by hypothesis. However,

$$\|T(u)\|_Y = \left\| \frac{M+1}{\|T(x)\|_Y} T(x) \right\|_Y = M + 1 > M.$$

In assignment 4, you will also be proving that any linear operator $X \to Y$ between *finite dimensional* normed spaces is continuous/bounded. If you follow the given hint, you will be using the correspondence between linear operators and matrices. However, there exists a proof that only requires that the domain space X be finite dimensional. That is, any linear operator $X \to Y$ with X being finite dimensional is necessarily bounded, regardless of the dimension of Y. This is not necessarily the case when $\dim(X) = \infty$, as the next example shows.

Example 8.2. Let X be the space of real valued differentiable functions on $[0, 1]$ equipped with the supremum norm $\|\cdot\|_\infty$. Since X contains all monomials of arbitrary degree, it is clear that X is infinite dimensional over $\mathbb{R}$. Now, consider the linear functional[13]

$$d_0 : X \to \mathbb{R}, \quad d_0(f) := f'(0).$$

Since differentiation is a linear process, this is a linear functional on X. However, d_0 is not continuous (and hence not bounded) on X. To see this, consider the sequence

$$f_n(x) := \frac{\sin(n^2 x)}{n}$$

which is obviously smooth (infinitely differentiable) for each $n \in \mathbb{N}$. Moreover,

$$\|f_n\|_\infty = \sup_{x\in[0,1]} \left|\frac{\sin(n^2 x)}{n}\right| \leq \frac{1}{n} \xrightarrow{n\to\infty} 0,$$

which means that $f_n \to 0$ in X. Observe also that $d_0(0) = 0$. However,

$$d_0(f_n) = f_n'(0) = \frac{n^2 \cos(n^2 0)}{n} = n$$

which does not converge to $d_0(0)$ as $n \to \infty$. Hence, we have that $f_n \to 0$ whilst $d_0(f_n) \not\to d_0(0)$ as $n \to \infty$. We infer that d_0 fails the sequential criterion for continuity whence d_0 is *not* continuous.

Given normed spaces $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ over $\mathbb{R}$ and a mapping $T : X \to Y$, we define the *kernel* of T to be the subset of X consisting of all points on which T is equal to 0, i.e.

$$\ker T := \{x \in X : T(x) = 0\}.$$

In the case of linear functionals, there is an important relation between the kernel and the continuity of the functional. In fact, we have the following:

Example 8.3. Let $(X, \|\cdot\|)$ be a normed space and $f : X \to \mathbb{R}$ a non-zero linear functional, meaning f is not everywhere vanishing. Then, f is *not* continuous on X if and only if $\ker f$ is dense in X.

[13] Given a vector space X over $\mathbb{R}$, a linear functional is a linear function $f : X \to \mathbb{R}$. These objects have their own "special name" as they are a central object of study in functional analysis. There is nothing special about their definition however; they are simply real valued linear mappings defined on X.

Proof. Suppose first that f is not continuous $X \to \mathbb{R}$. Then, by our earlier discussions and your assignment, this implies that $\|f\|_{X,\mathbb{R}} = \infty$. That is, there exists a sequence (x_n) in X with $\|x_n\| = 1$ for all $n \in \mathbb{N}$ such that $|f(x_n)| \to \infty$. In fact, since $|f(x_n)| \to \infty$, we may (after omitting at most finitely many terms) assume without loss of generality that $f(x_n) \neq 0$ for all $n \in \mathbb{N}$. Given any $x \in X$, we consider the sequence

$$z_n := x - \frac{f(x)}{f(x_n)} x_n$$

and notice that $z_n \to x$ as $n \to \infty$ because

$$\|z_n - x\| = \frac{|f(x)|}{|f(x_n)|} \|x_n\| = \frac{|f(x)|}{|f(x_n)|} \xrightarrow{n\to\infty} 0.$$

However, because f is linear,

$$f(z_n) = f(x) - \frac{f(x)}{f(x_n)} f(x_n) = f(x) - f(x) = 0.$$

Thus, (z_n) is a sequence in $\ker f$ converging to $x \in X$. Since $x \in X$ was arbitrary, we have shown that any point in the space X may be approximated by sequences in $\ker f$, which implies that $\ker f$ is dense in X.

Conversely, suppose that $\ker f$ is dense in X. Then, f cannot be continuous. Indeed, if f were continuous, then for any $x \in X$ we could find by density a sequence (z_n) in $\ker f$ such that $z_n \to x$ as $n \to \infty$. But then, continuity would imply

$$f(x) = \lim f(z_n) = \lim 0 = 0$$

by the sequential characterization of continuity. Thus, $f(x) = 0$ for all $x \in X$ which contradicts our assumption that $f \not\equiv 0$. Consequently, if $\ker f$ is dense and f is non-zero, then f is discontinuous. □

From here we can deduce a characterization of continuity in terms of the kernel.

Example 8.4. Let X be a normed space and $f : X \to \mathbb{R}$ a linear functional. Then, f is continuous/bounded if and only if $\ker f$ is closed.

Proof. Suppose first f is continuous $X \to \mathbb{R}$. Then, $\ker f = f^{-1}(\{0\})$ is closed in X since it is the continuous preimage of a closed set in $\mathbb{R}$.

Conversely, suppose $\ker f$ is closed. If $\ker f = X$ then $f(x) = 0$ for all $x \in X$ whence f is constant, and in particular continuous. Thus, we are reduced to the situation where $\ker f$ is a proper subset of X, in which case f is not everywhere vanishing. If f were not continuous, then the previous example implies that $\ker f$ is dense in X, meaning $\overline{\ker f} = X$. However, since $\ker f$ was assumed to be closed, this would imply $\ker f = \overline{\ker f} = X$ implying $f \equiv 0$. This contradiction means that $\ker f$ cannot be dense in this case, which of course means by the last example that f is continuous. □

8.5 The Space of Linear Operators and Extensions (Very Optional)

Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces and denote by $\|\cdot\|_{X,Y}$ the "norm" from the previous section (and, of course, your 4th assignment). Define $\mathcal{L}(X, Y)$ to be the real vector space of all bounded/continuous linear mappings $X \to Y$. Then, one easily show that $\|\cdot\|_{X,Y}$ is a valid norm on $\mathcal{L}(X, Y)$. Below, we show that $\mathcal{L}(X, Y)$ is itself a Banach space whenever Y is Banach. We would like to point out that we are mostly including this statement/proof for practice purposes since it utilizes an argument highly reminiscent of that which you employed to show that $C([a, b])$ is complete when endowed with the supremum norm (see Assignment 3). Consequently, some students may find this proof instructive or helpful review.

Proposition 8.8. *Let $(X, \|\cdot\|_X)$ be a normed space and $(Y, \|\cdot\|_Y)$ be a Banach Space. Then, $\mathcal{L}(X, Y)$ is a Banach Space.*

Proof. Let (T_n) be a Cauchy sequence in $\mathcal{L}(X, Y)$ with respect to the operator norm $\|\cdot\|_{X,Y}$. Now, given $x \in X$ with $x \neq 0$, it follows that

$$\|T_n(x) - T_m(x)\|_Y = \|(T_n - T_m)(x)\|_Y = \left\| \|x\|_X \, (T_n - T_m)\left(\frac{x}{\|x\|_X}\right) \right\|_Y \tag{8.2}$$

$$= \|x\|_X \left\| (T_n - T_m)\left(\frac{x}{\|x\|_X}\right) \right\|_Y \tag{8.3}$$

$$\leq \|T_n - T_m\|_{X,Y} \, \|x\|_X \, . \tag{8.4}$$

Since $T_n(0) = T_m(0)$ by linearity, this inequality holds trivially when $x = 0$. Hence, we obtain

$$\|T_n(x) - T_m(x)\|_Y \leq \|T_n - T_m\|_{X,Y} \, \|x\| \, , \quad \forall x \in X. \tag{8.5}$$

Notice that, since each T_n is linear, the sequence $(T_n(0))_{n\in\mathbb{N}}$ is the constant 0-sequence in Y and is hence convergent. If $x \neq 0$, then given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $\|T_n - T_m\|_{X,Y} < \varepsilon / \|x\|_X$ for all $n, m \geq N$. Consequently, by (8.4)

$$\|T_n(x) - T_m(x)\|_Y \leq \|T_n - T_m\|_{X,Y} \, \|x\|_X < \varepsilon$$

for all $n, m \geq N$. That is, $(T_n(x))_{n\in\mathbb{N}}$ is Cauchy in Y. Being that Y is complete, this sequence must also be convergent. Summarizing, we see that $(T_n(x))_{n\in\mathbb{N}}$ is convergent in Y for all $x \in X$ whence we may define

$$T : X \to Y, \quad T(x) := \lim_{n\to\infty} T_n(x).$$

We now claim that T is linear. Indeed, given $u, v \in X$ and $\alpha, \beta \in \mathbb{R}$ we have that

$$T(\alpha u + \beta v) = \lim_{n\to\infty} T_n(\alpha u + \beta v) = \lim_{n\to\infty} \alpha T_n(u) + \lim_{n\to\infty} \beta T_n(v)$$
$$= \alpha T(u) + \beta T(v).$$

Moreover, T is itself a bounded linear operator $X \to Y$. Indeed, since (T_n) is Cauchy in $\mathcal{L}(X,Y)$ with respect to the norm $\|\cdot\|_{X,Y}$, there must exist $C > 0$ such that $\|T_n\|_{X,Y} \leq C$ for all $n \in \mathbb{N}$. Thus, given $x \in X$ with $\|x\| = 1$, we have

$$\|T_n(x)\|_Y \leq \|T_n\|_{X,Y} \leq C$$

for all $n \in \mathbb{N}$. Since norms are continuous and $T_n(x) \to T(x)$ as $n \to \infty$, we infer that

$$\|T(x)\|_Y = \lim_{n\to\infty} \|T_n(x)\|_Y \leq C.$$

Hence, T is a bounded element and belongs to $\mathcal{L}(X,Y)$. Finally, to see that $T_n \to T$ in $\mathcal{L}(X,Y)$, let $x \in X$ be a unit vector. Using that (T_n) is Cauchy, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$ one has

$$\|T_n(x) - T_m(x)\|_Y \leq \|T_n - T_m\|_{X,Y} < \frac{\varepsilon}{2}.$$

Because $T_m(x) \to T(x)$ as $m \to \infty$ and norms are continuous, we may take the limit on both sides of the above as $m \to \infty$, thereby yielding the inequality

$$\|T_n(x) - T(x)\|_Y \leq \frac{\varepsilon}{2}$$

for all $n \geq N$. Finally, taking the supremum over all unit vectors $\|x\| = 1$, it follows that $\|T_n - T\|_{X,Y} \leq \varepsilon/2 < \varepsilon$ for all $n \geq N$ which shows that $T_n \to T$ in $\mathcal{L}(X,Y)$. □

Let (X, d_X) and (Y, d_Y) be metric spaces and $A \subseteq X$. In question 4 of assignment 4, you are asked to explore the methodology of extending a uniformly continuous function $f : A \to X$ to a new uniformly continuous function $\overline{f} : \overline{A} \to Y$ such that $\overline{f} = f$ on A. As it turns out, a very similar procedure allows one to extend linear operators in certain situations. Again, by way of providing additional practice relating to arguments you have previously encountered on assignments, we give this result below.

Theorem 8.9. *Let $(X, \|\cdot\|_X)$ be a normed vector space and $(Y, \|\cdot\|_Y)$ a Banach space. Suppose that D is a dense subspace of X and let $T : D \to Y$ be a bounded linear operator. Then, T has a unique bounded linear extension $T' : X \to Y$ such that $T'(x) = T(x)$ for all $x \in D$.*

Remark 8.6. Since linearity requires that T be defined on a subspace of X (and not merely a subset of X), this theorem is completely useless if X is finite dimensional. Indeed, this is because any dense subspace of a finite dimensional normed space X is automatically the entire space itself, which means that T is defined on the entire space by default. Indeed, let X be a finite dimensional vector space and $D \subseteq X$ a dense subspace. Being dense in X, we have that $\overline{D} = X$. However, since D is a subspace of X, it follows that D is finite dimensional in its own right. By your assignment 4, this means that D is closed in X whence $D = \overline{D} = X$. Ergo, D is indeed the entire space X.

Proof. Uniqueness is automatic since you have shown that any two continuous functions agreeing on a dense subset of a metric space must agree everywhere (see your midterm exam!). Thus, we need only construct a single extension T' satisfying the requirements of the theorem.

To this end, let $x \in X$ be given. Since D is dense in X, we may select a sequence (d_n) from D such that $d_n \to d$ in X. Because T is a bounded linear operator $D \to X$, there exists $C > 0$ such that

$$\|T(z)\|_Y \leq C \|z\|_X, \quad \forall z \in D.$$

Consequently, for all $n, m \in \mathbb{N}$,

$$\|T(d_n) - T(d_m)\|_Y = \|T(d_n - d_m)\|_Y \leq C \|d_n - d_m\|_X$$

whence it follows that $(T(d_n))_{n\in\mathbb{N}}$ is Cauchy in Y since (d_n) is convergent and hence Cauchy in X. Since Y is complete, this means that $(T(d_n))_{n\in\mathbb{N}}$ is convergent in Y. Moreover, if (d_n') is any other sequence in D converging to x, as $n \to \infty$, we would have

$$\left\|T(d_n) - T(d_n')\right\|_Y = \left\|T(d_n - d_n')\right\|_Y \leq C \left\|d_n - d_n'\right\|_X \leq C \left(\|d_n - x\|_X + \left\|x - d_n'\right\|_X\right)$$

for all $n \in \mathbb{N}$ whence

$$\lim_{n\to\infty} T(d_n) = \lim_{n\to\infty} T(d_n'),$$

since both sequences converge by the previous argument. Thus, we may define

$$T'(x) := \lim_{n\to\infty} T(d_n)$$

where this limit is well defined and independent of choice of sequence (d_n) from D. Now, we can easily see that $T' : X \to Y$ is linear since, given any $u, v \in X$ and $\alpha, \beta \in \mathbb{R}$, we may choose sequences $(u_n), (v_n)$ in D converging to u and v, respectively, as $n \to \infty$. By definition of T', because $\alpha u_n + \beta v_n \in D$ converges to $\alpha u + \beta v$,

$$\begin{aligned} T'(\alpha u + \beta v) &= \lim_{n\to\infty} T(\alpha u_n + \beta v_n) \\ &= \alpha \lim_{n\to\infty} T(u_n) + \beta \lim_{n\to\infty} T(v_n) \\ &= \alpha T'(u) + \beta T'(v), \end{aligned}$$

where we have used the linearity of T on D and the definitions of $T'(u)$ and $T'(v)$. Consequently, T' is indeed a linear operator $X \to Y$. To see that T' is bounded, fix again $x \in X$ and let (d_n) be a sequence in D converging to x as $n \to \infty$. Then,

$$\|T'(x)\|_Y = \lim_{n\to\infty} \|T(d_n)\|_Y \leq \lim_{n\to\infty} C \|d_n\|_X = C \|x\|_X.$$

Finally, T' is an extension of T since for any $x \in D$ the constant sequence $d_n := x$ is a sequence in D converging to x, which forces

$$T'(x) = \lim_{n\to\infty} T(x) = T(x).$$

□

9 Ninth Tutorial – Infinite Series

The next two tutorials will be entirely devoted to series. In the subsequent exercises and propositions, we will focus on concrete examples and prove new tests in a similar way the tests you are asked to establish on assignment 5. In general, we shall establish a test and demonstrate how it may apply by way of certain examples. You are not expected to know the statements of these new tests nor must you learn their proofs. They are simply here for the purposes of demonstration and practice. However, I do believe that some of these tests help to shed light on what may be done when other tests fail (e.g. when the ratio test is inconclusive).

Example 9.1. Suppose that (a_n) is a sequence of non-negative real numbers such that at least one of the a_n is non-zero. Given $n \in \mathbb{N}$, let

$$b_n := \frac{a_1 + \cdots + a_n}{n}$$

denote the average of the first n-terms. Then, the series $\sum_{n=1}^{\infty} b_n$ always diverges.

Proof. By hypothesis, there exists a first index $K \in \mathbb{N}$ such that $a_K \neq 0$. Then, for all $n \geq K$, it follows that

$$b_n = \frac{a_1 + \cdots + a_n}{n} \geq \frac{a_K}{n}$$

whence, for all $N \geq K$,

$$\sum_{n=1}^{N} b_n = \sum_{n=K}^{N} b_n \geq \sum_{n=K}^{N} \frac{a_K}{n} = a_K \sum_{n=K}^{N} \frac{1}{n},$$

where the right hand side is unbounded since the partial sums of the harmonic series diverge to infinity. Consequently, the partial sums of the series $\sum_{n=1}^{\infty} b_n$ are unbounded and therefore the series in question may not converge. □

We now present a lemma involving the natural logarithm, which will be integral (pardon the pun) to the example that follows. Of course, part of the lemma is well known to us from our construction of e in Analysis 1. However, we reiterate a portion of the argument for the sake of completeness. Feel free to skip the reading of this proof, if you wish.

Lemma 9.1. *For all $n \in \mathbb{N}$, there holds*

$$\frac{1}{n+1} < \ln\left(1 + \frac{1}{n}\right) < \frac{1}{n}. \tag{9.1}$$

Proof. Define $x_n := \left(1 + \frac{1}{n}\right)^n$ and $y_n := \left(1 + \frac{1}{n}\right)^{n+1}$, for any $n \in \mathbb{N}$. Then, we know from Analysis 1 that the sequence (x_n) is strictly increasing and converges to e as $n \to \infty$. Especially, $x_n < e$ for

al $n \in \mathbb{N}$. Observe also that (y_n) converges to e by the limit laws since

$$\begin{aligned}\lim y_n = \lim\left[\left(1+\frac{1}{n}\right)^n\left(1+\frac{1}{n}\right)\right] &= \lim\left(1+\frac{1}{n}\right)^n \lim\left(1+\frac{1}{n}\right)\\ &= e\cdot 1\\ &= e.\end{aligned}$$

Furthermore, (y_n) is strictly decreasing. To see this, let $n \in \mathbb{N}$ and consider y_n/y_{n+1}:

$$\begin{aligned}\frac{y_n}{y_{n+1}} = \frac{\left(1+\frac{1}{n}\right)^{n+1}}{\left(1+\frac{1}{n+1}\right)^{n+2}} &= \left(1+\frac{1}{n+1}\right)^{-1}\frac{\left(1+\frac{1}{n}\right)^{n+1}}{\left(1+\frac{1}{n+1}\right)^{n+1}}\\ &= \left(1+\frac{1}{n+1}\right)^{-1}\left(\frac{1+\frac{1}{n}}{1+\frac{1}{n+1}}\right)^{n+1}\\ &= \left(1+\frac{1}{n+1}\right)^{-1}\left(\frac{(n+1)^2}{n(n+2)}\right)^{n+1}\\ &= \left(1+\frac{1}{n+1}\right)^{-1}\left(\frac{n^2+2n+1}{n^2+2n}\right)^{n+1}\\ &= \left(1+\frac{1}{n+1}\right)^{-1}\left(1+\frac{1}{n^2+2n}\right)^{n+1}.\end{aligned}$$

Therefore, by Bernoulli's inequality, we find that

$$\begin{aligned}\frac{y_n}{y_{n+1}} \geq \left(1+\frac{1}{n+1}\right)^{-1}\left(1+\frac{n+1}{n^2+2n}\right) &> \left(1+\frac{1}{n+1}\right)^{-1}\left(1+\frac{1}{n+1}\right)\\ &= 1.\end{aligned}$$

where we have used that $(n+1)^2 > n^2+2n$. Summarizing, it follows that y_n is a strictly decreasing sequence converging to e as $n \to \infty$. Thus, $y_n > e$ for all $n \in \mathbb{N}$. Combining this with our previous statements, we infer that

$$x_n = \left(1+\frac{1}{n}\right)^n < e < \left(1+\frac{1}{n}\right)^{n+1} = y_n, \quad \forall n \in \mathbb{N}.$$

Taking logarithms (and using that the logarithm is strictly increasing) of all sides, we obtain

$$n\ln\left(1+\frac{1}{n}\right) < 1 < (n+1)\ln\left(1+\frac{1}{n}\right)$$

which is equivalent to the assertion of the lemma. □

Equipped with the lemma, we are now prepared for our intended example.

Example 9.2. It is known that the harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

diverges. That is, the partial sums $\sum_{n=1}^{N} \frac{1}{n} \to \infty$ as $N \to \infty$. The purpose of this example is to examine the "rate" at which this divergence takes place. To this end, we consider the sequence $(S_N)_{N \in \mathbb{N}}$ defined by putting

$$S_N := \sum_{n=1}^{N} \frac{1}{n} - \ln N,$$

for each $N \in \mathbb{N}$. We will show that (S_N) converges.

Proof. We claim that (S_N) is a decreasing sequence that is bounded below by 0. Thus, it will follow from the Monotone Convergence Theorem that (S_N) converges to a real number. First, observe that $S_N \geq 0$ for all $N \in \mathbb{N}$. Indeed, by the last lemma, we have

$$\begin{aligned} S_N = \sum_{n=1}^{N} \frac{1}{n} - \ln N &> \sum_{n=1}^{N} \ln\left(1 + \frac{1}{n}\right) - \ln N \\ &= \sum_{n=1}^{N} \ln\left(\frac{n+1}{n}\right) - \ln N \\ &= \sum_{n=1}^{N} [\ln(n+1) - \ln(n)] - \ln N \\ &= \ln(N+1) - \ln(1) - \ln(N) \\ &= \ln(N+1) - \ln(N) \\ &> 0. \end{aligned}$$

Thus, (S_N) is a sequence of positive real numbers. To see that (S_N) is decreasing, we argue as follows:

$$\begin{aligned} S_{N+1} - S_N &= \sum_{n=1}^{N+1} \frac{1}{n} - \ln(N+1) - \sum_{n=1}^{N} \frac{1}{n} + \ln(N) \\ &= \frac{1}{N+1} - [\ln(N+1) - \ln(N)] \\ &= \frac{1}{N+1} - \ln\left(1 + \frac{1}{N}\right) \\ &< 0. \end{aligned}$$

Thus, by our previous remarks, (S_N) is indeed convergent. □

Remark 9.1. Let (S_N) be the sequence defined in the previous example and put $\gamma := \lim_{N\to\infty} S_N$. This number γ is called the *Euler–Mascheroni* constant. This quantity appears in analytic number theory as it is related to the Riemann zeta function. It also appears in certain asymptotic expansions of the Γ-function and is related to the Laplace transform. To this day, it is unknown whether γ is rational or irrational (although it appears most number theorists believe it to be irrational). Moreover, it is unknown whether γ is algebraic or transcendental.[14]

9.1 The Limit Comparison Test

As an exercise, let us derive the limit comparison test, which serves as a "limit version" of the familiar comparison test. It is strictly weaker that the comparison test, but can sometimes be more elegant to apply.

Proposition 9.2 (Limit Comparison Test)**.** *Let (a_n) be a non-negative sequence and (b_n) a strictly positive sequence such that*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L.$$

Then,

1. *If $0 < L < \infty$ then $\sum a_n$ converges if and only if $\sum b_n$ converges.*
2. *If $L = 0$ then $\sum a_n$ converges whenever $\sum b_n$ converges.*
3. *If $L = \infty$, then $\sum a_n$ diverges whenever $\sum b_n$ diverges.*

Proof.

1. Choose $\varepsilon > 0$ such that $L - \varepsilon > 0$ and observe that there exists $N \in \mathbb{N}$ such that

$$\left| \frac{a_n}{b_n} - L \right| < \varepsilon, \quad \forall n \geq N.$$

   Equivalently, one has

$$L - \varepsilon < \frac{a_n}{b_n} < L + \varepsilon, \quad \forall n \geq N$$

   whence it follows that

$$(L - \varepsilon)b_n < a_n < (L + \varepsilon)b_n, \quad \forall n \geq N.$$

   It therefore follows from the comparison test that $\sum a_n$ converges if and only if $\sum b_n$ converges.

[14] A real number α is called algebraic if there exists a polynomial P, with integer coefficients, such that $P(\alpha) = 0$. A number $\alpha \in \mathbb{R}$ is called transcendental if no such polynomial exists.

2. In the event that $L = 0$, for $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that

$$0 \leq \frac{a_n}{b_n} < 1, \quad \forall n \geq N.$$

Thus, $a_n < b_n$ for all n large whence by comparison we see that $\sum a_n$ converges whenever $\sum b_n$ does.

3. Since $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$, there exists $N \in \mathbb{N}$ such that $a_n/b_n > 1$ for all $n \geq N$. By comparison, the assertion follows.

□

Example 9.3. Consider the series

$$1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} + \frac{1}{8} - \frac{1}{9} + \cdots$$

Namely, we consider the harmonic series where each 3rd term is negative. Let (S_N) denote the sequence of partial sums and observe that S_{3N} is given by

$$S_{3N} = \sum_{n=1}^{N} \left(\frac{1}{3n-2} + \frac{1}{3n-1} - \frac{1}{3n} \right).$$

The terms of this sum are

$$\frac{1}{3n-2} + \frac{1}{3n-1} - \frac{1}{3n} = \frac{9n^2 - 2}{3n(3n-2)(3n-1)}.$$

Note that these terms are non-negative for every $n \in \mathbb{N}$. Moreover,

$$\lim \frac{\frac{9n^2-2}{3n(3n-2)(3n-1)}}{\frac{1}{n}} = \lim \frac{9n^3 - 2n}{3n(3n-2)(3n-1)} = \frac{1}{3} > 0.$$

Using the Limit Comparison Test with the divergent harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$, we infer that the series

$$\sum_{n=1}^{\infty} \frac{9n^2 - 2}{3n(3n-2)(3n-1)}$$

is divergent. Thus (S_{3N}) diverges and and it follows that (S_N) diverges as well.

9.2 Raabe's Test

We begin by proving a test for absolute convergence commonly known as Raabe's test, which is similar in nature to Bertrand's test you are asked to prove on your assignment. Although this test is not in itself examinable, it's proof provides decent practice and serves as a useful exercise. Moreover, it can be used to analyze the convergence of the p-series $\sum_{n=1}^{\infty} n^{-p}$ for $p \neq 1$. One should think of Raabe's test as a possible tool to be used when the ratio test is inconclusive, i.e. when $a_{n+1}/a_n \to 1$ as $n \to \infty$.

Theorem 9.3 (Raabe's Test)**.** *Let (x_n) be a sequence of non-zero real numbers.*

(1) Suppose there exists $a > 1$ and $K \in \mathbb{N}$ such that

$$\left|\frac{x_{n+1}}{x_n}\right| \leq 1 - \frac{a}{n}, \quad \forall n \geq K. \tag{9.2}$$

Then, the series $\sum_{n=1}^{\infty} x_n$ converges absolutely.

(2) If there exists $a \leq 1$ and $K \in \mathbb{N}$ such that

$$\left|\frac{x_{n+1}}{x_n}\right| \geq 1 - \frac{a}{n}, \quad \forall n \geq K, \tag{9.3}$$

the series $\sum_{n=1}^{\infty} x_n$ does not converge absolutely.

Remark 9.2. Observe that it is implicit in (9.2) that $K > 1$. In fact, we must have $K > a$.

Proof.

(1) By manipulating (9.2), we find that

$$n\,|x_{n+1}| \leq (n-1)\,|x_n| - (a-1)\,|x_n|\,, \quad \forall n \geq K.$$

This implies that

$$(n-1)\,|x_n| - n\,|x_{n+1}| \geq (a-1)\,|x_n|\,, \quad \forall n \geq K. \tag{9.4}$$

Consequently, given $N \geq K$, we obtain (by summing (9.4) for $n = K, \ldots, N$) that

$$\begin{aligned}(K-1)\,|x_K| - N\,|x_{N+1}| = \sum_{n=K}^{N} \left((n-1)\,|x_n| - n\,|x_{n+1}|\right) &\geq \sum_{n=K}^{N} (a-1)\,|x_n| \\ &= (a-1)\sum_{n=K}^{N} |x_n|\,.\end{aligned}$$

In particular, for all $N \geq K$ we have

$$(a-1)\sum_{n=K}^{N}|x_n| \leq (K-1)\,|x_K|$$

which implies that the sums $\sum_{n=K}^{N}|x_n|$ are bounded for all $N \geq K$. Thus, the partial sums of the series $\sum_{n=1}^{\infty}|x_n|$ are bounded and this series must converge absolutely.

(2) By rearranging (9.3), we have

$$n\,|x_{n+1}| \geq (n-a)\,|x_n| \geq (n-1)\,|x_n|\,, \quad \forall n \geq K,$$

where we have used that $a \leq 1$ in the last step. Consequently, the sequence $(n\,|x_{n+1}|)$ is increasing for $n \geq K$. Especially, letting $c := K\,|x_{K+1}| > 0$, we have that

$$n\,|x_{n+1}| \geq c > 0, \quad \forall n \geq K$$

whence

$$|x_{n+1}| \geq \frac{c}{n}, \quad \forall n \geq K.$$

By comparison with the harmonic series, it follows that $\sum_{n=1}^{\infty}|x_n|$ diverges since

$$\sum_{n=K+1}^{N}|x_n| = \sum_{n=K}^{N-1}|x_{n+1}| \geq \sum_{n=K}^{N-1}\frac{c}{n}$$

for all $N \geq K+1$.

□

9.2.1 The Limit Raabe Test

Sometimes it can be tedious (or difficult) to derive the inequalities required to apply Raabe's test. Thankfully, there is a "limit version" of this test which is easier to use in practice, although the test itself is weaker than the full Raabe's test. In other words, the following is to Raabe's test what the limit comparison test is to the comparison test.

Corollary 9.4 (Limit Raabe's Test)**.** *Let (x_n) be a sequence of non-zero real numbers and assume the limit*

$$a := \lim\left(n\left(1-\frac{|x_{n+1}|}{|x_n|}\right)\right)$$

exists. If $a > 1$, then the series $\sum_{n=1}^{\infty}x_n$ converges absolutely. If $a < 1$, then the series $\sum_{n=1}^{\infty}x_n$ does not converge absolutely.

Proof. First, let us handle the case $a > 1$. Choose $b \in (1, a)$ and observe that there exists $K \in \mathbb{N}$ such that

$$\left| n\left(1 - \frac{|x_{n+1}|}{|x_n|}\right) - a \right| < a - b$$

for all $n \geq K$. Equivalently, for all such n,

$$b - a < n\left(1 - \frac{|x_{n+1}|}{|x_n|}\right) - a < a - b$$

which implies

$$b < n\left(1 - \frac{|x_{n+1}|}{|x_n|}\right), \quad \forall n \geq K.$$

Rearranging, we obtain that

$$\frac{|x_{n+1}|}{|x_n|} < 1 - \frac{b}{n}, \quad \forall n \geq K$$

which satisfies the requirement of Raabe's-Test-(1).

Next we show the given series does not converge absolutely if $a < 1$. Indeed, there exists $K \in \mathbb{N}$ such that

$$\left| n\left(1 - \frac{|x_{n+1}|}{|x_n|}\right) - a \right| < 1 - a$$

for all $n \geq K$. This is equivalent to having

$$a - 1 < n\left(1 - \frac{|x_{n+1}|}{|x_n|}\right) - a < 1 - a, \quad \forall n \geq K.$$

Especially,

$$\frac{|x_{n+1}|}{|x_n|} \geq 1 - \frac{1}{n}, \quad \forall n \geq K$$

from which the claim follows by Raabe's-Test-(2). □

9.2.2 *Applications of Raabe's Test*

Example 9.4. Discuss the convergence of $\displaystyle\sum_{n=1}^{\infty} \frac{(2n)!}{4^n (n!)^2}$.

Solution. Let us begin by computing

$$\begin{aligned}
\frac{|x_{n+1}|}{|x_n|} &= \frac{(2n+2)!}{4^{n+1}((n+1)!)^2} \cdot \frac{4^n (n!)^2}{(2n)!} \\
&= \frac{(2n+2)! 4^n (n!)^2}{(2n)! 4^{n+1}((n+1)!)^2} \\
&= \frac{(2n+1)(2n+2)}{4(n+1)^2}.
\end{aligned}$$

Clearly, this converges to 1 as $n \to \infty$ and so the ratio-test is inconclusive. Furthermore, I don't see an obvious approach via comparison. As a result, let us see if the Limit Raabe Test applies. For each $n \in \mathbb{N}$, we calculate

$$\begin{aligned} n\left(1 - \frac{|x_{n+1}|}{|x_n|}\right) &= n\left(1 - \frac{(2n+1)(2n+2)}{4(n+1)^2}\right) \\ &= n\left(\frac{4(n+1)^2 - (2n+1)(2n+2)}{4(n+1)^2}\right) \\ &= n\left(\frac{4n^2 + 8n + 4 - (4n^2 + 6n + 2)}{4(n+1)^2}\right) \\ &= n\left(\frac{2n+2}{4(n+1)^2}\right) \\ &= \frac{n}{2(n+1)}. \end{aligned}$$

This converges to $\frac{1}{2}$ as $n \to \infty$, and so Corollary 9.4 implies that the series $\sum_{n=1}^{\infty} x_n$ does not converge absolutely. Since $x_n > 0$ for all $n \in \mathbb{N}$, this means that $\sum_{n=1}^{\infty} x_n$ is divergent. □

9.3 Analytic Functions

Definition 9.1. Let $U \subseteq \mathbb{R}$ be an open set. A function $f : U \to \mathbb{R}$ is called (real) analytic at a point $x_0 \in U$ if f can be expressed as a power series in a ball centered at x_0. More precisely, f is analytic at x_0 if there exists $R > 0$ and a sequence $(a_n)_{n \geq 0}$ of real numbers such that

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

for all $x \in B(x_0, R) \subseteq U$, with the series converging absolutely on $B(x_0, R)$. We say that f is analytic in U if it is analytic at each point in U. The set consisting of all (real) analytic functions $U \to \mathbb{R}$, denoted $C^{\omega}(U)$, forms a vector space over $\mathbb{R}$ with the usual notions of addition and scalar multiplication.

Remark 9.3. Having seen in the lectures that power series are infinitely differentiable within their radius of convergence, it is automatic from the definition that an analytic function is infinitely differentiable.[15] Thus, denoting by $C^{\infty}(U)$ the space of all smooth (i.e. infinitely differentiable) functions $U \to \mathbb{R}$, we have that $C^{\omega}(U) \subset C^{\infty}(U)$. This inclusion is indeed strict, for there exist

[15] Indeed, if f is analytic in U, then for each $x_0 \in U$ we can find $R > 0$ such that $f(x)$ is equal to a power series $\sum_{n=0}^{\infty} a_x^n$ in $B(x_0, R)$. Since the power series, which is equal to f, is infinitely differentiable in its radius of convergence, we see that f is also infinitely differentiable there. As this holds for each $x_0 \in U$, we infer that f is smooth in U.

smooth functions that are *not* (real) analytic. For example, the function

$$f : \mathbb{R} \to \mathbb{R}, \quad x \mapsto \begin{cases} e^{-1/x} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0 \end{cases}$$

can be shown to be infinitely differentiable on $\mathbb{R}$. In fact, by induction, one can show that $f^{(n)}(0) = 0$ for all $n \in \mathbb{N}$. However, f cannot be written as a power series in any interval about the point 0.

Remark 9.4. When it comes to complex functions (i.e. functions $\mathbb{C} \to \mathbb{C}$) the story is strikingly different. As it turns out, a function $U \subseteq \mathbb{C} \to \mathbb{C}$ is (complex) analytic if and only if it is (complex) differentiable in U. Especially, as soon as a function $f : U \subseteq \mathbb{C} \to \mathbb{C}$ has (complex) derivatives of the first order, it is automatically infinitely differentiable *and* may be expressed as a power series locally. It is precisely because of this stark contrast between the complex case and the real case that mathematicians have adopted distinguishing nomenclature when referring to complex differentiable functions (indeed, such functions are called *holomorphic*). Holomorphic functions are the main object of study in Math 249/466.

Example 9.5. Determine the radius of convergence of the series $\sum_{n=0}^{\infty} a_n x^n$, assuming there exist constants $p, q \in \mathbb{R}$ such that $0 < p \leq |a_n| \leq q$ for all $n \in \mathbb{N}$.

Solution. The radius of convergence R for the series $\sum_{n=0}^{\infty} a_n x^n$ is given by

$$R = \frac{1}{\limsup\left(|a_n|^{1/n}\right)}.$$

Given $n \in \mathbb{N}$, observe that $p^{1/n} \leq |a_n|^{1/n} \leq q^{1/n}$ by the monotonicity of $x \mapsto \sqrt[n]{x}$. Hence,

$$\limsup p^{1/n} \leq \limsup |a_n|^{1/n} \leq \limsup q^{1/n}.$$

As seen in analysis 1, we have $\lim p^{1/n} = \lim q^{1/n} = 1$. Thus, the above becomes

$$1 = \lim p^{1/n} \leq \limsup |a_n|^{1/n} \leq \lim q^{1/n} = 1$$

which implies that the radius of convergence is $\frac{1}{1} = 1$. □

9.3.1 The Identity Theorem (Time Permitting)

Continuing with our study of analytic functions (that is, those functions that can be expressed locally as a power series), our aim in this section is to establish the famous Identity Theorem. Loosely speaking, this theorem states that if an analytic function $\mathbb{R} \to \mathbb{R}$ vanishes on a sequence of distinct points with an accumulation point, then f vanishes everywhere. As a consequence, we will deduce that two analytic functions $\mathbb{R} \to \mathbb{R}$ that agree on some open interval must agree

everywhere. This is a special property intrinsic to analytic functions, and being infinitely differentiable is not enough to make such an assertion. For example, the function described in Remark 9.3 is infinitely differentiable on $\mathbb{R}$ and equal to 0 on the open interval $(-\infty, 0)$ and yet does not vanish on all of $\mathbb{R}$.

We begin with the following lemma, which was proven in Analysis 1. We reiterate the argument for the proof of completeness, but you may skip it if you recall the result.

Lemma 9.5. *Let X be a non-empty proper subset of $\mathbb{R}$ (i.e. $X \subseteq \mathbb{R}$ with $X \neq \mathbb{R}$ and X non-empty). Then, ∂X is non-empty.*

Proof. Let $x \in X$ and $y \in X^{\mathsf{c}}$; not that X^{c} is non-empty since $X \neq \mathbb{R}$. Then, $x \neq y$ and so either $x < y$ or $y < x$. We only treat the case $x < y$ explicitly, as a similar argument applies to $y < x$ (we comment on this at the end). Now, if $x < y$, we consider the set

$$S := [x, y] \cap X$$

which is non-empty as it contains x. Moreover, $S \subseteq [x, y]$ implies that S is bounded. Therefore, $\sigma := \sup S$ exists. Note that $x \in S$ implies that $x \leq \sigma$. We claim that $\sigma \in \partial X$, which would imply $\partial X \neq \varnothing$. To this end, we let $\varepsilon > 0$ be arbitrary and venture forth to prove

$$B(\sigma, \varepsilon) \cap X \neq \varnothing \quad \text{and} \quad B(\sigma, \varepsilon) \cap X^{\mathsf{c}} \neq \varnothing.$$

Because σ is the least upper bound of S, there must exist $s_\varepsilon \in S$ such that

$$\sigma - \varepsilon < s_\varepsilon \leq \sigma < \sigma + \varepsilon$$

whence $s_\varepsilon \in B(\sigma, \varepsilon)$. However, $s_\varepsilon \in S$ implies that $s_\varepsilon \in X$ and so we obtain that $s_\varepsilon \in B(\sigma, \varepsilon) \cap X$. Hence, $B(\sigma, \varepsilon) \cap X \neq \varnothing$.

It remains to show that $B(\sigma, \varepsilon) \cap X^{\mathsf{c}} \neq \varnothing$. To show this, we first observe that y is an upper bound of S and hence $\sigma \leq y$. If $\sigma = y$, then $y \in B(\sigma, \varepsilon) \cap X^{\mathsf{c}}$ and we are done. Otherwise, $\sigma < y$ and we may choose $\delta > 0$ so small that $\sigma + \delta < y$. Without loss of generality, we may also assume that $\delta < \varepsilon$. Observe also that

$$x \leq \sigma < \sigma + \delta < y \implies \sigma + \delta \in [x, y].$$

Then, $\sigma + \delta \notin X$, because this would imply that $\sigma + \delta \in [x, y] \cap X = S$ contradicting $\sigma = \sup S$. As $\sigma + \delta \notin X$, and

$$\sigma - \varepsilon < \sigma + \delta < \sigma + \varepsilon,$$

it follows that $\sigma + \delta \in B(\sigma, \varepsilon) \cap X^{\mathsf{c}}$. Thus, we have proven that for each $\varepsilon > 0$, the open ball $B(\sigma, \varepsilon)$ intersects both X and X^{c}, whence $\partial X \neq \varnothing$.

In the case $y < x$, we argue similarly, but using $\iota := \inf([y, x] \cap X)$ instead of σ. □

Corollary 9.6. *The only clopen subsets of $\mathbb{R}$ are itself and the empty-set.*[16]

[16] In other words, $\mathbb{R}$ is connected.

Proof. Suppose that $X \subset \mathbb{R}$ is non-empty and both closed and open. If $X \neq \mathbb{R}$, then ∂X is non-empty (see notes or previous lemma). However, being that X is clopen, we have $X = \mathring{X}$ and $X = \overline{X}$ whence $\partial X = \overline{X} \setminus \mathring{X} = X \setminus X = \varnothing$. This contradictions shows that $X = \mathbb{R}$, as was asserted. □

We now state and prove the Identity Theorem.

Theorem 9.7 (Identity Theorem)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be analytic and suppose there exists a sequence (x_k) of points in $\mathbb{R}$ converging to some $x_0 \in \mathbb{R}$ such that $x_k \neq x_0$ and $f(x_k) = 0$ for all $k \in \mathbb{N}$. Then, $f(x) = 0$ for all $x \in \mathbb{R}$.*

Proof. First, we show that f vanishes in some open ball containing x_0. Being that f is analytic on $\mathbb{R}$, there exists $R > 0$ and real numbers $(a_n)_{n\geq 0}$ such that

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

for all $x \in \mathbb{R}$ satisfying $|x - x_0| < R$. That is, f has a power-series expansion in a neighbourhood of x_0. We claim that $f \equiv 0$ on $B(x_0, R)$. If not, then at least one of the coefficients a_n above must be non-zero. In particular, there exists a minimal $N \in \mathbb{N}_0$ such that $a_N \neq 0$. Then, for all $x \in B(x_0, R)$, we have that

$$\begin{aligned} f(x) = \sum_{n=N}^{\infty} a_n (x - x_0)^n &= (x - x_0)^N \sum_{n=N}^{\infty} a_n (x - x_0)^{n-N} \\ &= a_N (x - x_0)^N \sum_{n=N}^{\infty} \frac{a_n}{a_N} (x - x_0)^{n-N} \\ &= a_N (x - x_0)^N \left(1 + \sum_{n=N+1}^{\infty} \frac{a_n}{a_N} (x - x_0)^{n-N} \right) \\ &= a_N (x - x_0)^N \left(1 + g(x) \right), \end{aligned}$$

where we have defined $g(x) := \sum_{n=N+1}^{\infty} \frac{a_n}{a_N} (x - x_0)^{n-N}$ for $x \in B(x_0, R)$. Notice that g is well defined since it is defined by a series with the same radius of convergence as that representing f. In particular, g is continuous on $B(x_0, R)$ and $g(x_0) = 0$. Since $x_k \to x_0$, we may find $K \in \mathbb{N}$ such that for all $k \geq K$ there holds

$$|g(x_k)| < 1, \quad \forall k \geq K$$

whence $-1 < g(x_k) < 1$ and $1 + g(x_k) > 0$ for all $k \geq K$. On the other hand,

$$0 = f(x_k) = a_N (x_k - x_0)^N \left(1 + g(x_k) \right)$$

which forces $1 + g(x_k) = 0$ since $a_N \neq 0$ and $x_k \neq x_0$. This contradiction shows that all the coefficients $(a_n)_{n\geq 0}$ in the power series expansion of f around x_0 must be 0, i.e. $f(x) = 0$ for all $x \in B(x_0, R)$.

We now claim that $f \equiv 0$ on all of $\mathbb{R}$. To this end, let $Z \subseteq \mathbb{R}$ denote the *interior* of the set of points where f equals 0. Notice that Z is non-empty since it contains the point x_0 by virtue of the argument above. In addition, Z is open by definition. Finally, Z is closed by the sequential criterion for closed sets. Indeed, suppose that (z_k) is a sequence of points in Z converging to some $z_0 \in \mathbb{R}$. If z_0 appears in the terms of the sequence (z_k), we automatically have $z_0 \in Z$ and are done. Otherwise, $z_k \neq z_0$ for all $k \in \mathbb{N}$ and $f(z_k) = 0$ for all $k \in \mathbb{N}$. By the argument used above, it follows that $f = 0$ for all $x \in B(z_0, R)$ for some $R > 0$ whence $z_0 \in Z$. Thus, Z is clopen and non-empty. By the last lemma, we conclude that $Z = \mathbb{R}$. □

Corollary 9.8. *Let $f, g : \mathbb{R} \to \mathbb{R}$ be two real analytic functions. If there exists a non-empty open set $U \subseteq \mathbb{R}$ such that $f = g$ on U, then $f = g$ on $\mathbb{R}$.*

Proof. Choose a sequence of distinct points (x_k) in U converging to some $x_0 \in U$. Then, apply the Identity Theorem to $h(x) = f(x) - g(x)$. □

Topologically, the zero set of an analytic function is quite interesting. In fact, you are likely already quite familiar with these properties, as they are properties satisfied by the most familiar functions we have power series for (e.g. e^x, $\sin x$, $\cos x$, etc.). Looking at e^x for example (a function we know is analytic since it's power series converges everywhere), we note that e^x has no zeroes on $\mathbb{R}$. As for $\sin x$, we know it has a power series representation at each point (it's Taylor series) and is analytic. While $\sin x$ does have zeroes (at $k\pi$ for each $k \in \mathbb{Z}$), we observe that these zeroes are countable and separated (in the sense that for each zero z of $\sin x$, we can find a ball $(z - \varepsilon, z + \varepsilon)$ on which f has no other zeroes). The same can be said for $\cos x$. It turns out that these are phenomena which are realized by every non-trivial analytic function.

Example 9.6. Let $f : \mathbb{R} \to \mathbb{R}$ be analytic and non-trivial (f is not the constant zero function). Then, for each zero $z_0 \in \mathbb{R}$ of f, there exists $\varepsilon > 0$ such that z_0 is the only zero of f on $(z_0 - \varepsilon, z_0 + \varepsilon)$.

Proof. If f has no zeroes at all, the statement is vacuously true. Otherwise, let z be a zero of f. Suppose for a contradiction that the statement is false, so for each $\varepsilon > 0$, the interval $(z - \varepsilon, z + \varepsilon)$ contains a zero of f different from z. In particular, for each $k \in \mathbb{N}$, the interval $(z - \frac{1}{k}, z + \frac{1}{k})$ must contain a zero z_k of f, with $z_k \neq z$. However, since $z_k \in (z - \frac{1}{k}, z + \frac{1}{k})$, we have

$$|z_k - z_0| < \frac{1}{k}$$

for all $k \in \mathbb{N}$ which implies $\lim z_k = z_0$. But, because $f(z_k) = 0$ for each $k \in \mathbb{N}$, this would yield (by the Identity Theorem) that $f \equiv 0$ on $\mathbb{R}$. This is a contradiction since f was assumed to be non-trivial. □

Example 9.7. Let $f : \mathbb{R} \to \mathbb{R}$ be analytic and non-trivial. Then, on each compact set $K \subset \mathbb{R}$, the function f has at most finitely many zeroes. Deduce from this that f has countably many zeroes on $\mathbb{R}$.

Proof. First we show f has only finitely many zeroes on any compact set $K \subset \mathbb{R}^n$. Notice that f has no zeroes on K, the statement is still satisfied so we may assume that f has at least one zero on K. Let now $K_0 = f^{-1}(\{0\}) \cap K$ be the set of zeroes f achieves in the set K, i.e.

$$K_0 := f^{-1}(\{0\}) \cap K = \{x \in K : f(x) = 0\}.$$

Note that K_0 is a closed subset of the compact set K, since $f^{-1}(\{0\})$ is closed by continuity and K is compact (and hence closed). Thus, K_0 is itself compact. Moreover, for each $z \in K_0$, the previous example shows there exists $\varepsilon_z > 0$ such that $B_z := (z-\varepsilon_z, z+\varepsilon_z)$ contains no zeroes of f other than z. Now, each B_z is open, and $K_0 \subseteq \bigcup_{z\in K_0} B_z$, which means that $\{B_z\}_{z\in K_0}$ is an open cover of K_0. Since K_0 is compact, we may extract a finite subcover, i.e. there exist finitely many $z_1, \ldots, z_k \in K_0$ such that

$$K_0 \subseteq B_{z_1} \cup \cdots \cup B_{z_k}.$$

Since K_0 consists of precisely those zeroes of f in K, and using the fact that each B_{z_j} above contains only a single zero of f (the point z_j itself), it follows that K_0 is finite. That is, f has only finite many zeroes in K.

Next, we argue that f has countably many zeroes in $\mathbb{R}$, which amounts to proving the set $f^{-1}(\{0\})$ is countable. For each $k \in \mathbb{N}$, let $\Lambda_k := [-n, n]$ and notice that Λ_k is compact by Heine-Borel. Moreover,

$$\mathbb{R} = \bigcup_{k\in\mathbb{N}} \Lambda_k$$

whence

$$\begin{aligned} f^{-1}(\{0\}) = f^{-1}(\{0\}) \cap \bigcup_{k\in\mathbb{N}} \Lambda_k &= \bigcup_{k\in\mathbb{N}} f^{-1}(\{0\}) \cap \Lambda_k \\ &= \bigcup_{k\in\mathbb{N}} \{x \in \Lambda_k : f(x) = 0\}. \end{aligned}$$

By the previous part above, since each Λ_k is compact, f can only achieve finitely many zeroes on Λ_k. That is, the set $\{x \in \Lambda_k : f(x) = 0\}$ is finite. Then, $f^{-1}(\{0\})$ is the countable union of finite sets, and is itself countable. This completes the proof. □

10 Tenth Tutorial

This tutorial is mostly devoted to introductory problems involving the Riemann-Stieltjes integral. However, we first provide an exercise utilizing Taylor's Theorem. We also point out that several of the examples provided within these next two tutorials are technically versions of problems from your sixth assignment. However, given that the final exam is fast approaching and this last assignment is ungraded, it may be instructive and/or helpful to cover some of these problems together.

10.1 Applications of Taylor's Theorem

We now discuss how convexity is related to the sign of the second derivative, provided is exists. First, let us reiterate the definition of convexity.

Definition 10.1 (Convex Function)**.** Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ a function. We say that f is *convex* on I if one has

$$f((1-t)x_1 + tx_2) \leq (1-t)f(x_1) + tf(x_2)$$

for all $x_1, x_2 \in I$ and all $t \in [0,1]$.

The next result is an extension of Problem 1 from assignment 6. It's statement is likely familiar to you from your first calculus course (think of the second derivative test).

Theorem 10.1 (Convexity Characterization of Twice Differentiable Functions)**.** *Let $I \subseteq \mathbb{R}^n$ be an open interval and $f : I \to \mathbb{R}$ twice differentiable on I. Then, f is convex on I if and only if $f''(x) \geq 0$ for all $x \in I$.*

Proof. First, suppose that f is convex on I. Thus, for all x_1, x_2 belonging to I and all $t \in [0,1]$, there holds that

$$f((1-t)x_1 + tx_2) \leq (1-t)f(x_1) + tf(x_2). \tag{10.1}$$

Now, observe that by l'Hôpital's rule,

$$f''(a) = \lim_{h \to 0} \frac{f(a+h) - 2f(a) + f(a-h)}{h^2}, \tag{10.2}$$

for all $a \in I$. Indeed, by an application of L'Hôpital's rule, we have

$$\begin{aligned}
\lim_{h \to 0} \frac{f(a+h) - 2f(a) + f(a-h)}{h^2} &= \lim_{h \to 0} \frac{f'(a+h) - f'(a-h)}{2h} \\
&= \lim_{h \to 0} \frac{f'(a+h) - f(a) + f(a) - f'(a-h)}{2h} \\
&= \frac{1}{2}\left(\lim_{h \to 0} \frac{f'(a+h) - f'(a)}{h} + \lim_{h \to 0} \frac{f'(a) - f'(a-h)}{h} \right) \\
&= \frac{1}{2}\left(\lim_{h \to 0} \frac{f'(a+h) - f'(a)}{h} + \lim_{h \to 0} \frac{f'(a-h) - f'(a)}{-h} \right) \\
&= \frac{1}{2}\left(\lim_{h \to 0} \frac{f'(a+h) - f'(a)}{h} + \lim_{h \to 0} \frac{f'(a+h) - f'(a)}{h} \right) \\
&= \frac{f''(a) + f''(a)}{2} \\
&= f''(a).
\end{aligned}$$

Let $a \in I$ be given. Being that f satisfies (10.1) on I, it follows that for all h small enough so that $a+h, a-h \in I$,

$$f(a) = f\left(\frac{a+h}{2} + \frac{a-h}{2}\right) \leq \frac{f(a+h)}{2} + \frac{f(a-h)}{2}$$

where we have applied (10.1) with $t = \frac{1}{2}$. Equivalently,

$$f(a+h) - 2f(a) + f(a-h) \geq 0$$

for all h sufficiently small. Since $h^2 > 0$ for all $h \neq 0$, it follows that the quantities appearing in the limit (10.2) are always non-negative whence

$$f''(a) = \lim_{h \to 0} \frac{f(a+h) - 2f(a) + f(a-h)}{h^2} \geq 0.$$

As $a \in I$ was arbitrary, this proves the forward implication of our assertion.

Conversely, we assume that $f''(x) \geq 0$ for all $x \in I$ and show that f is convex on I. To this end, we fix $x_1, x_2 \in I$. Because (10.1) holds trivially if $x_1 = x_2$, or if $t \in \{0, 1\}$, we may assume without loss of generality that $x_1 \neq x_2$ and that $t \in (0, 1)$. Next, consider the point

$$x_0 := (1-t)x_1 + tx_2$$

which lies between x_1 and x_2. By Taylor's theorem applied to the function f with base point x_0, we see that there exists some $\xi_1 \in I$ between x_1 and x_0 such that

$$f(x_1) = f(x_0) + f'(x_0)(x_1 - x_0) + \frac{f''(\xi_1)}{2}(x_1 - x_0)^2.$$

Analogously, applying Taylor's theorem again, there exists $\xi_2 \in I$ between x_0 and x_2 such that

$$f(x_2) = f(x_0) + f'(x_0)(x_2 - x_0) + \frac{f''(\xi_2)}{2}(x_2 - x_0)^2.$$

Consequently,

$$\begin{aligned}
(1-t)f(x_1) + tf(x_2) &= f(x_0) + f'(x_0)\left((1-t)x_1 + tx_2 - x_0\right) \\
&\quad + (1-t)\frac{f''(\xi_1)}{2}(x_1 - x_0)^2 + t\frac{f''(\xi_2)}{2}(x_2 - x_0)^2 \\
&\geq f(x_0) + f'(x_0)\left((1-t)x_1 + tx_2 - x_0\right) \\
&= f(x_0) + f'(x_0)\left(x_0 - x_0\right) \\
&= f(x_0) \\
&= f((1-t)x_1 + tx_2).
\end{aligned}$$

□

10.2 Upper and Lower Sums

Entering into the topic of integration, let us reiterate a global overview of our construction for the Riemann-Stieltjes integral. There is no new information here for the most part, but it is often helpful to review these definitions with the purpose of internalizing them. Note that this is also discussed in greater detail within the lecture notes and course textbooks.

Definition 10.2 (Partition)**.** Let $I = [a, b]$ be a closed and bounded interval. A *partition* $\mathcal{P}$ of the interval I is a non-empty finite subset $\{x_0, \ldots, x_n\}$ of $[a, b]$ such that

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.$$

The *mesh* of the partition $\mathcal{P}$ is defined to be the length of the largest subinterval created by $\mathcal{P}$ and is denoted $\|\mathcal{P}\|$. Despite the suggestive notation, the mesh is *not* a norm.

Definition 10.3. Let $f : [a, b] \to \mathbb{R}$ be a bounded function and let $\mathcal{P} = (x_0, x_1, \ldots, x_n)$ be a partition of the interval I. Let $\alpha : [a, b] \to \mathbb{R}$ be an increasing function (sometimes called an *integrator*). We define the *lower sum* of f, with respect to the partition $\mathcal{P}$ and integrator function α, as follows:

$$L(f; \mathcal{P}, \alpha) := \sum_{i=1}^{n} m_i(\alpha(x_i) - \alpha(x_{i-1})) = \sum_{i=1}^{n} m_i \Delta\alpha(x_i), \tag{10.3}$$

where we put $\Delta\alpha(x_i) := (\alpha(x_i) - \alpha(x_{i-1}))$ and

$$m_i := \inf_{x \in [x_{i-1}, x_i]} f(x).$$

Similarly, one defines the *upper sum* of f (with respect to the partition $\mathcal{P}$ and integrator α) as:

$$U(f; \mathcal{P}) := \sum_{i=1}^{n} M_i \Delta\alpha(x_i), \tag{10.4}$$

where

$$M_i := \sup_{x \in [x_{i-1}, x_i]} f(x).$$

Notice that without the assumption that f is bounded, we cannot guarantee that the the supremum M_i of f on the subinterval $[x_{i-1}, x_i]$ will exist, and similarly for m_i. Consequently, a bounded function is *required* to make sense of our definition of upper and lower sums.

Notation: In the special case $\alpha(x) = x$, we shall write $U(f; \mathcal{P})$ instead of $U(f; P, \alpha)$ (similarly for lower sums).

Remark 10.1. I would like to take a moment to contrast these upper and lower (Darboux) sums to the usual Riemann sums you likely encountered in your standard calculus course stream. A Riemann sum is a finite sum of the form

$$\sum_{i=1}^{n} f(x_i^*)(\alpha(x_i) - \alpha(x_{i-1})) = \sum_{i=1}^{n} f(x_i^*)\Delta\alpha(x_i) = \sum_{i=1}^{n} f(x_i^*)\Delta x_i,$$

where each $x_i^* \in [x_{i-1}, x_i]$ and $\alpha(x) = x$. Notice that here, each Δx_i is scaled (i.e. multiplied) by some value the function f takes on the subinterval $[x_{i-1}, x_i]$. In our case, however, we instead use the supremum and infimum of the chosen function on the given subinterval $[x_{i-1}, x_i]$. This has the advantage that we do not need to worry about all possible values f can take on the subinterval $[x_{i-1}, x_i]$. However, it is important to keep in mind that m_i and M_i may not be achieved by f and thus these upper and lower sums need not be Riemann sums. Conversely, traditional Riemann sums can be defined for functions that are a priori not assumed to be bounded. In fact, in this theory, one can instead deduce that a Riemann integrable function is bounded. However, they are perfectly equivalent.

Example 10.1. Let $\alpha(x) = x$ and consider the function $f : [0, 1] \to \mathbb{R}$,

$$f(x) = \begin{cases} \frac{1}{2} & x = 0 \\ x & 0 < x < 1 \\ \frac{1}{2} & x = 1. \end{cases}$$

With $\mathcal{P} = \{x_0, x_1, x_2\} := \left\{0, \frac{1}{2}, 1\right\}$, we have

$$\begin{aligned} U(f; \mathcal{P}) &= \sum_{i=1}^{2} M_i(x_i - x_{i-1}) \\ &= M_1\left(\frac{1}{2} - 0\right) + M_2\left(1 - \frac{1}{2}\right) \\ &= \frac{1}{2}M_1 + \frac{1}{2}M_2. \end{aligned}$$

Here,

$$\begin{aligned} M_1 = \sup_{x \in [0,1/2]} f(x) &= \sup\left(\{f(x) : x = 0\} \cup \left\{f(x) : 0 < x \leq \frac{1}{2}\right\}\right) \\ &= \sup\left(\left\{\frac{1}{2}\right\} \cup (0, 1/2]\right) \\ &= \sup\left((0, 1/2]\right) \\ &= \frac{1}{2} \end{aligned}$$

and

$$\begin{aligned} M_2 = \sup_{x\in[1/2,1]} f(x) &= \sup\left(\{f(x) : 1/2 \le x < 1\} \cup \{f(x) : x = 1\}\right) \\ &= \sup\left([1/2, 1)\right) \\ &= 1. \end{aligned}$$

It follows that

$$U(f;\mathcal{P}) = \frac{1}{2}M_1 + \frac{1}{2}M_2 = \frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot 1 = \frac{3}{4}.$$

Similarly, we have

$$L(f;\mathcal{P}) = \sum_{i=1}^{2} m_i(x_i - x_{i-1}) = \frac{1}{2}m_1 + \frac{1}{2}m_2.$$

Utilizing our earlier calculations, we see that

$$m_1 = \inf_{x\in[0,1/2]} f(x) = \inf\left((0, 1/2]\right) = 0$$

and

$$m_2 = \inf_{x\in[1/2,1]} f(x) = \inf\left([1/2, 1)\right) = 1/2.$$

We can now compute

$$L(f;\mathcal{P}) = \frac{1}{2}\cdot 0 + \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}.$$

Given two partitions $\mathcal{P}$ and $\mathcal{Q}$ of the same interval, we say that $\mathcal{Q}$ is a *refinement* of $\mathcal{P}$ provided $\mathcal{P} \subseteq \mathcal{Q}$. That is, $\mathcal{Q}$ is called a refinement of $\mathcal{P}$ if the subintervals formed by $\mathcal{Q}$ can be obtained by "breaking apart" the subintervals made formed by $\mathcal{P}$. In particular, $\mathcal{P} \cup \mathcal{Q}$ is a refinement of both $\mathcal{P}$ and $\mathcal{Q}$ called their *common refinement*.

Although it may seem a tedious point, understanding how $L(f;\mathcal{P})$ and $U(f;\mathcal{P})$ behave with respect to refinements is fundamental to conceptualizing the motivation and meaning behind the Riemann-Stieltjes integral. As such, we take the time to reiterate the following result from class:

Lemma 10.2. *Let $f : [a, b] \to \mathbb{R}$ be bounded and let $\mathcal{P}, \mathcal{Q}$ be partitions of $[a, b]$. If $\mathcal{P} \subseteq \mathcal{Q}$, then one has*

(i) $L(f;\mathcal{P}) \le L(f;\mathcal{Q})$;

(ii) $U(f;\mathcal{Q}) \le U(f;\mathcal{P})$.

In short, this lemma communicates two very important pieces of information. First, it asserts that as we refine our partition of $[a, b]$ by increasingly many subintervals or endpoints, the lower sums *increase*. Analogously, as we refine our partitions, the upper sums *decrease*. Furthermore, if the function f in question is positive and $\alpha(x) = x$, one can see that the lower sums $L(f;\mathcal{P})$ are

an attempt and getting a low-estimate for the area under the curve of f, whilst the upper sums are an overestimate of the "true area" beneath the curve of f. This perspective is futher supported by the following corollary, which shows that $L(f;\mathcal{P}) \leq U(f;\mathcal{P})$ for any two partitions $\mathcal{P}, \mathcal{Q}$ of the same interval $[a,b]$.

Corollary 10.3. *Let $f : [a,b] \to \mathbb{R}$ be a bounded function and $\alpha : [a,b] \to \mathbb{R}$ increasing. Then, for any partitions $\mathcal{P}$ and $\mathcal{Q}$ of $[a,b]$, there holds that $L(f;\mathcal{P},\alpha) \leq U(f;\mathcal{Q},\alpha)$.*

Proof. Consider the common refinement $\mathcal{P} \cup \mathcal{Q}$ of both $\mathcal{P}$ and $\mathcal{Q}$; by virtue of the previous lemma, we have that

$$L(f;\mathcal{P},\alpha) \leq L(f;\mathcal{P}\cup\mathcal{Q},\alpha) \leq U(f;\mathcal{P}\cup\mathcal{Q},\alpha) \leq U(f;\mathcal{Q},\alpha).$$

□

Included below is an optional exercise regarding partitions and upper/lower sums. For convenience, I have also included a full solution to this exercise below.

Exercise 10.1. Consider the function $f(x) = x^2+x-1$ defined on the interval $[2,4]$ with $\alpha(x) = x$.

(a) Given the partition $\mathcal{P} = \{2, 5/2, 3, 4\}$, evaluate $L(f;\mathcal{P})$ and $U(f;\mathcal{P})$.

(b) Let $\mathcal{Q}$ be the refinement of $\mathcal{P}$ obtained by adjoining $11/4$. Evaluate $L(f;\mathcal{Q})$ and $U(f;\mathcal{Q})$.

(c) Since $\mathcal{Q}$ is a refinement of $\mathcal{P}$, can we conclude that $\|\mathcal{Q}\| < \|\mathcal{P}\|$?

(d) As a sanity check, verify Corollary 10.3 in this setting. That is, verify that $L(f;\mathcal{P}) \leq L(f;\mathcal{Q})$ and $U(f;\mathcal{P}) \geq U(f;\mathcal{Q})$.

Solution.

(a) Given the partition $\mathcal{P} = \{2, 5/2, 3, 4\}$, we evaluate $L(f;\mathcal{P})$ and $U(f;\mathcal{P})$.

In order to evaluate m_i and M_i for $i = 1, \ldots, 3$, notice that $f(x)$ is differentiable with derivative $f'(x) = 2x + 1$. In particular, f is increasing on it's domain. Thus, on any interval $[a,b] \subseteq [2,4]$ we have

$$\sup_{x\in[a,b]} f(x) = f(b) \quad \text{and} \quad \inf_{x\in[a,b]} f(x) = f(a).$$

With this in mind, we can evaluate the upper and lower sums.

$$\begin{aligned}
U(f;\mathcal{P}) &= \sum_{i=1}^{3} M_i \Delta x_i \\
&= \frac{1}{2} \sup_{x\in[2,5/2]} f(x) + \frac{1}{2} \sup_{x\in[5/2,3]} f(x) + \sup_{x\in[3,4]} f(x) \\
&= \frac{1}{2} f(5/2) + \frac{1}{2} f(3) + f(4) \\
&= \frac{1}{2} \cdot \frac{31}{4} + \frac{1}{2} \cdot 11 + 19 \\
&= \frac{227}{8} = 28.275
\end{aligned}$$

Similarly,

$$\begin{aligned}
L(f;\mathcal{P}) &= \sum_{i=1}^{3} m_i \Delta x_i \\
&= \frac{1}{2} \inf_{x\in[2,5/2]} f(x) + \frac{1}{2} \inf_{x\in[5/2,3]} f(x) + \inf_{x\in[3,4]} f(x) \\
&= \frac{1}{2} f(2) + \frac{1}{2} f(5/2) + f(3) \\
&= \frac{1}{2} \cdot 5 + \frac{1}{2} \cdot \frac{31}{4} + 11 \\
&= \frac{139}{8} = 17.375
\end{aligned}$$

(b) Next, consider the refinement $\mathcal{Q} = \{2, 5/2, 11/4, 3, 4\}$ of $\mathcal{P}$. We can evaluate

$$\begin{aligned}
U(f;\mathcal{Q}) &= \sum_{i=1}^{4} M_i \Delta x_i \\
&= \frac{1}{2} \sup_{x\in[2,5/2]} f(x) + \frac{1}{4} \sup_{x\in[5/2,11/4]} f(x) + + \frac{1}{4} \sup_{x\in[11/4,3]} f(x) + \sup_{x\in[3,4]} f(x) \\
&= \frac{1}{2} f(5/2) + \frac{1}{4} f(11/4) + \frac{1}{4} f(3) + f(4) \\
&= \frac{1}{2} \cdot \frac{31}{4} + \frac{1}{4} \cdot \frac{149}{16} + \frac{1}{4} \cdot 11 + 19 \\
&= \frac{1789}{64} = 27.953125
\end{aligned}$$

Similarly,

$$\begin{aligned} L(f;\mathcal{Q}) &= \sum_{i=1}^{4} m_i \Delta x_i \\ &= \frac{1}{2} \inf_{x\in[2,5/2]} f(x) + \frac{1}{4} \inf_{x\in[5/2,11/4]} f(x) + \frac{1}{4} \inf_{x\in[11/4,3]} f(x) + \inf_{x\in[3,4]} f(x) \\ &= \frac{1}{2} f(2) + \frac{1}{4} f(5/2) + \frac{1}{4} f(11/4) + f(3) \\ &= \frac{1}{2} \cdot 5 + \frac{1}{2} \cdot \frac{31}{4} + 11 \\ &= \frac{1137}{64} = 17.765625 \end{aligned}$$

(c) $\mathcal{Q}$ is a refinement of $\mathcal{P}$ but their mesh is the same: $\|\mathcal{P}\| = \|\mathcal{Q}\| = 1$. In general, observe that if $\mathcal{Q}$ is a refinement of $\mathcal{P}$ then $\|\mathcal{Q}\| \leq \|\mathcal{P}\|$.

(d) As expected from our last 2 lemmas, observe that

$$L(f;\mathcal{P}) \leq L(f;\mathcal{Q}) \leq U(f;\mathcal{Q}) \leq U(f;\mathcal{P}).$$

In particular,

$$0 \leq U(f;\mathcal{Q}) - L(f;\mathcal{Q}) \leq U(f;\mathcal{P}) - L(f;\mathcal{P}).$$

□

10.3 Upper and Lower Riemann-Stieltjes Integrals

Lemma 10.2 states that, as we refine our lower (resp. upper) sums, they increase (resp. decrease). Therefore, we introduce the following two quantities which should be thought of as the *final* upper and lower estimates for the integral f with integrator α.

Definition 10.4 (Upper and Lower Integrals)**.** Let $I = [a,b]$, $f : I \to \mathbb{R}$ be a bounded function, and $\alpha : [a,b] \to \mathbb{R}$ an increasing function. We will write $\mathscr{P}(I)$ to denote the collection consisting of all partitions $\mathcal{P}$ of I, i.e.

$$\mathscr{P}(I) := \{\mathcal{P} : \mathcal{P} \text{ is a partition of } I\}.$$

Then, we define the *lower integral* of f (with respect to the integrator α) as

$$L(f,\alpha) := \sup_{\mathcal{P}\in\mathscr{P}(I)} L(f;\mathcal{P},\alpha). \tag{10.5}$$

Analogously, the *upper integral* of f on $[a,b]$ (with integrator α) is defined by the rule

$$U(f,\alpha) := \inf_{\mathcal{P}\in\mathscr{P}(I)} U(f;\mathcal{P},\alpha). \tag{10.6}$$

Fix an arbitrary partition $\mathcal{Q}$ of I. By Lemma 10.3, given any partition $\mathcal{P}$ of I we must have

$$L(f;\mathcal{P},\alpha) \leq U(f;\mathcal{Q},\alpha).$$

Thus, $U(f;\mathcal{Q},\alpha)$ is an upper bound for

$$\{L(f;\mathcal{P},\alpha) : \mathcal{P} \in \mathscr{P}(I)\}.$$

As a supremum is a *least* upper bound, the supremum of $\{L(f;\mathcal{P},\alpha) : \mathcal{P} \in \mathscr{P}(I)\}$ is less than or equal to $U(f;\mathcal{Q},\alpha)$. In other words,

$$L(f,\alpha) \leq U(f;\mathcal{Q},\alpha).$$

But then, since $\mathcal{Q}$ is arbitrary, we see that $L(f,\alpha)$ is a lower bound for

$$\{U(f;\mathcal{Q},\alpha) : \mathcal{Q} \in \mathscr{P}(I)\}.$$

Ergo, $L(f,\alpha) \leq U(f,\alpha)$. Given the fundamental nature of this property, we choose to formally encode it in a lemma.

Lemma 10.4. *Given a bounded function $f : [a,b] \to \mathbb{R}$ and a increasing function $\alpha : [a,b] \to \mathbb{R}$, there holds $L(f,\alpha) \leq U(f,\alpha)$.*

In other words, our lower estimate for f is always less than or equal to our upper estimate for f. This realization is what sparks our definition of what it means for a function to be Riemann-Stieltjes integrable: namely, we declare a bounded function f to be integrable on $[a,b]$ if their lower and upper sums "meet in the middle".

10.4 The Riemann/Darboux-Stieltjes Integral

Definition 10.5. Let $I = [a,b]$, $f : I \to \mathbb{R}$ be a bounded function, and $\alpha : [a,b] \to \mathbb{R}$ an increasing function. We say that f is Riemann-Stieltjes integrable (with integrator α) if

$$U(f,\alpha) = L(f,\alpha).$$

In this case, we define $\int_a^b f\mathrm{d}\alpha = \int_a^b f(x)\mathrm{d}\alpha(x)$ to be this value:

$$\int_a^b f\mathrm{d}\alpha = U(f,\alpha) = L(f,\alpha).$$

For convenience, when $\alpha(x) = x$, we shall simply write $\int_a^b f$ to mean $\int_a^b f\mathrm{d}x$.

Interpretation. With the classical Riemann integral (i.e. the Riemann-Stieltjes integral with $\alpha(x) = x$), the upper and lower sums represent over and under estimates (respectively) for the area of the region between the curve of f and the x-axis, provided f does not change sign. As we refine the partition over which we are taking these approximations, these estimates should improve, giving rise to the $U(f)$ and $L(f)$, which should be thought of as the best overestimate and underestimate (respectively) for this area.[17] We then declare a function to be Riemann integrable if these two best estimates agree with each other, in which case this value is thought of as the "true area" under the curve of f.

For the general Riemann-Stieltjes integral, this interpretation remains very much true. The only difference, is the presence of the integrating factor α. Implicitly, in the standard Riemann integral construction, we are assuming that we have a natural way to measure the length of a line segment (i.e. the Δx_i's) and therefore can measure the area of a "small" rectangle with width Δx_i. In much the same way that we generalized the metric of $\mathbb{R}$ to consider spaces equipped with alternative notions of distance, defining the integral for an arbitrarily prescribed increasing function $\alpha : [a, b] \to \mathbb{R}$ can be motivated in a similar way. Depending on how α behaves, the Riemann-Stieltjes integral may consider certain points or sub-regions of the interval $[a, b]$ to be more "significant" when computing the "continuous sum" of $f on [a, b]$, thereby applying to a larger variety of practical situations (see the lecture notes). Notice, however, that we require α to be increasing for this to make sense since we don't want to end up with a rectangle in our approximation whose base has "negative length".

Remark 10.2. Suppose furthermore that there exists a partition $\mathcal{P}$ such that $L(f; \mathcal{P}, \alpha) = U(f; \mathcal{P}, \alpha)$. Can we conclude that g is Riemann-Stieltjes integrable with respect to α? Indeed, by Lemma 10.4, we have

$$L(f; \mathcal{P}, \alpha) \leq L(f, \alpha) \leq U(f, \alpha) \leq U(f; \mathcal{P}, \alpha).$$

Since $L(f; \mathcal{P}, \alpha) = U(f; \mathcal{P}, \alpha)$, we must in fact have equality. That is,

$$L(f; \mathcal{P}, \alpha) = L(f, \alpha) = U(f, \alpha) = U(f; \mathcal{P}, \alpha).$$

Thus, f is Riemann-Stieltjes integrable (with integrator α) and

$$\int_a^b f \mathrm{d}\alpha = L(f; \mathcal{P}, \alpha) = U(f; \mathcal{P}, \alpha).$$

Example 10.2. Consider the constant function $f : [a, b] \to \mathbb{R}$, $f(x) = c$. Let $\alpha(x) = x$ so that we recover the classical Riemann integral. With the partition $\mathcal{P} = \{a, b\}$, observe that

$$L(f; \mathcal{P}) = c(b - a) = U(f; \mathcal{P}).$$

Thus, by the previous remark, f is integrable and

$$\int_a^b f(x) \mathrm{d}x = c(b - a).$$

[17] We should be careful to point out that this is merely a way of thinking about $U(f)$ and $L(f)$. Be warned that these are not really "estimates" and may never be achieved over any partition of the interval.

Example 10.3. Consider the function

$$f(x) = \begin{cases} 1 & 0 \le x < 1 \\ 0 & x = 1. \end{cases}$$

We show that f is (classically) Riemann integrable and

$$\int_0^1 f(x)\mathrm{d}x = 1.$$

First, consider an arbitrary partition $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ of $[0,1]$. For any index $i = 1, \ldots, n$,

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x) = 1.$$

Thus,

$$U(f;\mathcal{P}) = \sum_{i=1}^{n} M_i \Delta x_i = \sum_{i=1}^{n} \Delta x_i = x_n - x_0 = 1.$$

It follows that

$$U(f) = \inf_{\mathcal{P} \in \mathscr{P}([0,1])} U(f;\mathcal{P}) = \inf_{\mathcal{P} \in \mathscr{P}([0,1])} 1 = 1.$$

Since $L(f) \le U(f)$, we must also have $L(f) \le 1$. To see that, in fact, $L(f) = 1$ consider an arbitrary number $0 < c < 1$. Then there must exist $x_1 \in (0,1)$ such that $c < x_1 < 1$. With the partition $\mathcal{Q} = \{0, x_1, 1\}$ we obtain

$$\begin{aligned} L(f;\mathcal{Q}) &= (x_1 - 0) \inf_{x \in [0, x_1]} f(x) + (1 - x_1) \inf_{x \in [x_1, 1]} f(x) \\ &= (x_1 - 0) \cdot 1 + (1 - x_1) \cdot 0 \\ &= x_1 > c. \end{aligned}$$

Thus,

$$L(f) \ge L(f;\mathcal{Q}) > c.$$

Since $c \in (0,1)$ was arbitrary, we infer[18] that $L(f) = 1$. Thus, f is integrable and

$$\int_a^b f\mathrm{d}x = U(f) = L(f) = 1.$$

Example 10.4. Fix $c \in (0,1)$ and consider the function

$$f(x) = \begin{cases} 1 & 0 \le x \le c \\ 0 & c < x \le 1. \end{cases}$$

[18] We know that 1 is an upper bound for $\{L(f;\mathcal{P}) : \mathcal{P} \in \mathscr{P}([0,1])\}$. Furthermore, any $c < 1$ is not an upper bound of this set. That is, 1 is the least upper bound, i.e. $L(f) = 1$.

We show that f is Riemann integrable and

$$\int_0^1 f(x)\mathrm{d}x = c.$$

To this end, let $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[0,1]$ such that $x_{j-1} = c$ for some index j. We make the following observations:

- For $i < j$, we have $M_i = m_i = 1$.
- Next, observe that $M_j = 1$ and $m_j = 0$ since, on the interval $[x_{j-1}, x_j] = [c, x_j]$, the functions f achieves both 0 and 1.
- Finally, for $i > j$, we see that $M_i = m_i = 0$.

Using these observations, we compute

$$\begin{aligned} L(f;\mathcal{P}) = \sum_{i=1}^{n} m_i \Delta x_i &= \sum_{i=1}^{j-1} m_i \Delta x_i + \sum_{i=j}^{n} m_i \Delta x_i \\ &= \sum_{i=1}^{j-1} \Delta x_i \\ &= x_{j-1} - x_0 = c. \end{aligned}$$

Now, suppose we have an arbitrary partition $\mathcal{Q}$ of $[0,1]$. If $c \in \mathcal{Q}$, then by our argument above, we must have $L(f;\mathcal{Q}) = c$. If instead $c \notin Q$, we can refine the partition $\mathcal{Q}$ to obtain a new partition $\mathcal{Q}'$ containing c. Then, since $\mathcal{Q}'$ is a refinement of $\mathcal{Q}$, we obtain

$$L(f;\mathcal{Q}) \leq L(f;\mathcal{Q}') = c$$

In short, one has $L(f;\mathcal{Q}) \leq c$ with equality when $c \in \mathcal{Q}$. It follows that c is the least upper bound of $\{L(f;\mathcal{Q}) : \mathcal{Q} \in \mathscr{P}([0,1])\}$, i.e. $L(f) = c$.

We now show that $U(f) = c$ (and thus $U(f) = L(f) = c$). First, since

$$L(f) \leq U(f),$$

we automatically have $c \leq U(f)$. Moreover, we claim that for any $\alpha > c$, we can find a partition $\mathcal{Q}$ such that

$$U(f;\mathcal{Q}) < \alpha.$$

To find such a partition $\mathcal{Q}$, pick $x_2 \in (0,1)$ such that $c < x_2 < \alpha$. Then, with $\mathcal{Q} = \{x_0, x_1, x_2, x_3\} = \{0, c, x_2, 1\}$, we obtain

$$\begin{aligned} U(f;\mathcal{Q}) &= (c-0)\sup_{[0,c]} f + (x_2 - c)\sup_{[c,x_2]} f + (1-x_2)\sup_{[x_2,1]} f \\ &= (c-0)\cdot 1 + (x_2 - c)\cdot 1 + (1-x_2)\cdot 0 \\ &= x_2 < \alpha. \end{aligned}$$

Thus,

$$U(f;\mathcal{Q}) < \alpha.$$

Since $U(f) \leq U(f;\mathcal{Q})$, it follows that $U(f) < \alpha$. Using that the inequality holds true for all $\alpha > c$, we infer that $U(f) \leq c$. Since we also know that $c \leq U(f)$, it follows that $U(f) = c$.

In conclusion, f is integrable and

$$\int_a^b f = U(f) = L(f) = c.$$

Proposition 10.5. *Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$ and $\alpha : [a,b] \to \mathbb{R}$ a strictly increasing function. If $f(x) \geq 0$ and*

$$\int_a^b f \mathrm{d}\alpha = 0,$$

then $f(x) = 0$ for all $x \in [a,b]$.

Proof. Given that $\int_a^b f \mathrm{d}\alpha = 0$, we must have $L(f,\alpha) = 0$ by definition. Since f is non-negative on $[a,b]$, it follows that, for any partition $\mathcal{P}$ of $[a,b]$, there holds

$$0 \leq L(f;\mathcal{P},\alpha) \leq L(f,\alpha) = 0.$$

Thus, $L(f;\mathcal{P},\alpha) = 0$ for any partition $\mathcal{P}$ of $[a,b]$. If there exists a point $z \in [a,b]$ such that $f(z) > 0$, then by continuity we can find $\delta > 0$ such that

$$f(x) > \frac{f(z)}{2}$$

for all $x \in [a,b] \cap (z-\delta, z+\delta)$. In particular, there exists a subinterval $[c,d] \subseteq [a,b]$ such that $f(x) > \frac{f(z)}{2}$ for all $z \in [c,d]$. Next, consider the partition $\mathcal{P}$ of $[a,b]$ given by $\mathcal{P} := \{a,c,d,b\}$ and observe that, since f is non-negative on $[a,b]$, we must have

$$L(f;\mathcal{P},\alpha) \geq \inf_{x\in[c,d]} f(x)\,(\alpha(d)-\alpha(c)) \geq \frac{f(z)}{2}\,(\alpha(d)-\alpha(c)) > 0$$

which is a contradiction. □

Proposition 10.6. *Let $f : \mathbb{R} \to \mathbb{R}$ be the Dirichlet function. If $\alpha : [a,b] \to \mathbb{R}$ is increasing and non-constant, then f is not Riemann-Stieltjes integrable on $[a,b]$.*

Proof. We will show that $U(f;\alpha) \neq L(f;\alpha)$. To this end, let $\mathcal{P} = \{x_0,\ldots,x_n\}$ be an arbitrary partition of $[a,b]$ and let $\{I_j\}_{j=1}^n$ be the subintervals of $[a,b]$ formed by this partition. Then, since $\mathbb{Q}$ and $\mathbb{R}\setminus\mathbb{Q}$ are dense in $\mathbb{R}$, it follows that

$$\sup_{x\in I_j} f(x) = 1, \quad \inf_{x\in I_j} f(x) = 0$$

for all indices $j = 1, \ldots, n$. Therefore,

$$U(f; \mathcal{P}, \alpha) = \sum_{j=1}^{n} \Delta\alpha(x_j) = \sum_{j=1}^{n} \big[\alpha(x_j) - \alpha(x_{j-1})\big] = \alpha(x_n) - \alpha(x_0) = \alpha(b) - \alpha(a).$$

This implies that $U(f; \alpha) = \alpha(b) - \alpha(a) > 0$. On the other hand,

$$L(f; \mathcal{P}, \alpha) = 0$$

whence $L(f; \alpha) = 0$. Consequently, f is not Riemann-Stieltjes integrable on $[a, b]$. □

Remark 10.3. The hypothesis that α be strictly increasing cannot be dropped. Indeed, if α is constant (and thus still increasing), then every bounded function $f : [a, b] \to \mathbb{R}$ is automatically Riemann-Stieltjes integrable with integral $\int_a^b f \mathrm{d}\alpha = 0$.

11 Final Tutorial

In this tutorial, we shall proceed with our treatment of the Riemann integral. More precisely, we will restate the important ε-criterion for Riemann integrability as well as its sequential counterpart. In doing so, we will also provide several worked out examples utilizing these newer integrability results.

Theorem 11.1. *Let $[a, b] \subset \mathbb{R}$ be an interval and let $f : [a, b] \to \mathbb{R}$ be a bounded function. Let $\alpha : [a, b] \to \mathbb{R}$ be an increasing function. Then, f is Riemann-Stieltjes integrable (with integrator α) on $[a, b]$ if and only if, for each $\varepsilon > 0$, there exists a partition $\mathcal{P}_\varepsilon$ of $[a, b]$ such that*

$$U(f; \mathcal{P}, \alpha) - L(f; \mathcal{P}, \alpha) < \varepsilon.$$

Remark 11.1. This integrability criterion has some very convenient advantages over the *definition* we sometimes use to deduce integrability. Indeed, since $U(f)$ and $L(f)$ take into consideration *all* possible partitions of $[a, b]$, one is somtimes forced to consider a tremendous variety of partitions in order to draw conclusions about f. However, when utilizing Theorem 11.1, it is enough to construct one partition $\mathcal{P}_\varepsilon$ for each $\varepsilon > 0$.

Naturally, Theorem 11.1 admits the following sequential analogue. The major advantage of this corollary is that the value of the integral in question is identified as the limit of a sequence, which can sometimes feel more natural to work with. That being said, one can simply send $\varepsilon \searrow 0$ in the ε-criterion to recover the value of the integral, in the exact same manner.

Corollary 11.2 (Sequential Criterion for Riemann-Stieltjes Integrability)**.** *Let $[a, b] \subset \mathbb{R}$ be an interval, $\alpha : [a, b] \to \mathbb{R}$ an increasing function, and $f : [a, b] \to \mathbb{R}$ a bounded function. Then,*

f is Riemann-Stieltjes integrable on $[a,b]$ (with integrator α) if and only if there exists a sequence $(\mathcal{P}_n)_{n\in\mathbb{N}}$ of partitions of $[a,b]$ such that

$$\lim_{n\to\infty}\left[U(f;\mathcal{P}_n,\alpha)-L(f;\mathcal{P}_n,\alpha)\right]=0.$$

In this case, we also have

$$\int_a^b f\mathrm{d}\alpha=\lim_{n\to\infty}U(f;\mathcal{P}_n,\alpha)=\lim_{n\to\infty}L(f;\mathcal{P}_n,\alpha).$$

Proof. If f is Riemann-Stieltjes integrable on $[a,b]$, with integrator α, then given $n\in\mathbb{N}$ there exists by the ε-criterion for integrability a partition $\mathcal{P}_n$ of $[a,b]$ such that

$$U(f;\mathcal{P}_n,\alpha)-L(f;\mathcal{P}_n,\alpha)<\frac{1}{n}$$

whence $\lim_{n\to\infty}\left[U(f;\mathcal{P}_n,\alpha)-L(f;\mathcal{P}_n,\alpha)\right]=0$.

Conversely, assume there exists a sequence $(\mathcal{P}_n)_{n\in\mathbb{N}}$ of partitions of $[a,b]$ such that

$$\lim_{n\to\infty}\left[U(f;\mathcal{P}_n,\alpha)-L(f;\mathcal{P}_n,\alpha)\right]=0.$$

Then, given $\varepsilon>0$, we can find $N\in\mathbb{N}$ such that

$$U(f;\mathcal{P}_n,\alpha)-L(f;\mathcal{P}_n,\alpha)<\varepsilon,\quad\forall n\geq N.$$

Especially, $U(f;\mathcal{P}_N,\alpha)-L(f;\mathcal{P}_N,\alpha)<\varepsilon$ which verifies the ε-criterion for integrability. Thus, $L(f,\alpha)=U(f,\alpha)$ and

$$L(f;\mathcal{P}_n,\alpha)\leq L(f,\alpha)=U(f,\alpha)\leq U(f;\mathcal{P}_n,\alpha),\quad\forall n\in\mathbb{N}.$$

This implies that

$$0\leq U(f;\mathcal{P}_n,\alpha)-U(f,\alpha)\leq U(f,\mathcal{P}_n,\alpha)-L(f,\mathcal{P}_n,\alpha),\tag{11.1}$$

$$0\leq L(f,\alpha)-L(f;\mathcal{P}_n,\alpha)\leq U(f;\mathcal{P}_n,\alpha)-L(f;\mathcal{P}_n,\alpha)\tag{11.2}$$

where the right hand sides of both equations converge to 0 as $n\to\infty$. Consequently,

$$\int_a^b f\mathrm{d}\alpha=\lim_{n\to\infty}U(f;\mathcal{P}_n,\alpha)=\lim_{n\to\infty}L(f;\mathcal{P}_n,\alpha).$$

□

We follow this up with an elementary proof that the signum function is classically Riemann integrable.

Example 11.1. Prove that the signum function

$$\operatorname{sgn} : \mathbb{R} \to \mathbb{R}, \quad \operatorname{sgn}(x) := \begin{cases} -1 & \text{if } x < 0, \\ 0 & \text{if } x = 0, \\ 1 & \text{if } x > 0 \end{cases}$$

is Riemann integrable on $[-1, 1]$ with $\int_{-1}^{1} \operatorname{sgn} \mathrm{d}x = 0$.

Proof. In the hopes of applying Corollary 11.2, given $n \in \mathbb{N}$, we consider the partition

$$\mathcal{P}_n := \left\{-1, -\frac{1}{n}, \frac{1}{n}, 1\right\}$$

and observe that

- $\sup_{x \in [-1, -\frac{1}{n}]} \operatorname{sgn}(x) = -1$ and $\inf_{x \in [-1, -\frac{1}{n}]} \operatorname{sgn}(x) = -1$;
- $\sup_{x \in [-\frac{1}{n}, \frac{1}{n}]} \operatorname{sgn}(x) = 1$ and $\inf_{x \in [-\frac{1}{n}, \frac{1}{n}],} \operatorname{sgn}(x) = -1$;
- $\sup_{x \in [\frac{1}{n}, 1]} \operatorname{sgn}(x) = 1$ and $\inf_{x \in [\frac{1}{n}, 1]} \operatorname{sgn}(x) = 1$;

Therefore, a straightforward calculation gives

$$U(\operatorname{sgn}; \mathcal{P}_n) = \left(1 - \frac{1}{n}\right)(-1) + \left(\frac{2}{n}\right) 1 + \left(1 - \frac{1}{n}\right)(1) = \frac{2}{n}$$

Similarly,

$$L(\operatorname{sgn}; \mathcal{P}_n) = \left(1 - \frac{1}{n}\right)(-1) + \left(\frac{2}{n}\right)(-1) + \left(1 - \frac{1}{n}\right)(1) = -\frac{2}{n}.$$

It follows that

$$\lim_{n \to \infty} \left[U(\operatorname{sgn}; \mathcal{P}_n) - L(\operatorname{sgn}; \mathcal{P}_n)\right] = \lim_{n \to \infty} \frac{4}{n} = 0.$$

Hence, by Corollary 11.2, the signum function $\operatorname{sgn}(x)$ is Riemann integrable on $[-1, 1]$. Furthermore,

$$\int_a^b \operatorname{sgn} = \lim_{n \to \infty} U(\operatorname{sgn}; \mathcal{P}_n) = \lim_{n \to \infty} \frac{2}{n} = 0.$$

□

11.1 A Dirac-Type Integral

Proposition 11.3. *Let $a < s < b$ and $\alpha_a < \alpha_s < \alpha_b$ be real numbers, and define the monotone increasing function*

$$\alpha : [a,b] \to \mathbb{R}, \quad \alpha(x) := \begin{cases} \alpha_a & \text{if } a \le x < s, \\ \alpha_s & \text{if } x = s, \\ \alpha_b & \text{if } s < x \le b. \end{cases}$$

Then, a bounded function $f : [a,b] \to \mathbb{R}$ is α-integrable on $[a,b]$ if and only if f is continuous at s. In this case:

$$\int_a^b f \mathrm{d}\alpha = f(s)(\alpha_b - \alpha_a). \tag{11.3}$$

Proof. We first show that f is α-integrable on $[a,b]$ if f is continuous at s, and establish the formula (11.3). To this end, given $\varepsilon > 0$, we invoke the continuity of f at s to choose $\delta > 0$ such that

$$|f(x) - f(s)| < \varepsilon,$$

for all $x \in [a,b]$ with $|x - s| \le \delta$. In fact, since s is an interior point of (a,b), we may assume $\delta > 0$ is small enough that $[s-\delta, s+\delta] \subset (a,b)$. Now, consider the partition $\mathcal{P} := \{a, s-\delta, s+\delta, b\}$ of $[a,b]$. For this partition, we compute

$$\begin{aligned} U(f;\mathcal{P},\alpha) = \left[\sup_{x\in[a,s-\delta]} f(x)\right](\alpha(s-\delta) - \alpha(a)) + \left[\sup_{x\in[s-\delta,s+\delta]} f(x)\right](\alpha(s+\delta) - \alpha(s-\delta)) \\ + \left[\sup_{x\in[s+\delta,b]} f(x)\right](\alpha(b) - \alpha(s+\delta)). \end{aligned}$$

Because $\alpha(s+\delta) = \alpha(b)$ and $\alpha(a) = \alpha(s-\delta)$, the above reduces to

$$U(f;\mathcal{P},\alpha) = \left[\sup_{x\in[s-\delta,s+\delta]} f(x)\right](\alpha_b - \alpha_a).$$

Now, on $[s-\delta, s+\delta]$, we have that $f(s) - \varepsilon < f(x) < f(s) + \varepsilon$ and so

$$f(s)(\alpha_b - \alpha_a) \le U(f;\mathcal{P},\alpha) \le (f(s)+\varepsilon)(\alpha_b - \alpha_a). \tag{11.4}$$

Similarly,

$$(f(s)-\varepsilon)(\alpha_b - \alpha_a) \le L(f;\mathcal{P},\alpha) \le f(s)(\alpha_b - \alpha_a). \tag{11.5}$$

Subtracting these equations yields that

$$U(f;\mathcal{P},\alpha) - U(f;\mathcal{P},\alpha) \le 2\varepsilon(\alpha_b - \alpha_a).$$

As $\varepsilon > 0$ was arbitrary, we conclude that f is α-integrable on $[a, b]$. Moreover,

$$(f(s) - \varepsilon)(\alpha_b - \alpha_a) \leq L(f; \mathcal{P}, \alpha) \leq \int_a^b f \mathrm{d}\alpha \leq U(f; \mathcal{P}, \alpha) \leq (f(s) + \varepsilon)(\alpha_b - \alpha_a).$$

Sending $\varepsilon \searrow 0$ yields (11.3).

Conversely, suppose f is α-integrable on $[a, b]$. We claim that f is continuous at s. Let $\varepsilon > 0$ be given and choose (by the integrability hypothesis) a partition $\mathcal{P}$ such that

$$U(f; \mathcal{P}, \alpha) - L(f; \mathcal{P}, \alpha) < \varepsilon \min\{\alpha_b - \alpha_s, \alpha_s - \alpha_a\}.$$

Without loss of generality, by refining $\mathcal{P}$ if necessary, we may assume without harm to the proof that $s \in \mathcal{P}$. Thus, we may enumerate the elements of $\mathcal{P} = \{x_0, \ldots, x_k, \ldots, x_n\}$ with $x_0 = a$, $x_k = s$, and $x_n = b$. By the same argument as above, and using that α is constant on any interval $[x_{i-1}, x_i]$ formed by $\mathcal{P}$ unless $i - 1 = k$ or $i = k$, we may compute the upper and lower sums:

$$U(f; \mathcal{P}, \alpha) = \left[\sup_{x \in [x_{k-1}, s]} f(x)\right](\alpha_s - \alpha_a) + \left[\sup_{x \in [s, x_{k+1}]} f(x)\right](\alpha_b - \alpha_s)$$
$$L(f; \mathcal{P}, \alpha) = \left[\inf_{x \in [x_{k-1}, s]} f(x)\right](\alpha_s - \alpha_a) + \left[\inf_{x \in [s, x_{k+1}]} f(x)\right](\alpha_b - \alpha_s)$$

For convenience, let us denote by

$$M_{f,1} := \sup_{x \in [x_{k-1}, s]} f(x), \quad m_{f,1} := \inf_{x \in [x_{k-1}, s]} f(x), \quad M_{f,2} := \sup_{x \in [s, x_{k+1}]} f, \quad m_{f,2} := \inf_{x \in [s, x_{k+1}]} f.$$

Then, it follows that, in this notation:

$$U(f; \mathcal{P}, \alpha) - L(f; \mathcal{P}, \alpha) = (M_{f,1} - m_{f,1})(\alpha_s - \alpha_a) + (M_{f,2} - m_{f,2})(\alpha_b - \alpha_s) < \varepsilon \min\{\alpha_b - \alpha_s, \alpha_s - \alpha_a\}.$$

In particular, this implies both

$$(M_{f,1} - m_{f,1}) < \frac{\varepsilon \min\{\alpha_b - \alpha_s, \alpha_s - \alpha_a\}}{\alpha_s - \alpha_a} \leq \varepsilon$$

and, similarly,

$$(M_{f,2} - m_{f,2}) < \varepsilon.$$

Now define $\delta := \min\{s - x_{k-1}, x_{k+1} - s\}$ and let $x \in [a, b]$ be such that $|x - s| < \delta$. Then either

$x \in [x_{k-1}, s]$ or $x \in [s, x_{k+1}]$. In the former case, notice[19] that

$$|f(x) - f(s)| \leq \sup_{x \in [x_{k-1}, x]} f(x) - \inf_{x \in [x_{k-1}, s]} f(x) = M_{f,1} - m_{f,1} < \varepsilon.$$

Similarly, if $x \in [s, x_{k+1}]$ we find that

$$|f(x) - f(s)| \leq \sup_{x \in [x_{k-1}, x]} f(x) - \inf_{x \in [x_{k-1}, s]} f(x) = M_{f,1} - m_{f,1} < \varepsilon.$$

Thus, for any x belonging to $[a, b]$ with $|x - s| < \delta$ must necessarily satisfy $|f(x) - f(s)| < \varepsilon$. As $\varepsilon > 0$ was arbitrary, this is precisely the statement that f is continuous at s. This completes the proof.

□

Remark 11.2. Although we only stated (and provided proof) for the result in the case where $s \in (a, b)$ is an interior point, very little changes if $s = a$ or $s = b$ in this result. More precisely, if $s = a$ and $\alpha(x) = \alpha_a = \alpha_s$ for $x = a$ with $\alpha(x) = \alpha_b$ for all $x \in (a, b]$, then α "jumps" at a and instead we have that f is α-integrable if and only if f is *right*-continuous at $a = s$, in which case:

$$\int_a^b f \mathrm{d}\alpha = f(a)(\alpha_b - \alpha_a).$$

Similarly, if $a < s = b$, then f is α-integrable if and only if f is *left* continuous at $s = b$, and the above formula continues to hold true in this case.

11.2 The Additivity Theorem

Lemma 11.4. *Let $I = [a, b] \subset \mathbb{R}$ be an interval and fix $x_0 \in I$. Let $\mathscr{P}_{x_0}(I)$ denote the set of all partitions of I which contain x_0 as a partition point. Then, for any bounded function $f : I \to \mathbb{R}$ and every increasing function $\alpha : I \to \mathbb{R}$, there holds*

$$U(f; \alpha) = \inf_{\mathcal{P} \in \mathscr{P}_{x_0}(I)} U(f; \mathcal{P}, \alpha),$$
$$L(f; \alpha) = \sup_{\mathcal{P} \in \mathscr{P}_{x_0}(I)} L(f; \mathcal{P}, \alpha),$$

[19] If $f : [a, b] \to \mathbb{R}$ is bounded, then for any $x, y \in [a, b]$ we automatically have that

$$|f(x) - f(y)| \leq \sup_{u \in [a,b]} f(u) - \inf_{u \in [a,b]} f(u).$$

Indeed, $\inf_{u \in [a,b]} f(u) \leq f(x) \leq \sup_{u \in [a,b]} f(u)$ and $\inf_{u \in [a,b]} f(u) \leq f(y) \leq \sup_{u \in [a,b]} f(u)$ whence

$$\inf_{u \in [a,b]} f(u) - \sup_{u \in [a,b]} f(u) \leq f(x) - f(y) \leq \sup_{u \in [a,b]} f(u) - \inf_{u \in [a,b]} f(u).$$

Thus, to compute the upper integral (respectively lower integral) it suffices to take the infimum (respectively supremum) over those partitions containing a point $x_0 \in [a, b]$ of our choice.

Proof. Clearly, $\mathscr{P}_{x_0}(I) \subseteq \mathscr{P}(I)$, where the latter denotes the set of all partitions of I. Therefore,

$$U(f;\alpha) = \inf_{\mathcal{P}\in\mathscr{P}(I)} U(f;\mathcal{P},\alpha) \leq \inf_{\mathcal{P}\in\mathscr{P}_{x_0}(I)} U(f;\mathcal{P},\alpha).$$

On the other hand, given $\mathcal{Q} \in \mathscr{P}(I)$, we can consider the refinement $\mathcal{Q}'$ obtained by adding x_0 to the partition $\mathcal{Q}$. Then,

$$\inf_{\mathcal{P}\in\mathscr{P}_{x_0}(I)} U(f;\mathcal{P},\alpha) \leq U(f;\mathcal{Q}',\alpha) \leq U(f;\mathcal{Q},\alpha).$$

Taking the infimum over all $\mathcal{Q} \in \mathscr{P}(I)$, we obtain

$$\inf_{\mathcal{P}\in\mathscr{P}_{x_0}(I)} U(f;\mathcal{P},\alpha) \leq U(f;\alpha)$$

whence

$$U(f;\alpha) = \inf_{\mathcal{P}\in\mathscr{P}_{x_0}(I)} U(f;\mathcal{P},\alpha).$$

A similar argument applies for the lower sums. □

Lemma 11.5. *Let $f : [a, b] \to \mathbb{R}$ be a bounded function and $\alpha : [a, b] \to \mathbb{R}$ increasing. Then, for any $c \in (a, b)$, we have*

$$\overline{\int_a^b} f\,\mathrm{d}\alpha = \overline{\int_a^c} f\,\mathrm{d}\alpha + \overline{\int_c^b} f\,\mathrm{d}\alpha.$$

Similarly,

$$\underline{\int_a^b} f\,\mathrm{d}\alpha = \underline{\int_a^c} f\,\mathrm{d}\alpha + \underline{\int_c^b} f\,\mathrm{d}\alpha.$$

Proof. We only prove the first identity, leaving the second as an exercise which may be handled analogously. First, observe that if we are given a partition $\mathcal{P}_1$ of $[a, c]$ and a partition $\mathcal{P}_2$ of $[c, b]$, then their union $\mathcal{P}_1 \cup \mathcal{P}_2$ is a partition of $[a, b]$ containing c as a partition point. Conversely, every partition $\mathcal{P}$ of $[a, b]$ having c as a partition point may be written in such a way. Moreover, if $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$ is a partition of $[a, b]$ such that $\mathcal{P}_1$ partitions $[a, c]$ and $\mathcal{P}_2$ partitions $[c, b]$, it is easy to see that

$$U(f;\mathcal{P},\alpha) = U(f;\mathcal{P}_1,\alpha) + U(f;\mathcal{P}_2,\alpha).$$

By the previous lemma, we infer that

$$\begin{aligned}\overline{\int_a^b} f\,\mathrm{d}\alpha &= \inf\{U(f;\mathcal{P},\alpha) : \mathcal{P} \text{ is a partition of } [a,b] \text{ with } c \in \mathcal{P}\}\\ &= \inf\{U(f;\mathcal{P},\alpha) : \mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2 \text{ s.t. } \mathcal{P}_1 \in \mathscr{P}([a,c]) \text{ and } \mathcal{P}_2 \in \mathscr{P}([c,b])\}\\ &= \inf\{U(f;\mathcal{P}_1,\alpha) + U(f;\mathcal{P}_2,\alpha) : \mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2 \text{ s.t. } \mathcal{P}_1 \in \mathscr{P}([a,c]) \text{ and } \mathcal{P}_2 \in \mathscr{P}([c,b])\}\\ &= \inf\{U(f;\mathcal{P}_1,\alpha) : \mathcal{P}_1 \in \mathscr{P}([a,c])\} + \inf\{U(f;\mathcal{P}_2,\alpha) : \mathcal{P}_2 \in \mathscr{P}([c,b])\}\\ &= \overline{\int_a^c} f\,\mathrm{d}\alpha + \overline{\int_c^b} f\,\mathrm{d}\alpha.\end{aligned}$$

This proves the first identity. The proof of the second identity is handled in the same manner and is left as an exercise. □

Theorem 11.6 (Additivity Theorem)**.** *Let $f : [a,b] \to \mathbb{R}$ be bounded and $\alpha : [a,b] \to \mathbb{R}$ monotone increasing. Then, f is Riemann-Stieltjes integrable on $[a,b]$ (with integrator α) if and only if f is Riemann-Stieltjes integrable on $[a,c]$ and $[b,c]$ for every $c \in (a,b)$. Moreover, in this case,*

$$\int_a^b f\,\mathrm{d}\alpha = \int_a^c f\,\mathrm{d}\alpha + \int_c^b f\,\mathrm{d}\alpha$$

for each $c \in (a,b)$.

Proof. Suppose f is Riemann-Stieltjes integrable (with respect to α) on $[a,b]$. Then, given a point $c \in (a,b)$ we have by the last lemma that

$$\int_a^b f\,\mathrm{d}\alpha = \underline{\int_a^b} f\,\mathrm{d}\alpha = \underline{\int_a^c} f\,\mathrm{d}\alpha + \underline{\int_c^b} f\,\mathrm{d}\alpha \le \overline{\int_a^c} f\,\mathrm{d}\alpha + \overline{\int_c^b} f\,\mathrm{d}\alpha = \overline{\int_a^b} f\,\mathrm{d}\alpha = \int_a^b f\,\mathrm{d}\alpha.$$

Therefore, this inequality is an equality, and

$$\underline{\int_a^c} f\,\mathrm{d}\alpha + \underline{\int_c^b} f\,\mathrm{d}\alpha = \overline{\int_a^c} f\,\mathrm{d}\alpha + \overline{\int_c^b} f\,\mathrm{d}\alpha.$$

Especially, this implies that

$$\overline{\int_a^c} f\,\mathrm{d}\alpha = \underline{\int_a^c} f\,\mathrm{d}\alpha + \underline{\int_c^b} f\,\mathrm{d}\alpha - \overline{\int_c^b} f\,\mathrm{d}\alpha \le \underline{\int_a^c} f\,\mathrm{d}\alpha.$$

Since we always have $\overline{\int_a^c} f\,\mathrm{d}\alpha \ge \underline{\int_a^c} f\,\mathrm{d}\alpha$, it follows from the above that

$$\underline{\int_a^c} f\,\mathrm{d}\alpha = \overline{\int_a^c} f\,\mathrm{d}\alpha$$

so that f is Riemann-Stieltjes integrable on $[a, c]$. Similarly, one can show that f is Riemann-Stieltjes integrable on $[b, c]$. At this point, it follows that

$$\int_a^b f\mathrm{d}\alpha = \overline{\int_a^b} f\mathrm{d}\alpha = \overline{\int_a^c} f\mathrm{d}\alpha + \overline{\int_c^b} f\mathrm{d}\alpha = \int_a^c f\mathrm{d}\alpha + \int_c^b f\mathrm{d}\alpha.$$

Conversely, suppose that f is Riemann-Stieltjes integrable on $[a, c]$ and $[c, b]$ for some $c \in (a, b)$. Then,

$$\overline{\int_a^b} f\mathrm{d}\alpha = \overline{\int_a^c} f\mathrm{d}\alpha + \overline{\int_c^b} f\mathrm{d}\alpha = \underline{\int_a^c} f\mathrm{d}\alpha + \underline{\int_c^b} f\mathrm{d}\alpha = \underline{\int_a^b} f\mathrm{d}\alpha$$

whence f is Riemann-Stieltjes integrable on $[a, b]$. □

11.3 Hölder's Inequality

Theorem 11.7. *Let $\alpha : [a, b] \to \mathbb{R}$ be monotone increasing and $f, g : [a, b] \to \mathbb{R}$ be Riemann-Stieltjes integrable on $[a, b]$ with respect to α. Then,*

$$\|fg\|_{1,\alpha} \leq \|f\|_{p,\alpha} \|g\|_{q,\alpha}$$

where $1 \leq p \leq q \leq \infty$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1$$

with the convention $1/\infty = 0$. Here,

$$\|f\|_{p,\alpha} = \begin{cases} \left(\int_a^b |f|^p \,\mathrm{d}\alpha\right)^{1/p} & \text{if } 1 \leq p < \infty, \\ \sup_{x\in[a,b]} |f(x)| & \text{if } p = \infty \end{cases}$$

Proof. By replacing f with $|f|$ and g with $|g|$, we may assume without loss of generality that $f, g \geq 0$. Now, let us first treat the case $p > 1$ (especially, $q < \infty$).

Letting $\mathcal{P}$ be an arbitrary partition of $[a, b]$, denote by $I_1, \ldots, I_n$ the subintervals of $[a, b]$ formed by $\mathcal{P}$. Write $m_{f,i}$, and $m_{fg,i}$ to denote the infimum of f and fg on each I_i, respectively. Moreover, denote by $M_{g,i}$ the supremum of g on each I_i. It is easy to check that $m_{fg,i} \leq m_{f,i} M_{g,i}$.

Then, by Hölder's inequality for finite sums, we have

$$\begin{aligned}
L(fg;\mathcal{P},\alpha) = \sum_{i=1}^{n} m_{fg,i}\Delta\alpha(x_i) &\le \sum_{i=1}^{n} m_{f,i}M_{g,i}\Delta\alpha(x_i)^{\frac{1}{p}+\frac{1}{q}} \\
&\le \sum_{i=1}^{n} m_{f,i}\Delta\alpha(x_i)^{\frac{1}{p}} M_{g,i}\Delta\alpha(x_i)^{\frac{1}{q}} \\
&\le \left(\sum_{i=1}^{n} m_{f,i}^{p}\Delta\alpha(x_i)\right)^{1/p}\left(\sum_{i=1}^{n} M_{g,i}^{q}\Delta\alpha(x_i)\right)^{1/q} \\
&= \left(\sum_{i=1}^{n} m_{f^p,i}\Delta\alpha(x_i)\right)^{1/p}\left(\sum_{i=1}^{n} M_{g^q,i}\Delta\alpha(x_i)\right)^{1/q} \\
&= L(f^p;\mathcal{P},\alpha)^{1/p}U(g^q;\mathcal{P},\alpha)^{1/q}.
\end{aligned}$$

Here we have used that $m_{f,i}^{p} = m_{f^p,i}$ and $M_{g,i}^{q} = M_{g^q,i}$, which can be verified easily. Now, let $\varepsilon > 0$ be arbitrary. Because the map $x \mapsto x^{1/q}$ is continuous at $\int_a^b g^q \mathrm{d}\alpha$, there exists $\delta > 0$ such that for all $x \ge 0$

$$\left|x - \int_a^b g^q \mathrm{d}\alpha\right| < \delta \implies \left|x^{1/q} - \left(\int_a^b g^q \mathrm{d}\alpha\right)^{1/q}\right| < \varepsilon.$$

By definition of the integral, we may select a partition $\mathcal{P}_1$ of $[a,b]$ such that

$$\int_a^b fg\mathrm{d}\alpha - \varepsilon < L(fg;\mathcal{P}_1,\alpha).$$

Analogously, we may choose a partition $\mathcal{P}_2$ of $[a,b]$ such that

$$\int_a^b g^q \mathrm{d}\alpha \le U(g^q;\mathcal{P}_2,\alpha) < \int_a^b g^q \mathrm{d}\alpha + \delta.$$

From our choice for $\delta > 0$, we have

$$\left|U(g^q;\mathcal{P},\alpha)^{1/q} - \left(\int_a^b g^q \mathrm{d}\alpha\right)^{1/q}\right| < \varepsilon.$$

Let now $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$ be the common refinement of $\mathcal{P}_1$ and $\mathcal{P}_2$. Then,

$$L(fg,\mathcal{P}_1,\alpha) \le L(fg;\mathcal{P},\alpha) \quad \text{and} \quad U(g^q;\mathcal{P},\alpha) \le U(g^q;\mathcal{P}_2,\alpha).$$

Returning to our previous estimates, we obtain

$$\begin{aligned}\int_a^b fg\mathrm{d}\alpha - \varepsilon < L(fg,\mathcal{P}_1,\alpha) &\leq L(fg;\mathcal{P},\alpha)\\ &\leq L(f^p;\mathcal{P},\alpha)^{1/p}U(g^q;\mathcal{P},\alpha)^{1/q}\\ &\leq \left(\int_a^b f^p\mathrm{d}\alpha\right)^{1/p} U(g^q;\mathcal{P}_2,\alpha)^{1/q}\\ &< \left(\int_a^b f^p\mathrm{d}\alpha\right)^{1/p}\left(\left(\int_a^b g^q\mathrm{d}\alpha\right)^{1/q}+\varepsilon\right)\end{aligned}$$

Thus, there holds

$$\int_a^b fg\mathrm{d}\alpha - \varepsilon < \left(\int_a^b f^p\mathrm{d}\alpha\right)^{1/p}\left(\left(\int_a^b g^q\mathrm{d}\alpha\right)^{1/q}+\varepsilon\right)$$

for all $\varepsilon > 0$. Sending $\varepsilon \searrow 0$, we conclude that

$$\int_a^b fg\mathrm{d}\alpha \leq \left(\int_a^b f^p\mathrm{d}\alpha\right)^{1/p}\left(\int_a^b g^q\mathrm{d}\alpha\right)^{1/q},$$

which is precisely the desired inequality.

If $p = 1$, then the asserted inequality is trivial by monotonicity of the integral:

$$\|fg\|_{1,\alpha} = \int_a^b fg\mathrm{d}\alpha \leq \int_a^b f\,\|g\|_{\infty,\alpha}\,\mathrm{d}\alpha = \|g\|_{\infty,\alpha}\int_a^b f\mathrm{d}\alpha = \|f\|_{1,\alpha}\,\|g\|_{\infty,\alpha}.$$

□

11.4 A Useful Change of Variables Formula

Theorem 11.8. *Let $\alpha : [a,b] \to \mathbb{R}$ be an increasing function and assume that α is differentiable on $[a,b]$ and that the derivative α' is classically Riemann integrable on $[a,b]$, i.e. $\alpha' \in \mathcal{R}([a,b])$. Then, f is Riemann-Stieltjes integrable (with respect to the integrator α) if and only if $f\alpha'$ is classically Riemann integrable on $[a,b]$. In this case, we also have*

$$\int_a^b f\mathrm{d}\alpha = \int_a^b f(x)\alpha'(x)\mathrm{d}x.$$

This is problem 3 on your (optional) assignment 6, and it is two part problem designed to guide you through the proof of this result. We approach this theorem exactly as you are instructed to within this problem. Indeed, we encode part (a) of this problem in the form of the following lemma:

Lemma 11.9. *Under the hypotheses of Theorem 11.8, for any bounded function $f : [a, b] \to \mathbb{R}$ and every partition $\mathcal{P}$ of $[a, b]$,*

$$|U(f; \mathcal{P}, \alpha) - U(f\alpha'; \mathcal{P})| \leq M \left(U(\alpha'; \mathcal{P}) - L(\alpha'; \mathcal{P})\right)$$

and

$$|L(f; \mathcal{P}, \alpha) - L(f\alpha'; \mathcal{P})| \leq M \left(U(\alpha'; \mathcal{P}) - L(\alpha'; \mathcal{P})\right),$$

where $M \geq 0$ is such that $|f(x)| \leq M$ for all $x \in [a, b]$.

Proof of Lemma. Let $\mathcal{P}$ be an arbitrary partition of $[a, b]$, and list the elements $\mathcal{P} = \{x_0, \ldots, x_n\}$ where $a = x_0 < x_1 < \cdots < x_n = b$. Let $I_1, \ldots, I_n$ denote the sub intervals of $[a, b]$ formed by $\mathcal{P}$ and define for any bounded function $h : [a, b] \to \mathbb{R}$ the quantities

$$M_{h,i} := \sup_{x \in I_i} h(x) \quad \text{and} \quad m_{h,i} := \inf_{x \in I_i} h(x).$$

In this notation, observe that

$$U(f\alpha'; \mathcal{P}) = \sum_{i=1}^{n} M_{f\alpha',i} \Delta x_i.$$

On the other hand,

$$U(f; \mathcal{P}, \alpha) = \sum_{i=1}^{n} M_{f,i} \Delta\alpha(x_i) = \sum_{i=1}^{n} M_{f,i} \Delta \left(\alpha(x_i) - \alpha(x_{i-1})\right).$$

Since α satisfies the requirements of the mean value theorem on $[a, b]$, for each interval $I_i = [x - i - 1, x_i]$ formed by our partition $\mathcal{P}$, there exists a point $c_i \in (x_{i-1}, x_i)$ for which

$$\Delta\alpha(x_i) = (\alpha(x_i) - \alpha(x_{i-1})) = \alpha'(c_i) \Delta x_i.$$

Substituting this into our previous identity, we obtain that

$$U(f; \mathcal{P}, \alpha) = \sum_{i=1}^{n} M_{f,i} \alpha'(c_i) \Delta x_i$$

whence

$$|U(f; \mathcal{P}, \alpha) - U(f\alpha'; \mathcal{P})| = \left| \sum_{i=1}^{n} M_{f,i} \alpha'(c_i) \Delta x_i - \sum_{i=1}^{n} M_{f\alpha',i} \Delta x_i \right| \tag{11.6}$$

$$= \left| \sum_{i=1}^{n} \left(M_{f,i} \alpha'(c_i) - M_{f\alpha',i}\right) \Delta x_i \right| \tag{11.7}$$

$$\leq \sum_{i=1}^{n} \left| M_{f,i} \alpha'(c_i) - M_{f\alpha',i} \right| \Delta x_i \tag{11.8}$$

$$= \sum_{i=1}^{n} \left| M_{f\alpha'(c_i),i} - M_{f\alpha',i} \right| \Delta x_i. \tag{11.9}$$

In this last line, we have used that $\alpha'(c_i) \geq 0$ (since α is increasing) implies that

$$M_{f\alpha'(c_i),i} = \sup_{x\in I_i}(f(x)\alpha'(c_i)) = \alpha'(c_i)\sup_{x\in I_i}(f(x)) = M_{f,i}\alpha'(c_i).$$

Recall [20]that given any two bounded functions $g, h : [a,b] \to \mathbb{R}$, we have

$$\left|\sup_{x\in I_i} g(x) - \sup_{x\in I_i} h(x)\right| \leq \sup_{x\in I_i}|g(x) - h(x)|.$$

Thus, by taking for each index $i = 1, \ldots, n$ the functions $g_i(x) := f(x)\alpha'(c_i)$ and $h(x) := f(x)\alpha'(x)$ we find through this inequality that

$$\begin{aligned}
\left|M_{f\alpha'(c_i),i} - M_{f\alpha'}, i\right| &= \left|\sup_{x\in I_i} g_i(x) - \sup_{x\in I_i} h_i(x)\right| \\
&\leq \sup_{x\in I_i}|g(x) - h(x)| \\
&= \sup_{x\in I_i}|f(x)\alpha'(c_i) - f(x)\alpha'(x)| \\
&= \sup_{x\in I_i}[|f(x)|\,|\alpha'(c_i) - \alpha'(x)|] \\
&\leq M\sup_{x\in I_i}|\alpha'(c_i) - \alpha'(x)| \\
&\leq M\left(M_{\alpha',i} - m_{\alpha',i}\right).
\end{aligned}$$

Notice that we are using here that, for each $x \in I_i$, we have

$$m_{\alpha',i} - M_{\alpha',i} \leq \alpha'(c_i) - \alpha'(x) \leq M_{\alpha',i} - m_{\alpha',i}.$$

[20]It is in fact elementary (in the sense that it only uses analysis 1 material!) to verify this inequality. Indeed, for any $x \in I_i$, we have that

$$g(x) = g(x) - h(x) + h(x) \leq |g(x) - h(x)| + h(x) \leq \sup_{x\in I_i}|g(x) - h(x)| + \sup_{x\in I_i} h(x)$$

whence we obtain that

$$\sup_{x\in I_i} g(x) - \sup_{x\in I_i} h(x) \leq \sup_{x\in I_i}|g(x) - h(x)|.$$

Similarly,

$$\sup_{x\in I_i} h(x) - \sup_{x\in I_i} g(x) \leq \sup_{x\in I_i}|g(x) - h(x)|.$$

Returning now to (11.6)-(11.9), it follows that

$$\begin{aligned}
|U(f;\mathcal{P},\alpha) - U(f\alpha';\mathcal{P})| &\leq \sum_{i=1}^{n} \left|M_{f\alpha'(c_i),i} - M_{f\alpha',i}\right| \Delta x_i \\
&\leq M \sum_{i=1}^{n} \left(M_{\alpha',i} - m_{\alpha',i}\right) \Delta x_i \\
&= M\left(U(\alpha',\mathcal{P}) - L(\alpha',\mathcal{P})\right).
\end{aligned}$$

The proof for the lower sum analogue is similar and left as an exercise. □

With this we now prove the theorem.

Proof of Theorem 11.8. First, we assume that f is Riemann-Stieltjes integrable on $[a,b]$ with respect to α. We claim that $f\alpha'$ is classically Riemann integrable as well. Indeed, first note that α' is bounded (it is assumed to be classically Riemann integrable) and so $f\alpha'$ is also bounded.

Let now $\varepsilon > 0$ be given and $M > 0$ be as in the statement of Theorem 11.8. Being that f is assumed to be α-integrable, there exists a partition P_ε on $[a,b]$ for which

$$0 \leq U(f;\mathcal{P}_\varepsilon,\alpha) - L(f;\mathcal{P}_\varepsilon,\alpha) < \frac{\varepsilon}{2}.$$

In addition, because α' is classically Riemann integrable, there is a partition P'_ε of $[a,b]$ such that

$$0 \leq U(\alpha';\mathcal{P}'_\varepsilon) - L(\alpha';\mathcal{P}'_\varepsilon) < \frac{\varepsilon}{4M}.$$

Letting $\mathcal{Q}_\varepsilon := \mathcal{P}_\varepsilon \cup \mathcal{P}'_\varepsilon$, we obtain a partition for which both of the above holds, i.e.

$$0 \leq U(f;\mathcal{Q}_\varepsilon,\alpha) - L(f;\mathcal{Q}_\varepsilon,\alpha) < \frac{\varepsilon}{2}, \quad 0 \leq U(\alpha';\mathcal{Q}_\varepsilon) - L(\alpha';\mathcal{Q}_\varepsilon) < \frac{\varepsilon}{4M}. \tag{11.10}$$

Then, by our lemma:

$$\begin{aligned}
U(f\alpha',\mathcal{Q}_\varepsilon) - L(f\alpha',\mathcal{Q}_\varepsilon) &= U(f\alpha',\mathcal{Q}_\varepsilon) - U(f;\mathcal{Q}_\varepsilon,\alpha) + U(f;\mathcal{Q}_\varepsilon,\alpha) - L(f;\alpha,\mathcal{Q}_\varepsilon) \\
&\quad + L(f;\mathcal{Q}_\varepsilon,\alpha) - L(f\alpha',\mathcal{Q}_\varepsilon) \\
&\leq 2M\left(U(\alpha',\mathcal{Q}_\varepsilon) - L(\alpha',\mathcal{Q}_\varepsilon)\right) + \left(U(f;\mathcal{P}_\varepsilon,\alpha) - L(f;\mathcal{P}_\varepsilon,\alpha)\right) \\
&< 2M\frac{\varepsilon}{4M} + \frac{\varepsilon}{2} \\
&= \varepsilon.
\end{aligned}$$

Hence, $f\alpha'$ is classically Riemann integrable. Moreover, all our work (including the lemma!) implies also that

$$\begin{aligned}\int_a^b f\alpha' \mathrm{d}x \leq U(f\alpha', \mathcal{Q}_\varepsilon) &< U(f; \mathcal{Q}_\varepsilon, \alpha) + \frac{\varepsilon}{4} \\ &< L(f; \mathcal{Q}_\varepsilon, \alpha) + \frac{3\varepsilon}{4} \\ &\leq \int_a^b f \mathrm{d}\alpha + \frac{3\varepsilon}{4} \\ &\leq U(f; \mathcal{Q}_\varepsilon, \alpha) + \frac{3\varepsilon}{4} \\ &< L(f; \mathcal{Q}_\varepsilon, \alpha) + \frac{5\varepsilon}{4} \\ &\leq L(f\alpha', \mathcal{Q}_\varepsilon) + \frac{3\varepsilon}{2} \\ &\leq \int_a^b f\alpha' \mathrm{d}x + \frac{3\varepsilon}{2}\end{aligned}$$

It follows that for each $\varepsilon > 0$ we have

$$\int_a^b f\alpha' \mathrm{d}x \leq \int_a^b f \mathrm{d}\alpha + \frac{5\varepsilon}{4} \leq \int_a^b f\alpha' \mathrm{d}x + \frac{3\varepsilon}{2}.$$

Sending $\varepsilon \searrow 0$, we infer that

$$\int_a^b f\alpha' \mathrm{d}x = \int_a^b f \mathrm{d}\alpha.$$

Finally, showing that f is α-integrable when $f\alpha'$ is classically Riemann integrable is done in the exact same way. □

11.5 The p-Series as an Improper Integral

We conclude by covering a very nice problem from your assignment 6, which effectively points out a connection between infinite series and integration theory. Concretely, this problem asks you to realize the famous p-series $\sum_{n=1}^{\infty} n^{-p}$ as an appropriate Riemann-Stieltjes integral. More generally, infinite series can also be understood as a Lebesgue integral with respect to the so-called "counting measure".

Example 11.2. Let $p > 1$ be a real number and define $\alpha(x) := \left\lfloor \frac{1}{-x} \right\rfloor$, where $\lfloor \cdot \rfloor$ denotes the floor function (see assignment 6 also). Prove that the improper integral

$$\int_0^1 x^p \mathrm{d}\alpha$$

exists and that

$$\int_0^1 x^p \,\mathrm{d}\alpha = \sum_{n=1}^{\infty} \frac{1}{n^p}.$$

Solution. Note first that we are forced to consider this as an improper integral, as $\alpha(0)$ is not defined. Now, to establish the validity of this improper integration, we must first ensure that for each $t > 0$ the function $f(x) := x^p$ is α-integrable on $[t, 1]$. However this follows readily from class results. Indeed, for each $t > 0$, being that f is continuous on $[0, 1] \supseteq [t, 1]$, we automatically have the α-integrability of f on $[t, 1]$.

Let now $t \in (0, 1)$ be arbitrary. Let $k_t := \alpha(t) = \left\lfloor \frac{1}{-t} \right\rfloor$. Then, $k_t \in \mathbb{Z}$ and $k_t \leq \frac{-1}{t} < k_t + 1$. Moreover, note that $k_t \neq 0$. Consequently,

$$-\frac{1}{k} \leq t < -\frac{1}{k_t + 1}.$$

Now, given any $x \in (0, 1]$ note that $\alpha(x) = k \in \mathbb{Z}$ provided

$$k \leq -\frac{1}{x} < k + 1 \iff -\frac{1}{k} \leq x < -\frac{1}{k+1}.$$

By the additivity theorem, we may write

$$\int_t^1 x^p \,\mathrm{d}\alpha = \int_t^{-1/(k_t+1)} x^p \,\mathrm{d}\alpha + \int_{-1/(k_t+1)}^{-1/(k_t+2)} x^p \,\mathrm{d}\alpha + \cdots + \int_{1/2}^1 x^p \,\mathrm{d}\alpha.$$

Notice that, by our earlier observations, α is constant on each interval appearing in the integrals above, with the exception of the right end-point where α makes a "jump". Moreover, the length (with respect to α) of each interval being integrated over is precisely 1. Thus, we are in a position to apply Proposition 11.3 (see Remark 11.2 also) to each of these integrals above. Doing so, we obtain that

$$\begin{aligned}
\int_t^1 x^p \,\mathrm{d}\alpha &= \int_t^{-1/(k_t+1)} x^p \,\mathrm{d}\alpha + \int_{-1/(k_t+1)}^{-1/(k_t+2)} x^p \,\mathrm{d}\alpha + \cdots + \int_{1/2}^1 x^p \,\mathrm{d}\alpha \\
&= \left(-\frac{1}{k_t+1}\right)^p + \left(-\frac{1}{k_t+2}\right)^p + \cdots + \left(\frac{1}{2}\right)^p + 1^p \\
&= \sum_{n=1}^{-(k_t+1)} \frac{1}{n^p}.
\end{aligned}$$

Finally, noticing that $k_t \to -\infty$ as $t \searrow 0$, we see that $-(k_t + 1) \to +\infty$ which implies that

$$\int_0^1 x^p \,\mathrm{d}\alpha = \lim_{t \searrow 0} \int_t^1 x^p \,\mathrm{d}\alpha = \lim_{t \searrow 0} \sum_{n=1}^{-(k_t+1)} \frac{1}{n^p} = \sum_{n=1}^{\infty} \frac{1}{n^p},$$

where the series on the right is convergent if and only if $p > 1$. □